Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: CALCULATION REPORT FOR SAND FILTER AND DISTRIBUTION BOX - 49W1 Raw Water Treatment Plant

Document Number:	**MD-216-49W1-EG-CV-CAL-1002**
Revision:	**D01**
Revision Date:	**08.03.2024**
Status:	**AS-BUILT**

Outotec Document Number:	900707	GAU01	AG11	PED05	49002	OU100732590	R3
L&T Document Number:	O19045	080	C	K	24890	002	R3

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	23.11.2021	Issued for approval	NanWu	MauNot	LuiPiz	
A02	12.01.2022	Issued for approval	NanWu	MauNot	LuiPiz	
C01	10.02.2022	Issued for construction	NanWu	MauNot	LuiPiz	
D01	08.03.2024	AS-BUILT - Cover updated, Content UnChanged	tadcla	---	tadcla	

TABLE OF CONTENTS

1. Intent 3
2. Scope 3
3. Material Property 3
4. Geometry Of The Structures 4
5. Load Calculations 5
   5.1. Design loads 5
      5.1.1. Dead load 5
      5.1.2. Operational load and Fluid load 5
      5.1.3. Wind load 6
      5.1.4. Seismic load 7
   5.2. Load combinations 9
6. Design basis 3
7. Analysis For Sand Filter 10
   7.1. Load cases in RFEM 11
   7.2. Global deformation and soil pressure results 14
   7.3. Analysis for short side walls 16
   7.4. Analysis for long side walls 24
   7.5. Analysis for intermediate walls 32
   7.6. Analysis for slab 38
8. Analysis for Distribution Box 43
   8.1. Load case in RFEM 44
   8.2. Global deformation results 50

References 64

Attachment 1 Input drawing from flootech 65

Attachment 2 Comment resolution sheet 66

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

1. INTENT

This calculation report illustrates the design process of sand filter and distribution box in Ma'aden Mansourah Massarah Gold Project located in Saudi Arabia.

2. SCOPE

A02

This calculation report is in accordance with structure layout drawing from Flootech, SBC-301 for loads and forces, SBC-304 for concrete structures design, ACI318 Building Code Requirements for Structural Concrete, ACI350R, Environmental Engineering Concrete Structures and project design criteria. The analysis is carried out with Dubal RFEM 5.21 program.

3. MATERIAL PROPERTY

Concrete compressive strength at 28 days: f_{ck} = 35 N/mm^2

Reinforcing bars ASTM A615- Grade 60, yield strength f_{yk} = 420 N/mm^2

4. DESIGN BASIS

A02

According to ACI 318, Exposure categories and classes: Freezing and thawing F0, Sulfate S0, In contact with water W0, Corrosion protection of reinforcement C0.
Max deformation and settlement should be limited to 50mm.
A crack width of 0.4 mm may be acceptable for appearance in most cases. However, the max crack width should be limited to 0.2mm for liquid-retaining structures according to ACI350R.
Vertical modulus of subgrade reaction K_s = 33,750kN/m^3 and horizontal modulus of subgrade reaction K_h = 155,000 kN/m^2 is applied at 0.6m below FGL is given in geotechnical report MD-216-3000-EG-CV-RPT-1008-C01, BH08.
Recommended allowable bearing capacity σ_s = 375kN/m^2. Therefore, the contact stress results shall not exceed the limit.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

5. GEOMETRY OF THE STRUCTURES

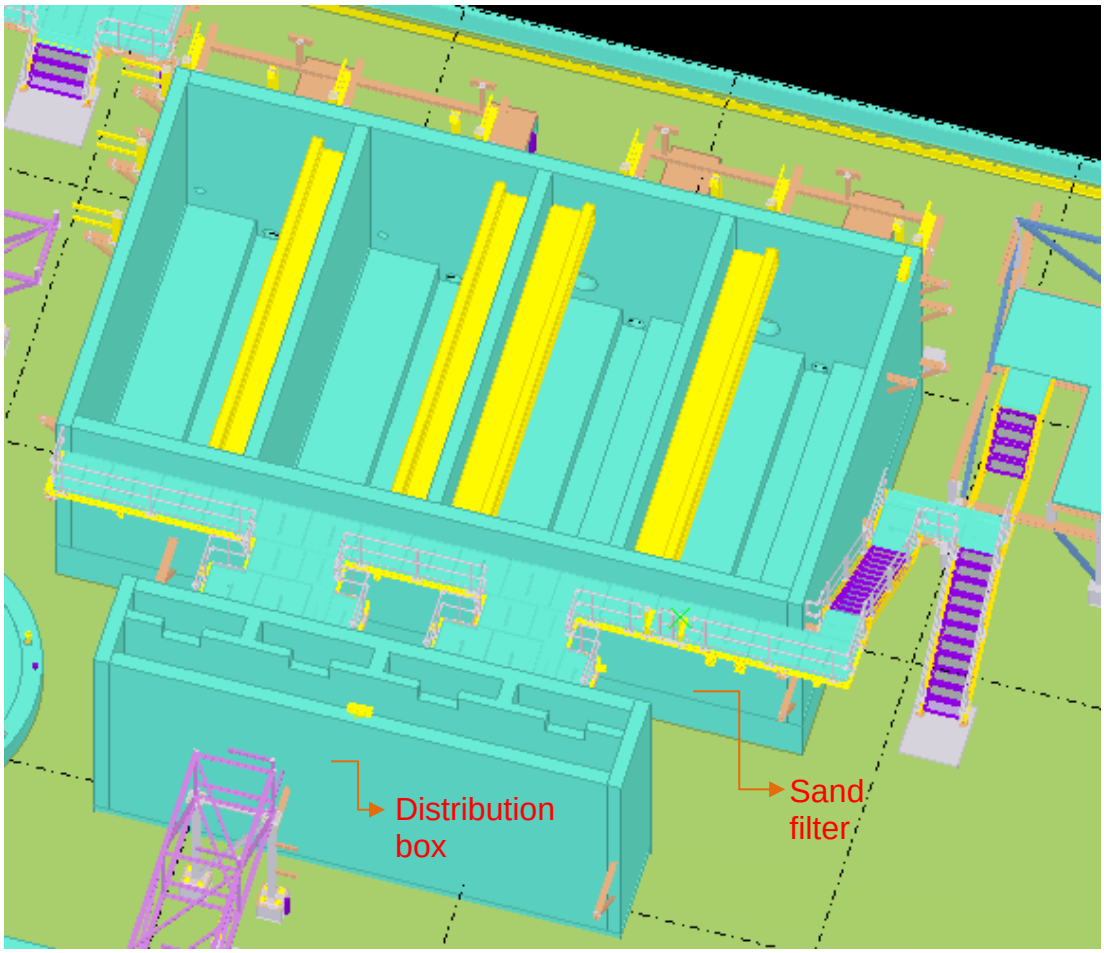


Figure 1 Structure description

A02

Distribution box is full of water in normal operation all the time. For sand filter, 3 internal filters are in operation and 1 in stand-by mode full of water. During air-backwash, water level in the filter is lower than in operation. Maintenance works are done when all of the filters are shut down and empty, because there is only one automatic valve to separate the filter from process.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

6. LOAD CALCULATIONS

6.1. Design loads

All Design Loads are indicated in SBC-301. Basic Load Cases are:

1	Dead load (D)	as Dead load
2	Operational load (F)	as Fluid load
3	Wind load in ±X and ±Y direction (W)	as Wind load
4	Seismic load (E)	as Seismic load

6.1.1. Dead load

Includes self-weight of the structure.

6.1.2. Operational load and Fluid load

for sand filter:

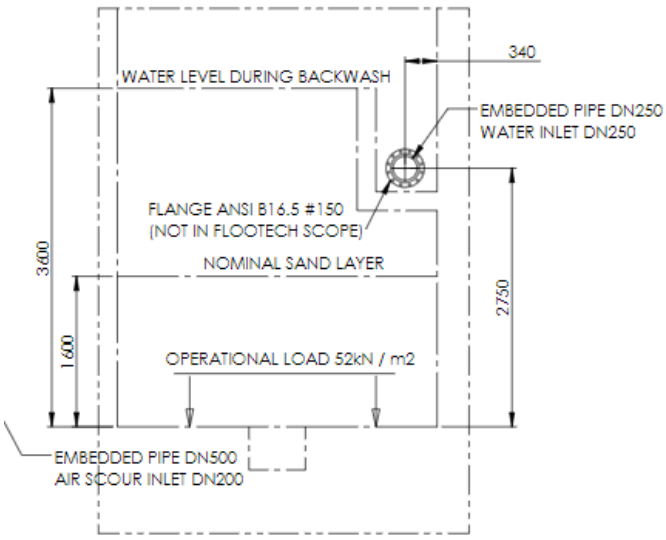


Figure 2 Load input description

Operational load uniformly distributed on the slab is given as 52kN/m^2.

Water weight is uniformly distributed on base slab. Hydrostatic pressure on all vertical walls is calculated and applied (γ_w = 10kN/m^3). Pressure distribution is triangular in nature and the extreme pressure $\gamma_0 H$ exists at the bottom of wall.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Sand weight $\gamma_s H_s$ = 20kN/m^3 x 1600mm=32kN/m^2 is applied as uniform load on slab.

for distribution box:

Water weight is uniformly distributed on slab and hydrostatic pressure on vertical walls.

6.1.3. Wind load

Wind load is calculated according to SBC-301, Section 7. Basic wind speed 43m/s and Site class C are given in project design criteria.

wind load is calculated at height z [m]

$z := 6$

Terrain Exposure Constants, Refer Table 7.2-1(Page7/43) of SBC-301

$z_g := 275 \quad \alpha := 9.5$

The velocity pressure exposure coefficient Kz, As per Table 7.2-2(Page-7/44)

$$K_z := \begin{cases} 2.01 \cdot \left(\frac{5}{z_g}\right)^{\frac{2}{\alpha}} & \text{if } z < 5 \\ 2.01 \cdot \left(\frac{z}{z_g}\right)^{\frac{2}{\alpha}} & \text{otherwise} \end{cases}$$

$K_z = 0.898$

Topographic factor Kzt, topographic effect not considered, section 6.4.3.1 of SBC-301,

$$K_{zt} = \left(1 + K_1 \cdot K_2 \cdot K_3\right)^2$$

$K_{zt} := 1.0$

Wind directionality factor Kd, Table 6.4-1(Page-6/7) of SBC-301

$K_d := 0.95$

Basic wind speed in [km/h], = 43m/s. Exposure category = C

$V_w := 155$

Importance factor. As per Table 6.5-1(Page-6/9) of SBC-301

$I := 1$

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Velocity Pressure qz evaluated at height z: Refer Section 7.2.8 of SBC-301

$$q_z := \left(0.0473 \cdot 10^{-3} \cdot K_z \cdot K_{zt} \cdot K_d \cdot V_w^{\,2} \cdot I\right) \frac{kN}{m^2} = 0.97 \cdot \frac{kN}{m^2}$$

Gust effect factor for a rigid structure. Refer Section 7.2.7

$$G_g := 0.85$$

internal pressure coefficient.

$$GC_{pi} = \pm 0.18$$

$GC_{pi} := 0.18$ use plus and minus value to show positive and negative internal pressure in calculation

Design Wind Loads in [kN/m2] on Rigid buildings of all height: Section 7.2.10.2.1 p.7/8

$$p = q \cdot G \cdot C_p - q_i \cdot (GC_{pi})$$

6.1.4. Seismic load

Seismic load is evaluated using response spectrum method, in horizontal (x,y) and vertical (z) directions, according to SBC 301, 2007. The sum of effective modal mass factor achieved over 95% in each direction.

0,2 sec spectral response accelaration

$$S_S := 0.10g$$

1,0 sec spectral response accelaration

$$S_1 := 0.02g$$

the structure is permitted to be assigned Seismic Design Category A. (Ch. 9.4.1)

value of F_a, based on site class B, defined in Project Design Criteria (Table 9.4.3a and b)

$$F_a := 1.0 \qquad F_v := 1.0$$

the maximum earthquake spectral response acceleration for short periods (9.4.3-1)

$$S_{MS} := F_a \cdot S_S = 0.981 \frac{m}{s^2}$$

the maximum earthquake spectral response acceleration at a period of 1-sec (9.4.3-2)

$$S_{M1} := F_v \cdot S_1 = 0.196 \frac{m}{s^2}$$

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

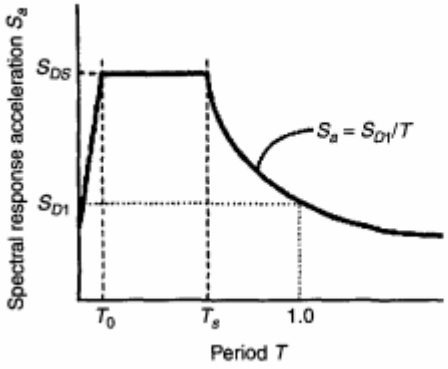
Design spectral Response Acceleration at short period

$$S_{DS} := \frac{2}{3} \cdot S_{MS} = 0.654 \frac{m}{s^2}$$

Design spectral Response Acceleration at 1-sec period

$$S_{D1} := \frac{2}{3} \cdot S_{M1} = 0.131 \frac{m}{s^2}$$

Design Response Spectrum, SBC 301, Section 9.4, p.9/14



$$T_0 := 0.2 \cdot \frac{S_{D1}}{S_{DS}} = 0.04$$

$$T_S := \frac{S_{D1}}{S_{DS}} = 0.2$$

the design spectral response acceleration, Ch. 9.4.5, page 9/13

$$S_a := \begin{cases} \left[S_{DS} \cdot \left(0.4 + 0.6 \cdot \frac{T_T}{T_0} \right) \right] & \text{if } 0 \le T_T \le T_0 \\ S_{DS} & \text{if } T_0 < T_T \le T_S \\ \left(\frac{S_{D1}}{T_T} \right) & \text{otherwise} \end{cases}$$

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

6.2. Load combinations

Ultimate Strength Design - Ultimate Load Combinations: SBC_301, Ch. 2.3)

Load Combination	Expression
101	1,40 (D+F)
102	1,20D + 0,80W
103	1,20D + 1,00E
104	0,90D + 1,60W
105	0,90D + 1,00E

Allowable Stress Design - Serviceability Load Combinations: SBC_301, Ch. 2.4)

Load Combination	Expression
201	D + F
202	D + F + W
203	D + F + 0,70E
204	0,60D + W
205	0,60D + 0,70E

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

7. ANALYSIS FOR SAND FILTER

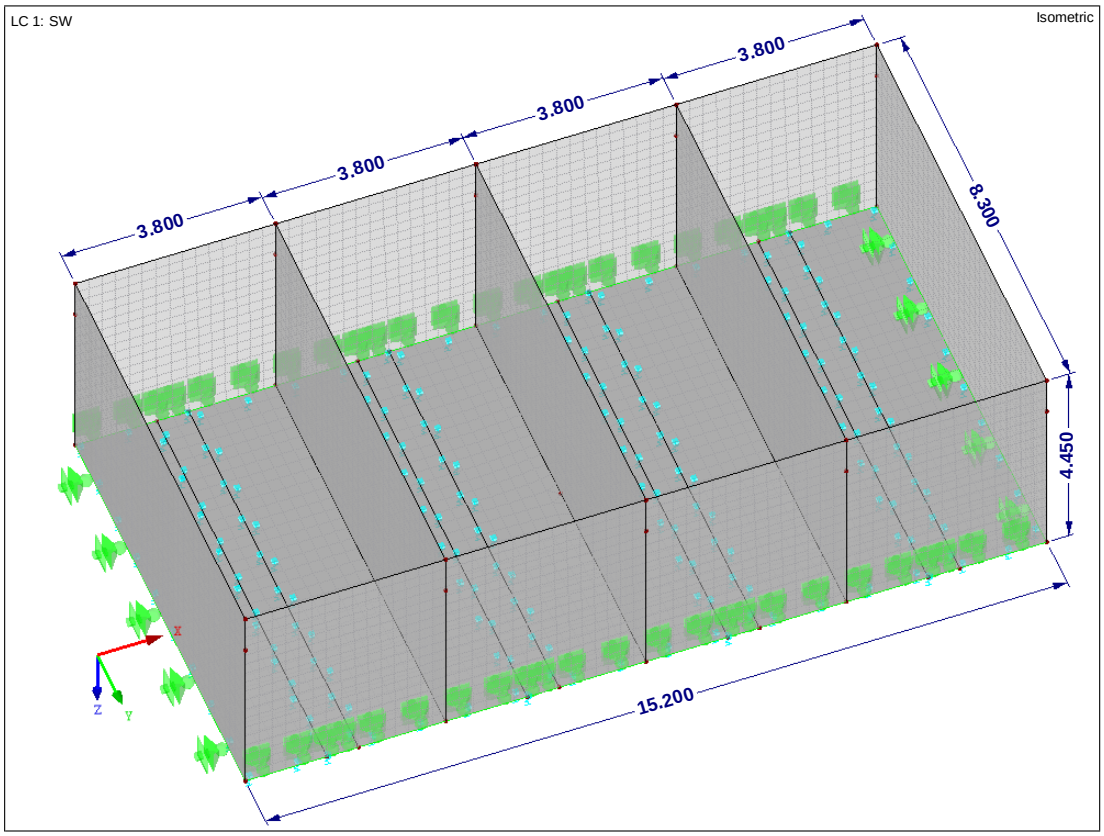


Figure 3 Structure dimension

The sand filter consists of side walls, intermediate walls, and base slab. All vertical walls are supported by the base slab, which will be supported by compacted gravel underneath. Vertical walls have a thickness of 300mm. Bottom slab has smallest thickness 300mm and max thickness 800mm.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

7.1. Load cases in RFEM

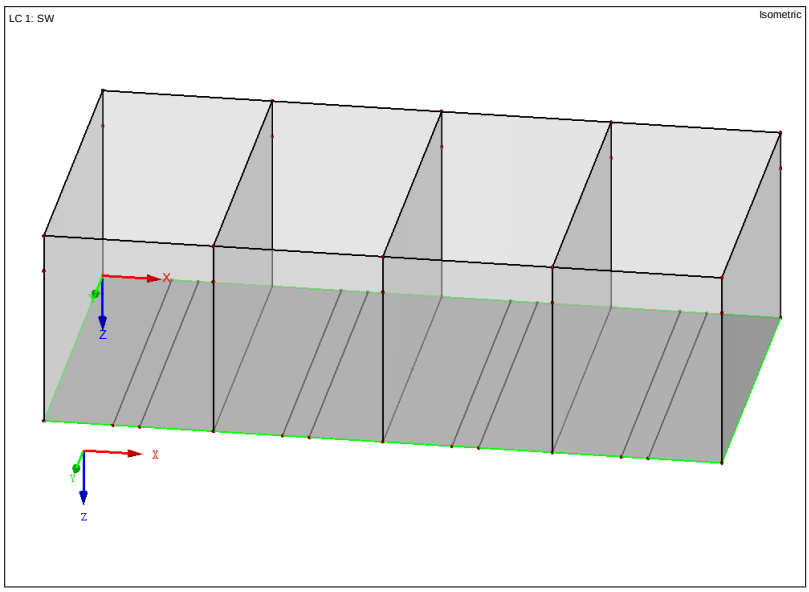


Figure 4 Self-weight and dead load

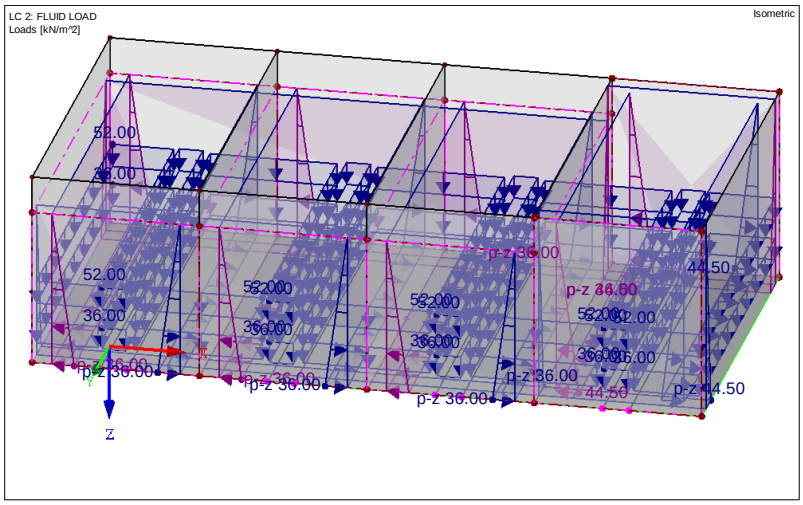


Figure 5 Operational load and fluid load

Operational level for fluid 3.6m and full wall height 4.45m. The maximum characteristic fluid pressure can be calculated as product of the Density and height 10 kN/m^3 * 3.6m = 36 kN/m^2.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

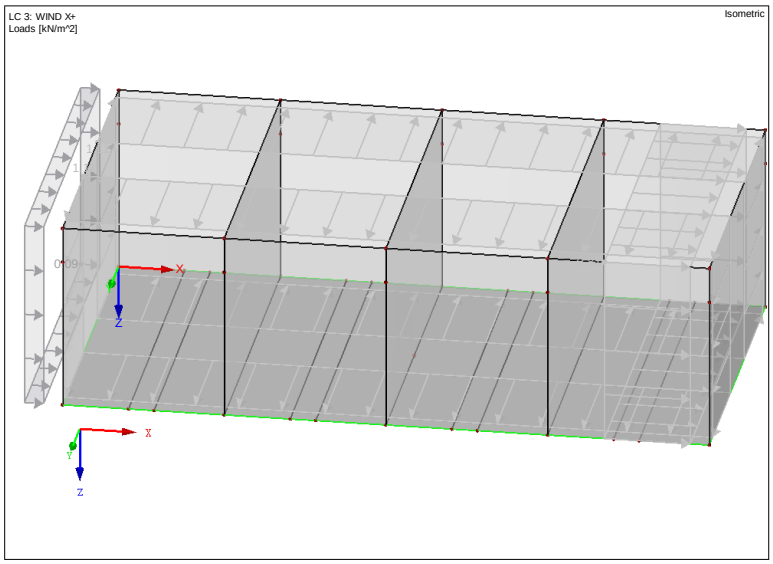


Figure 6 Wind load in X-direction neg

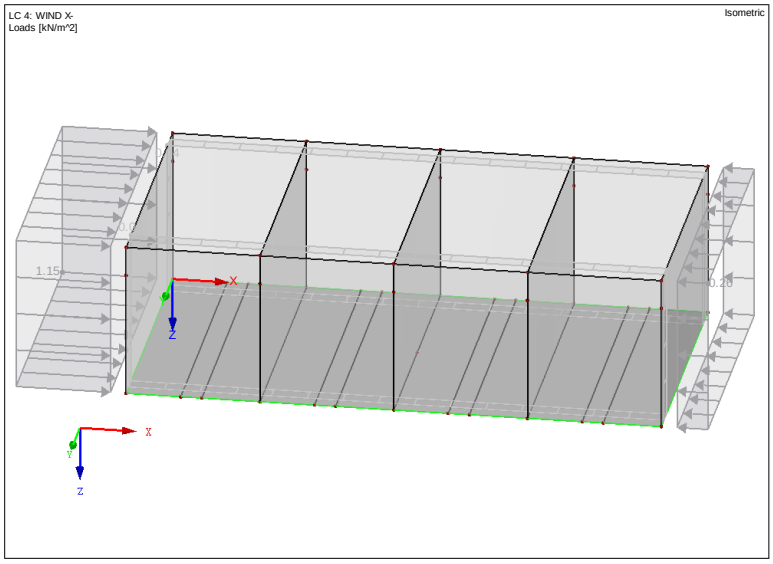


Figure 7 Wind load in X-direction pos

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

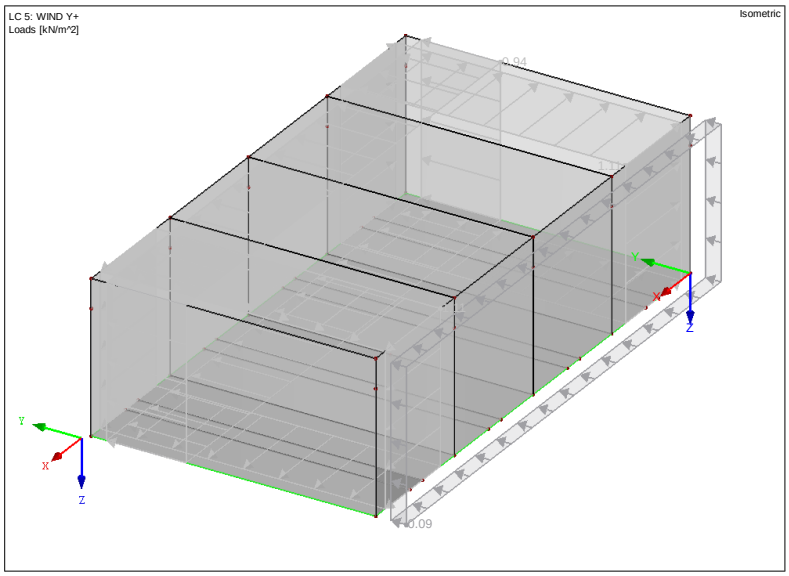


Figure 8 Wind load in Y-direction neg

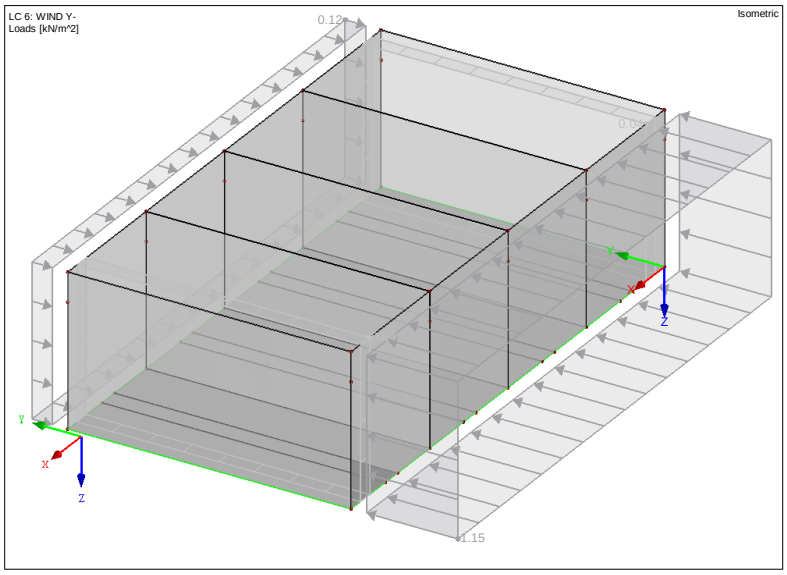


Figure 9 Wind load in Y-direction pos

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

7.2. Global deformation and soil pressure results

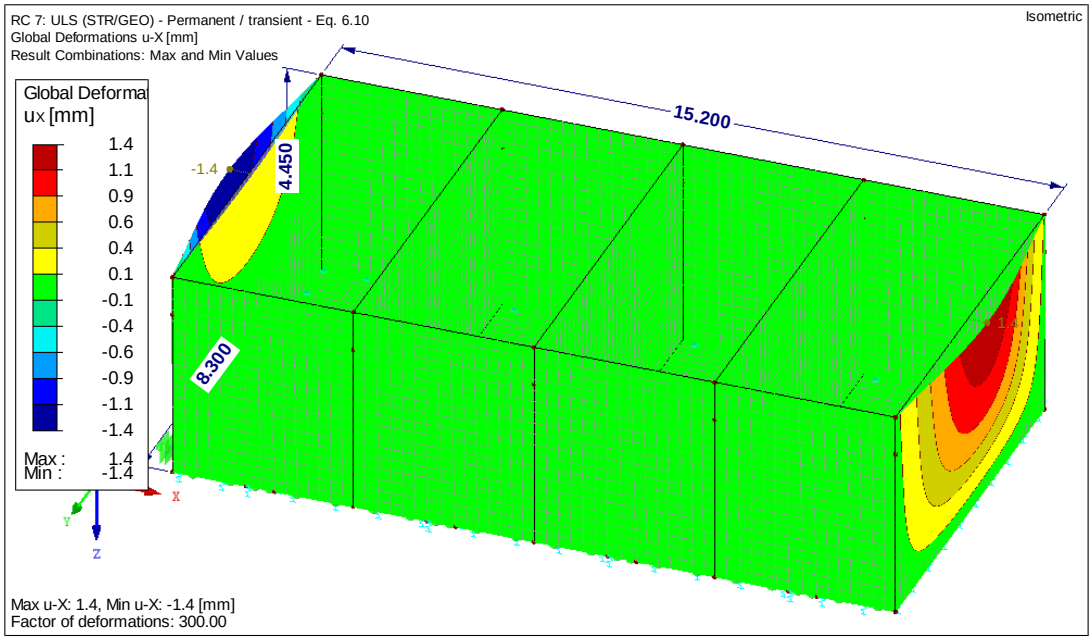


Figure 10 Global deformation in x-direction u_x

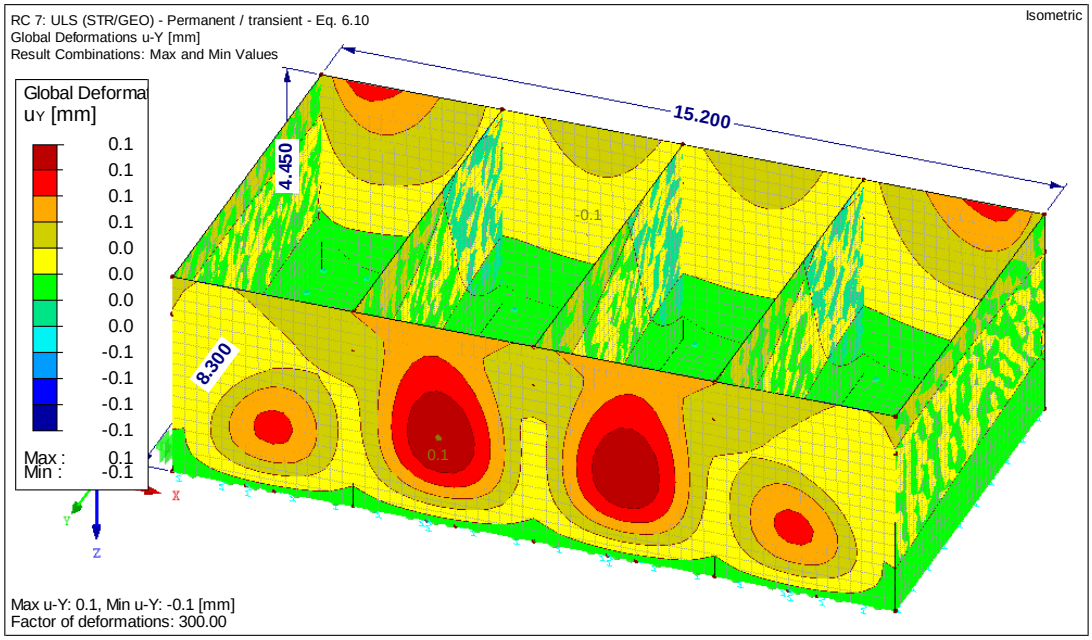


Figure 11 Global deformation in y-direction u_y

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

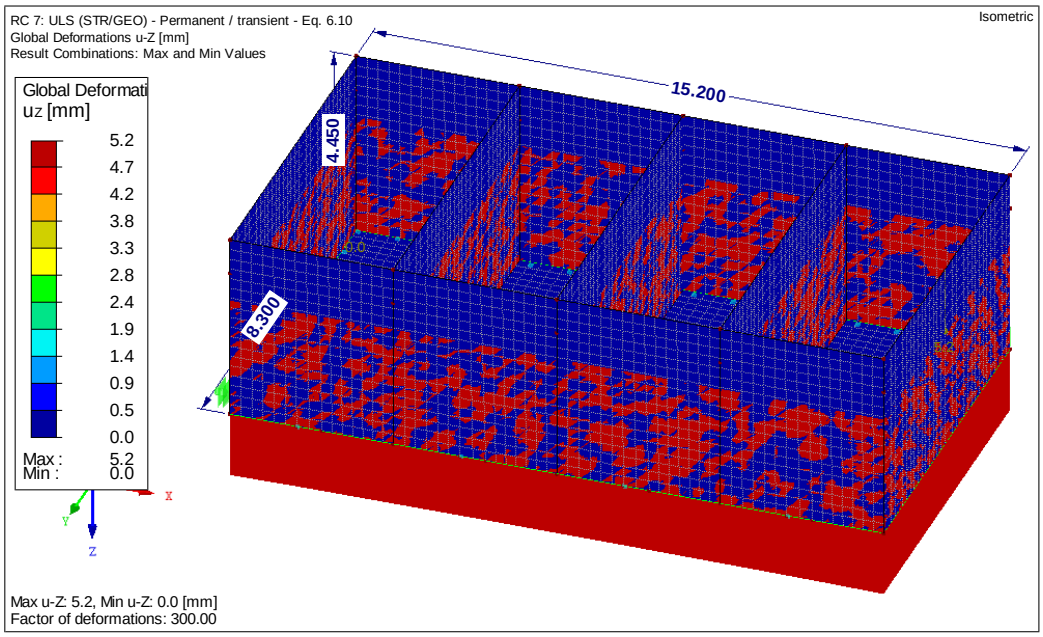


Figure 12 Global deformation in z-direction u_z

$u < u_{max}$ =50mm, ok!

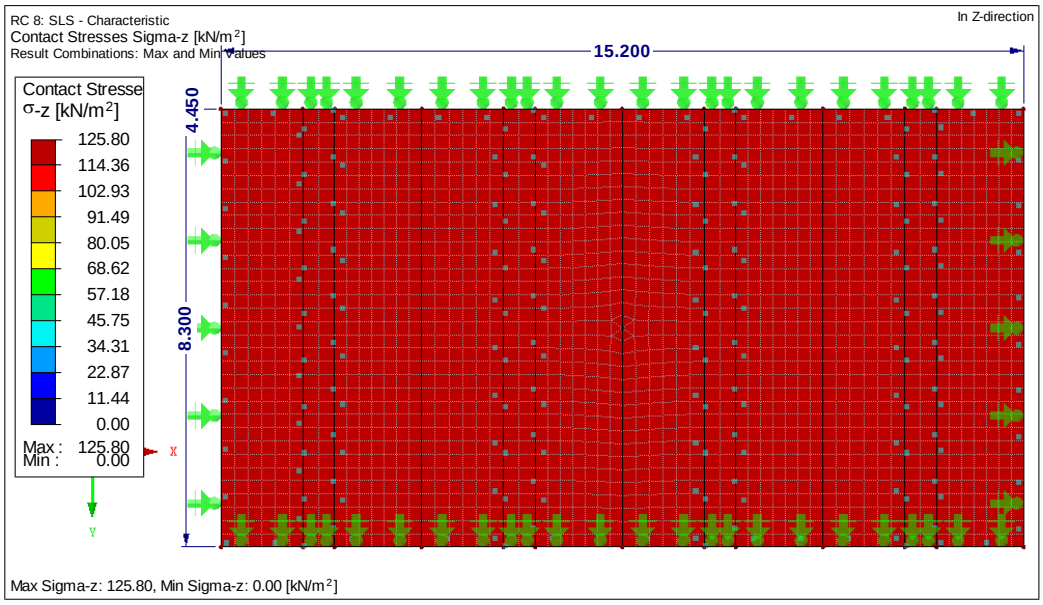


Figure 13 Max. contact stress result at ULS

$\sigma < \sigma_{max}$ = 375kN/m^2, ok!

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

7.3. Analysis for short side walls

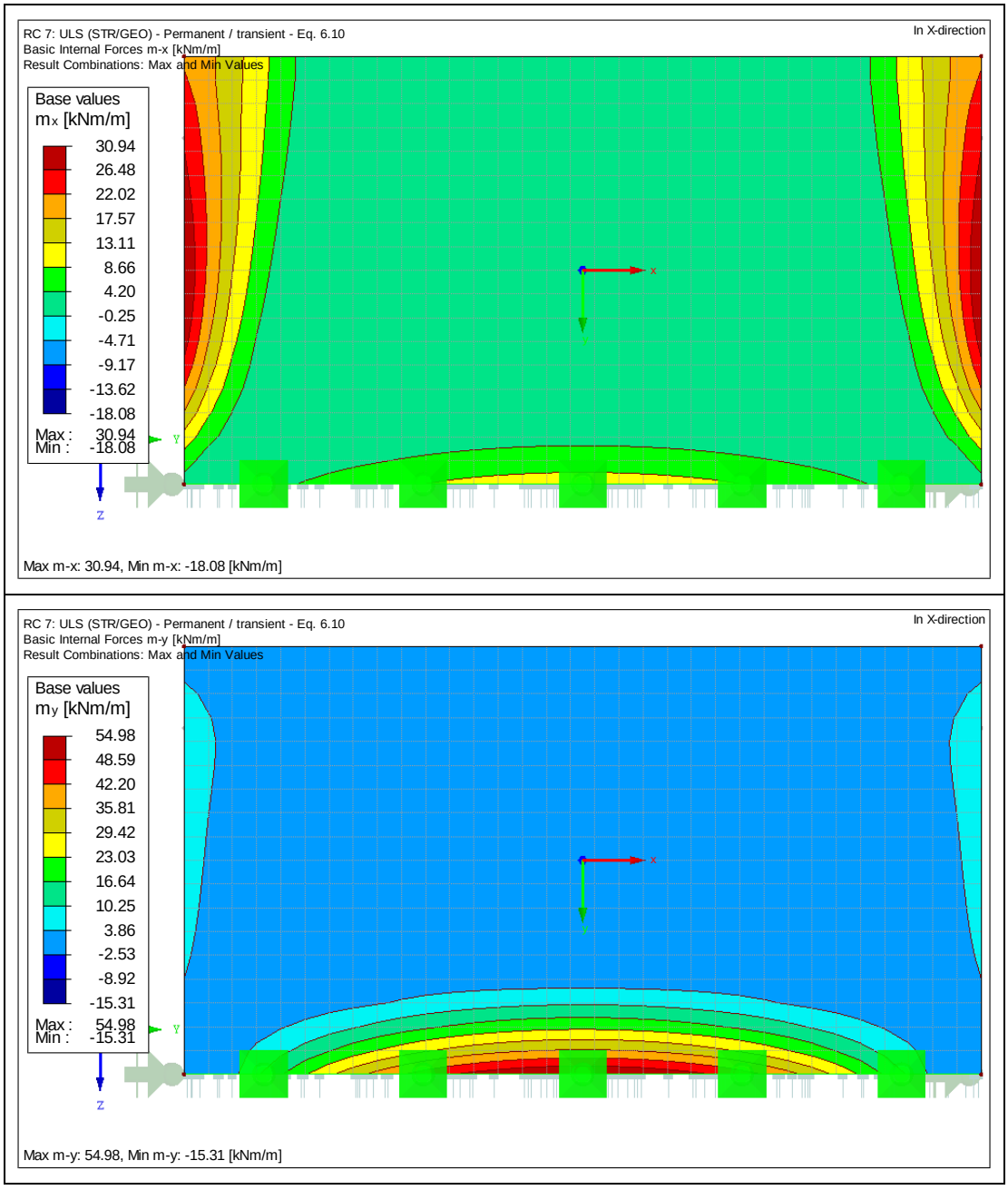
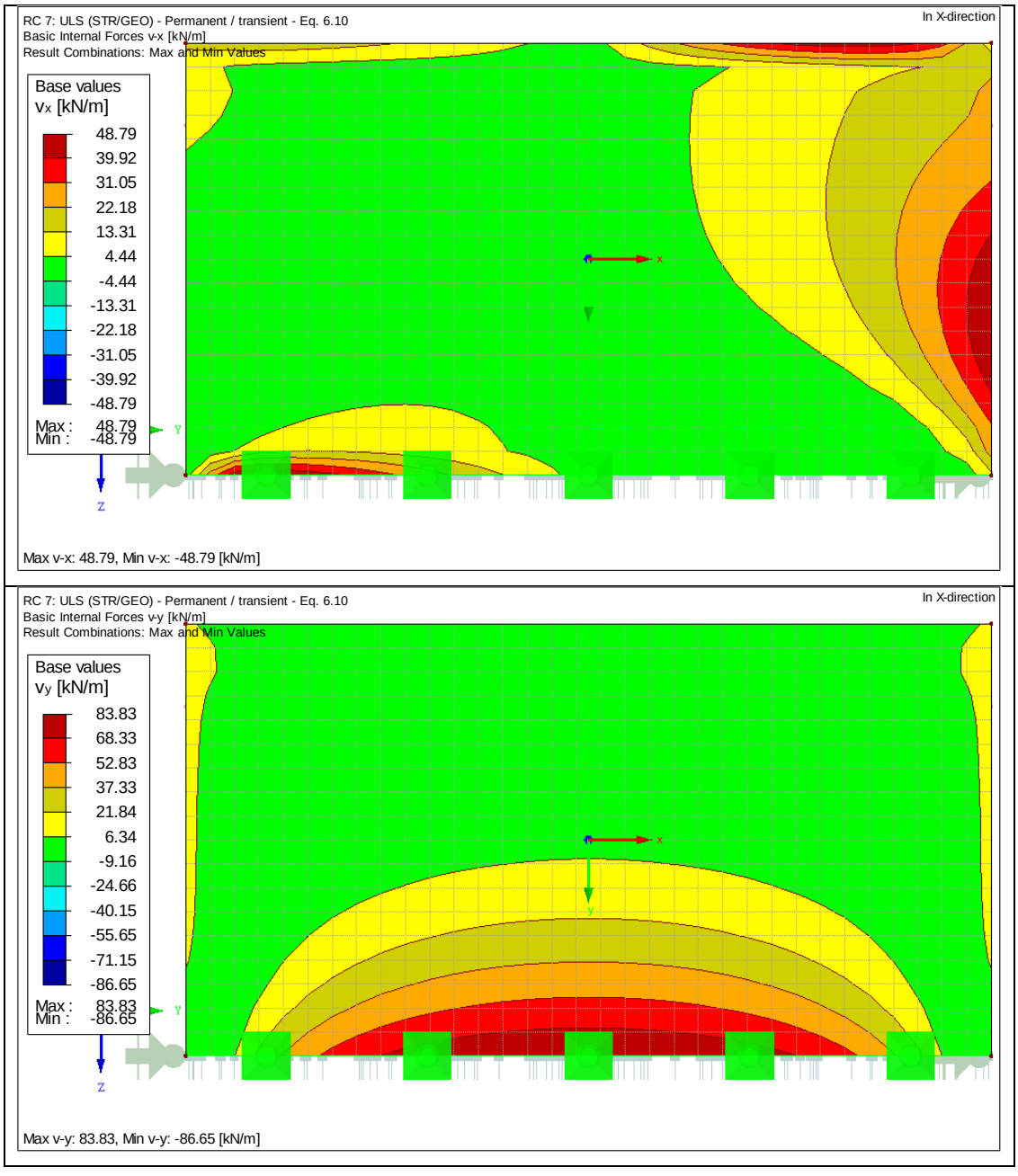


Figure 14 Bending moment results for short side walls

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



A02

Figure 15 Shear force result for short side walls

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

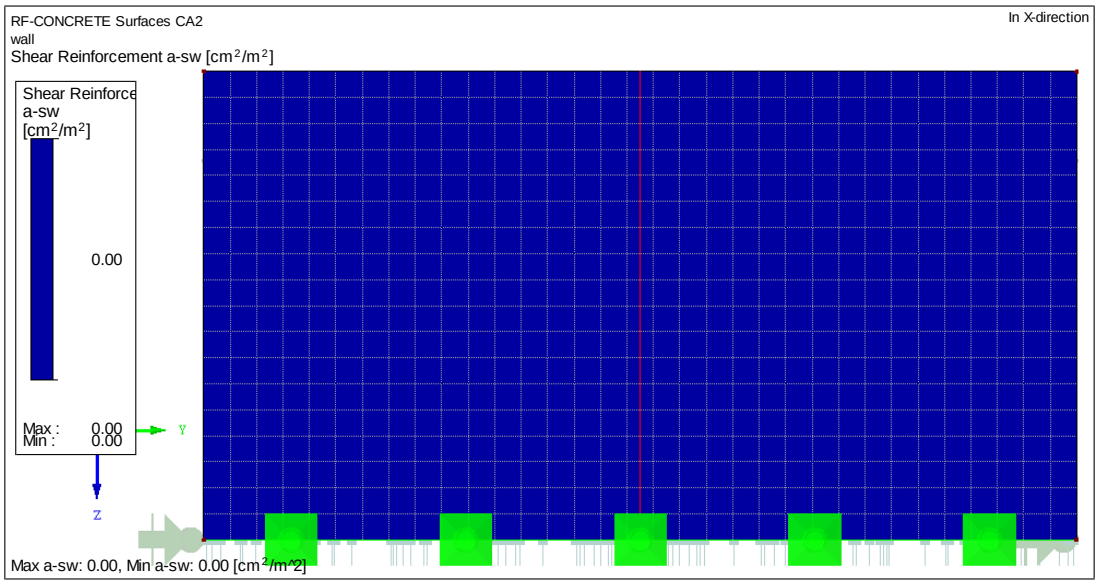


Figure 16 Required shear reinforcement area

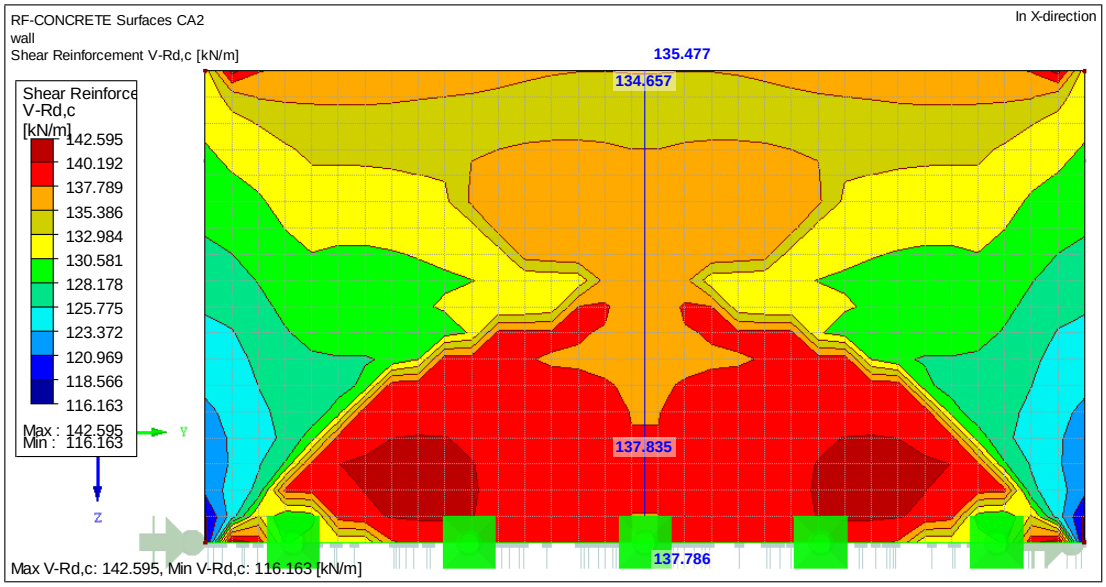


Figure 17 $V_{Rd,c}$ design shear resistance without shear reinforcement

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

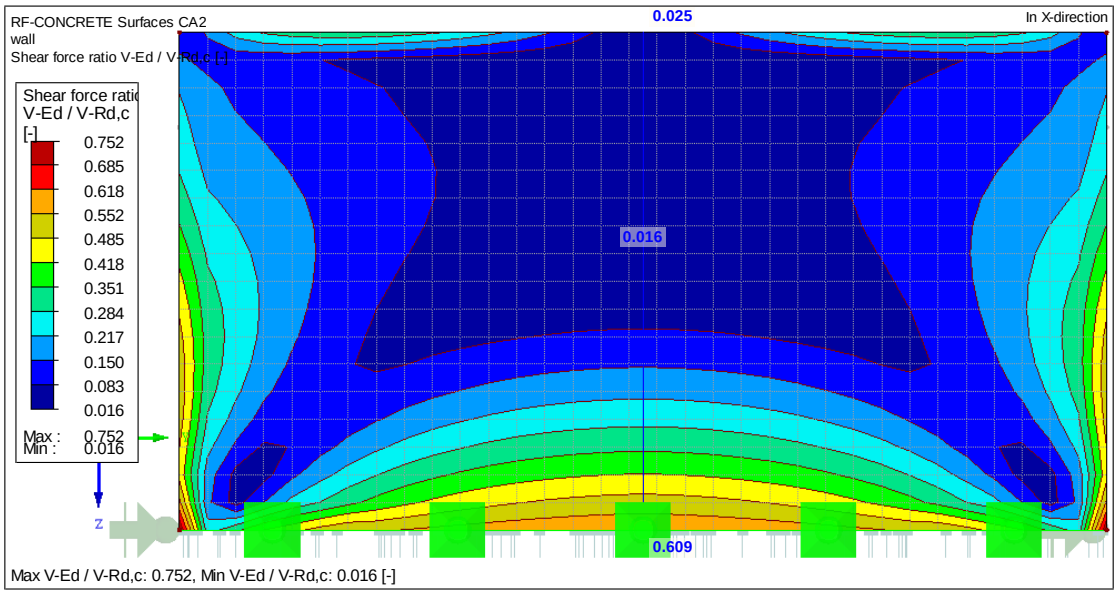
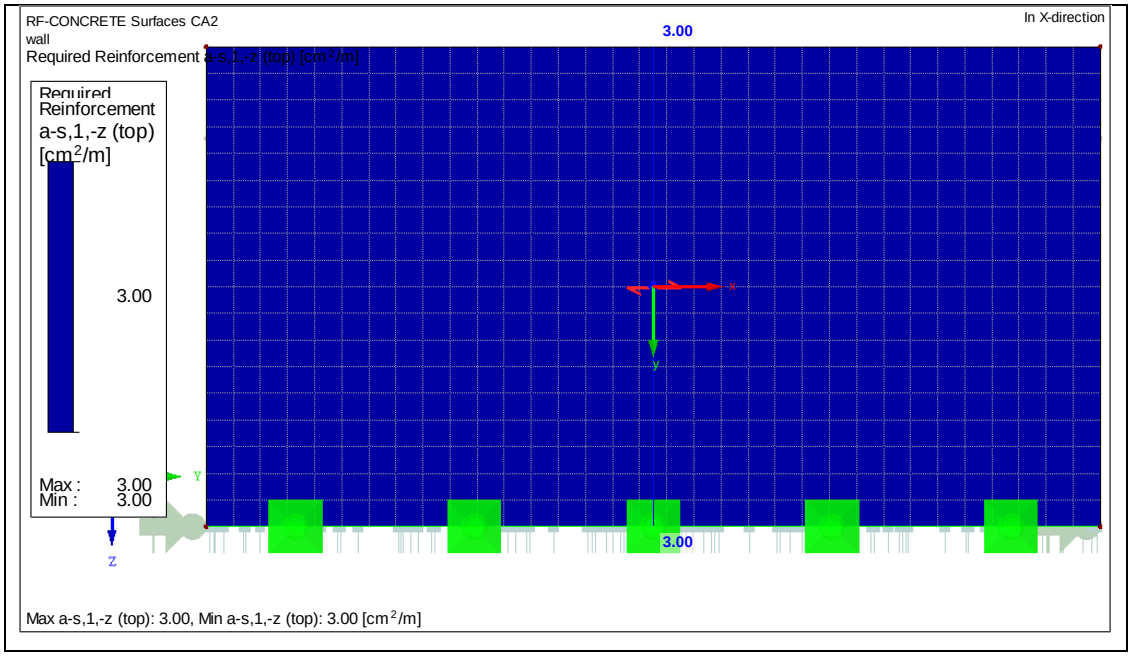


Figure 18 utility ratio $V_{Ed}/V_{Rd,c}$

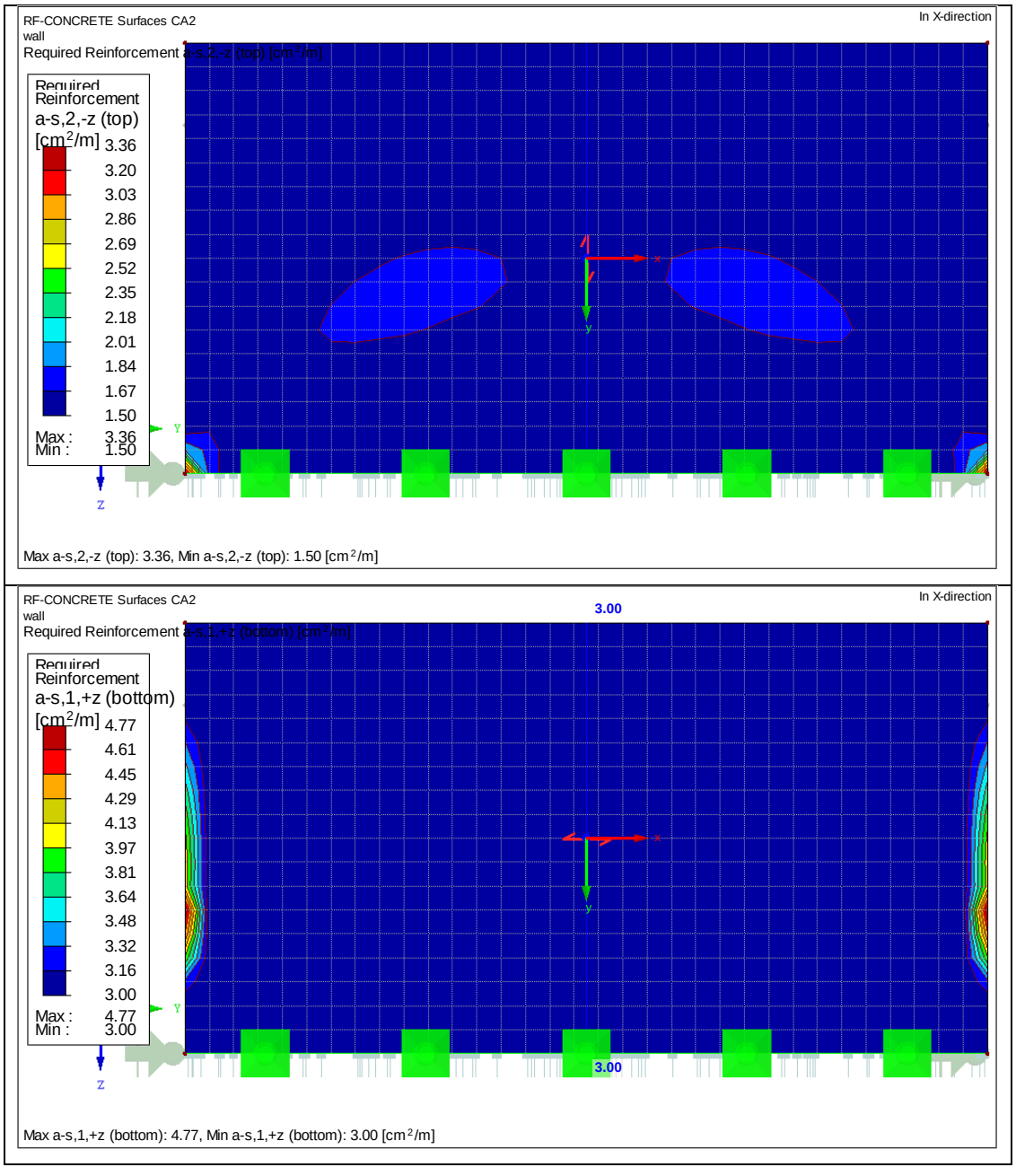
utility ratio= applied shear force/ shear resistance= $V_{Ed}/V_{Rd,c}$< 1.0, ok!



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

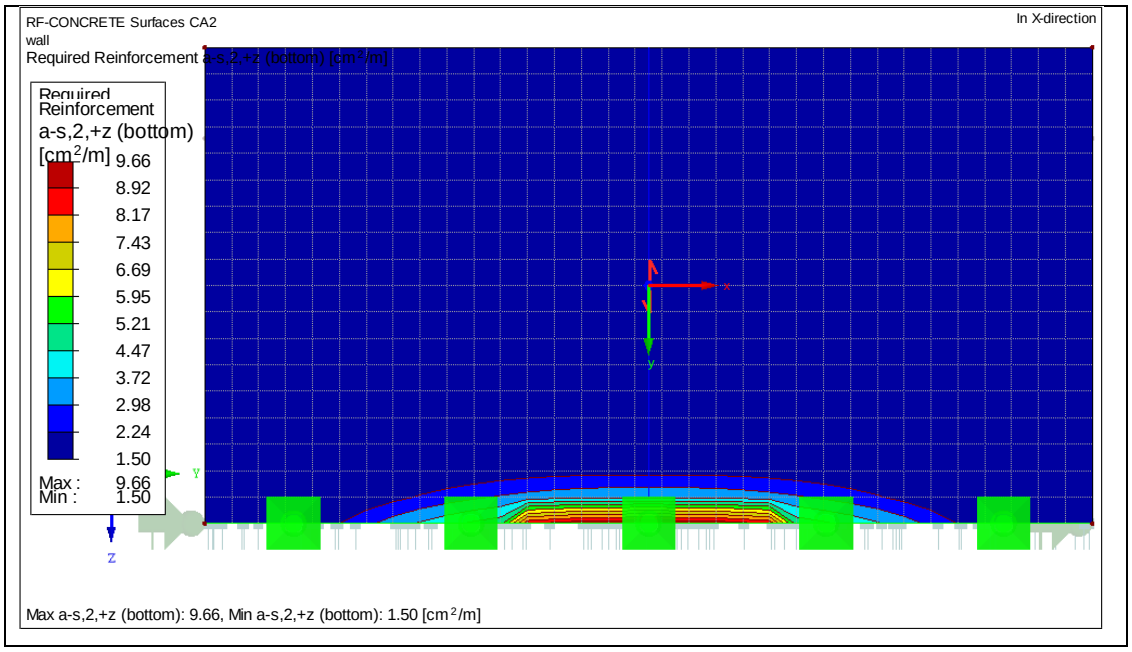
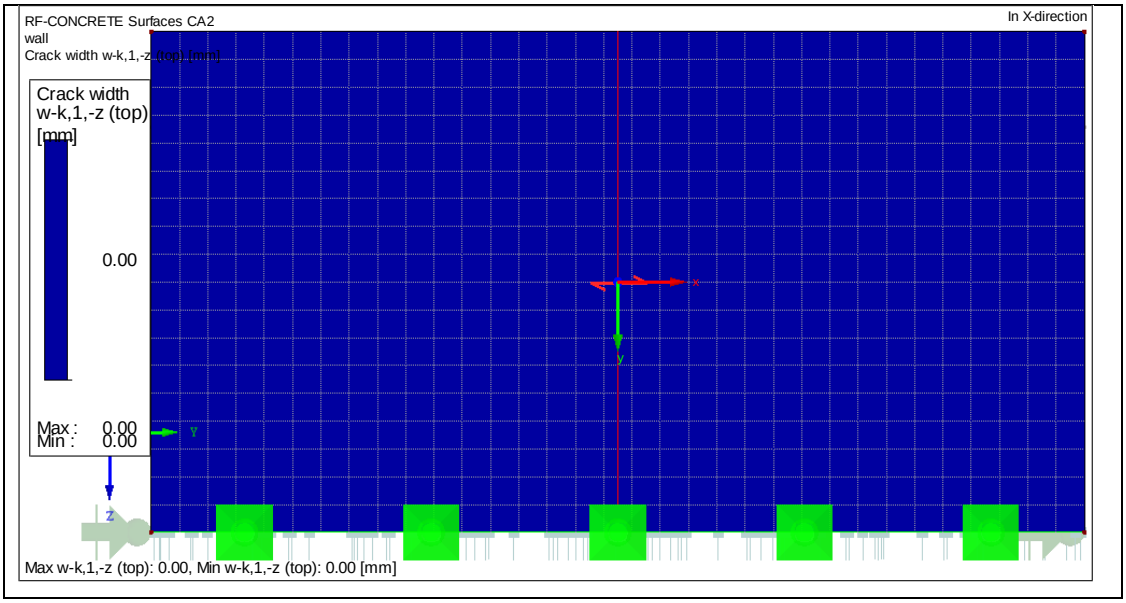


Figure 19 Required reinforcement area for short side walls

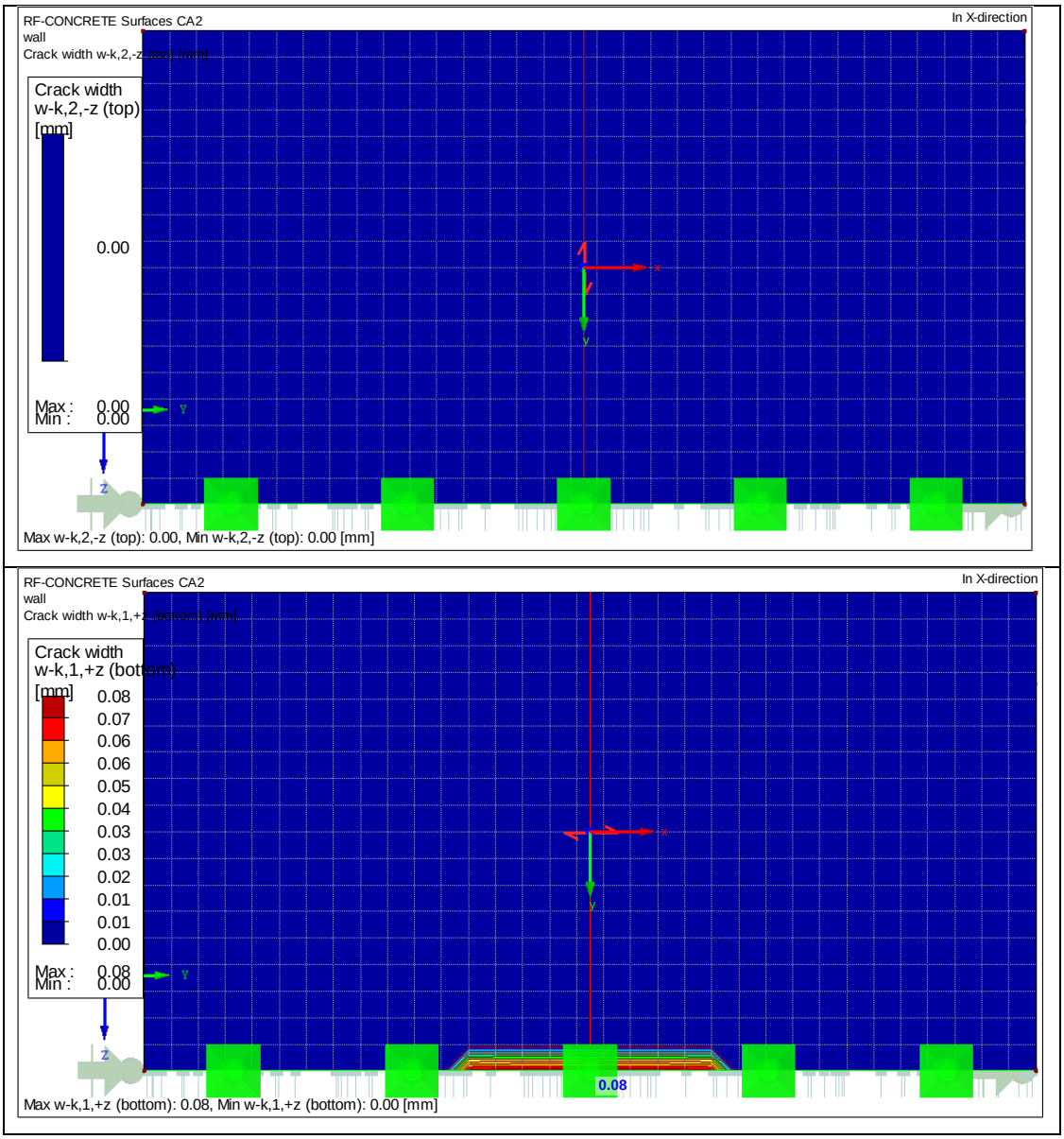
Provide basic reinforcement T12@150, A_{st} = 7.54 cm^2/m



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

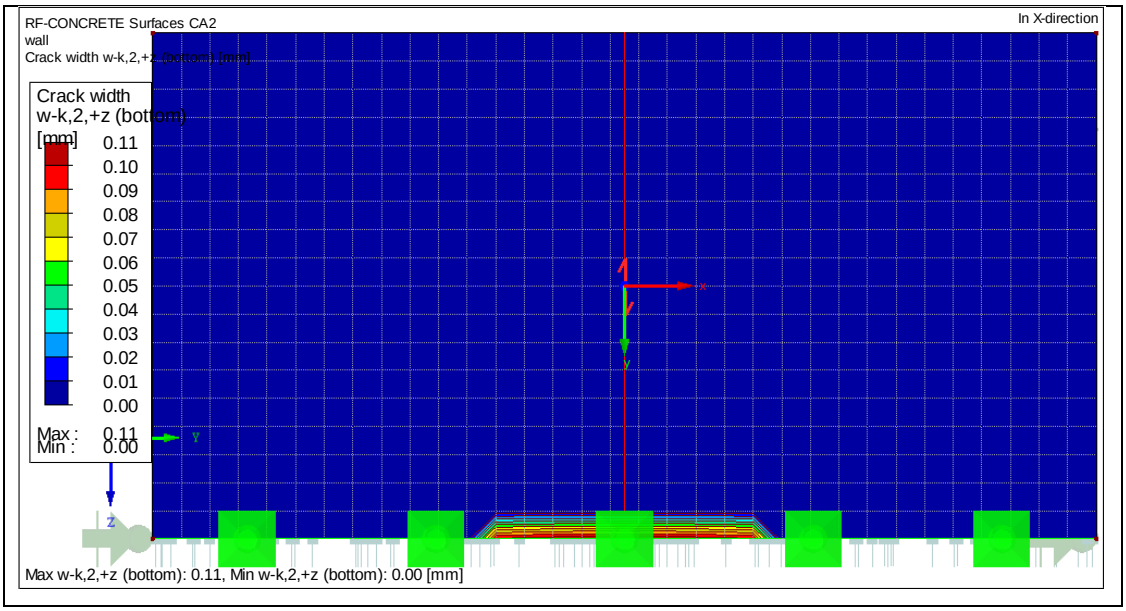


Figure 20 Cracking check

A02

$w_k < w_{k,max}$ = 0.20mm, ok!

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

7.4. Analysis for long side walls

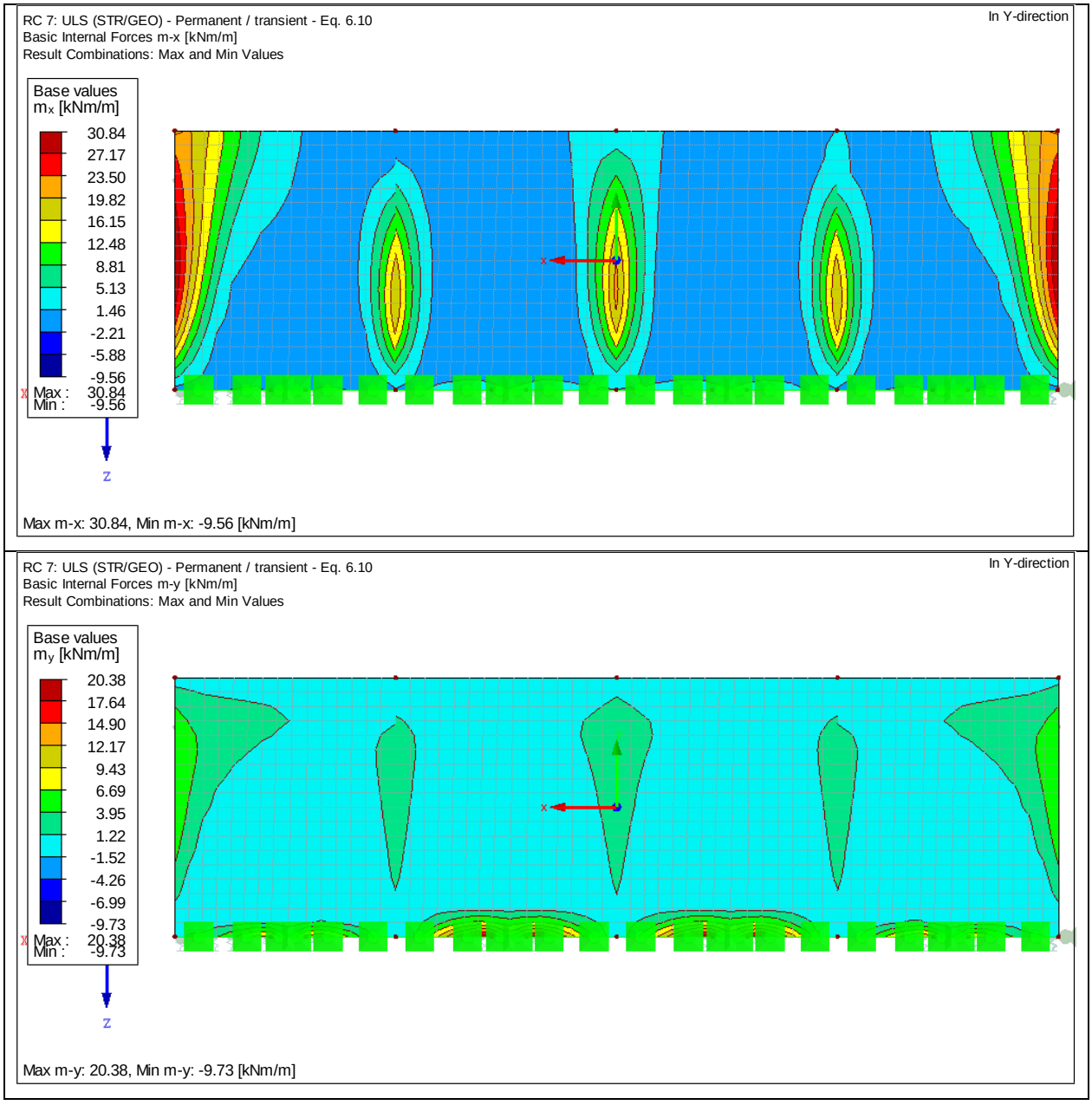
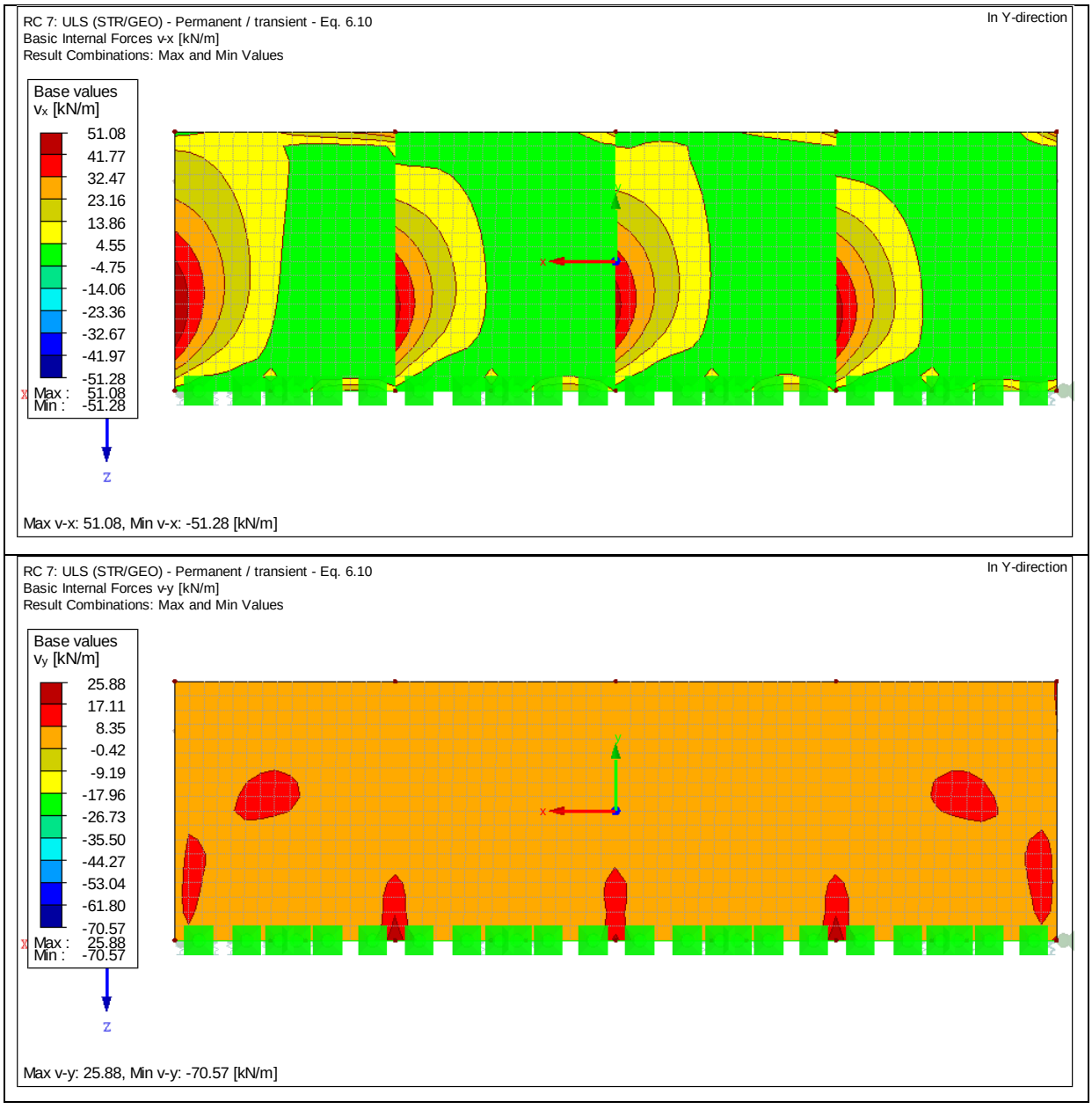


Figure 21 Bending moment results for long side walls

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



A02

Figure 22 Shear force results for long side walls

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

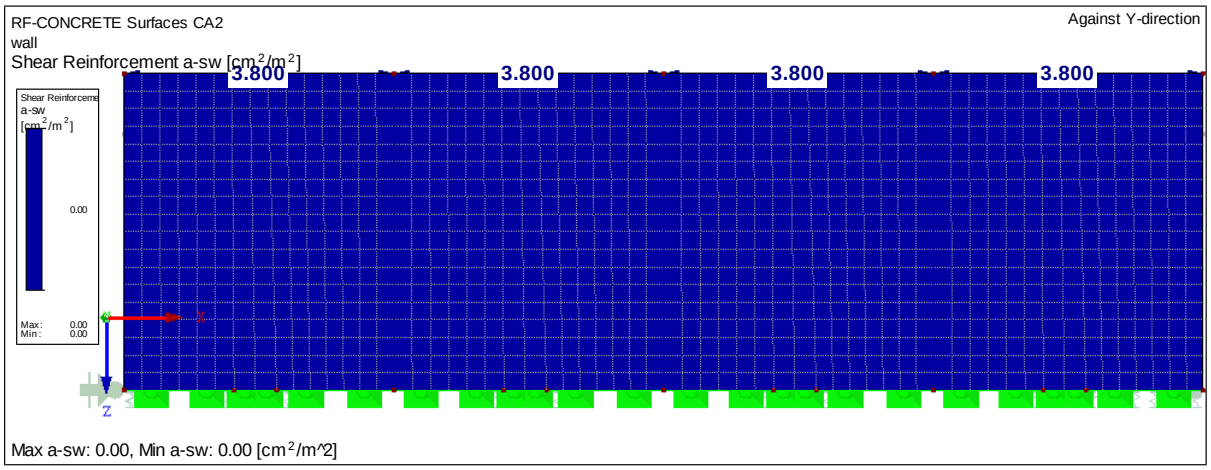


Figure 23 Required shear reinforcement area for long side walls

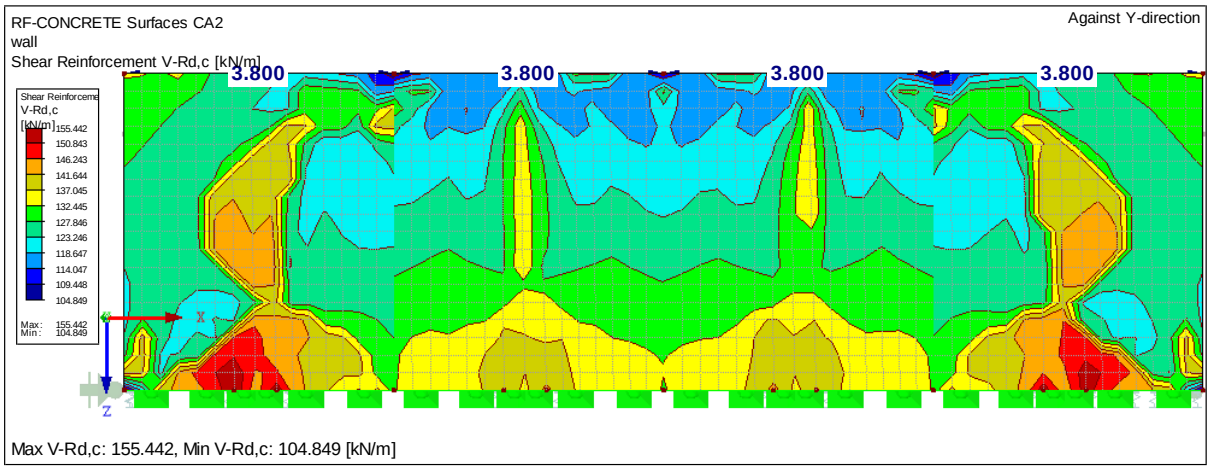


Figure 24 $V_{Rd,c}$ design shear resistance without shear reinforcement

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

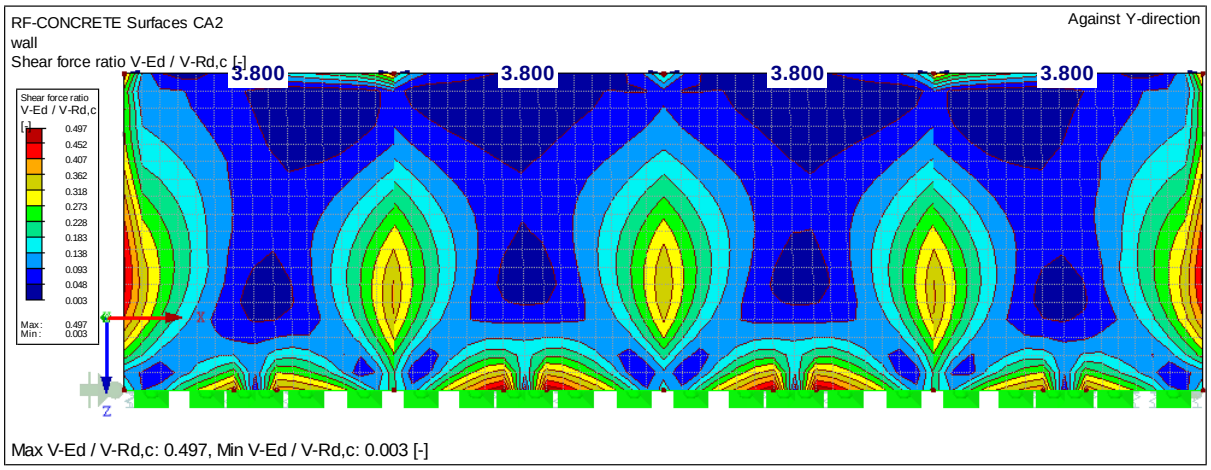
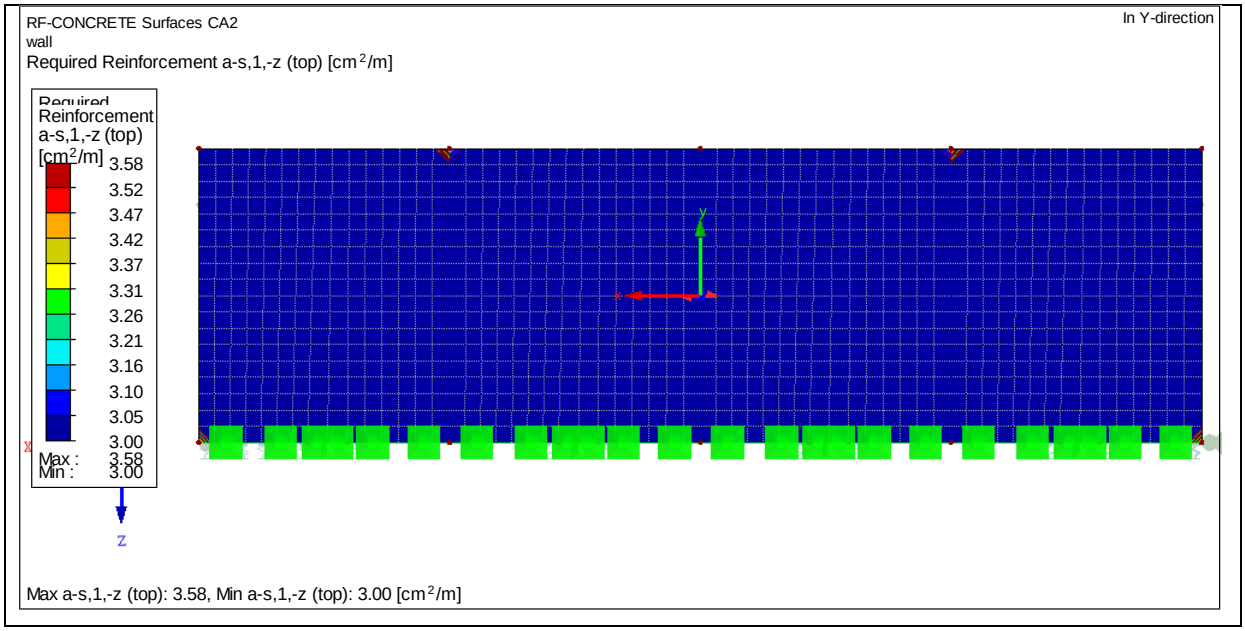


Figure 25 utility ratio $V_{Ed}/V_{Rd,c}$

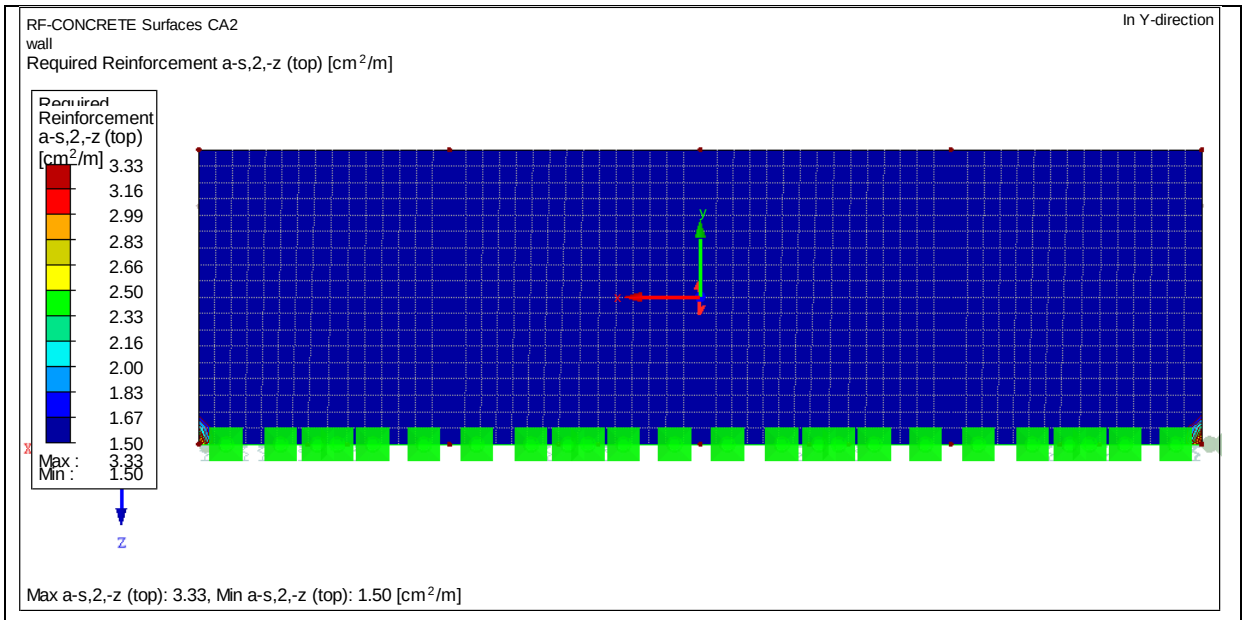
utility ratio= applied shear force/ shear resistance= $V_{Ed}/V_{Rd,c}$< 1.0, ok!

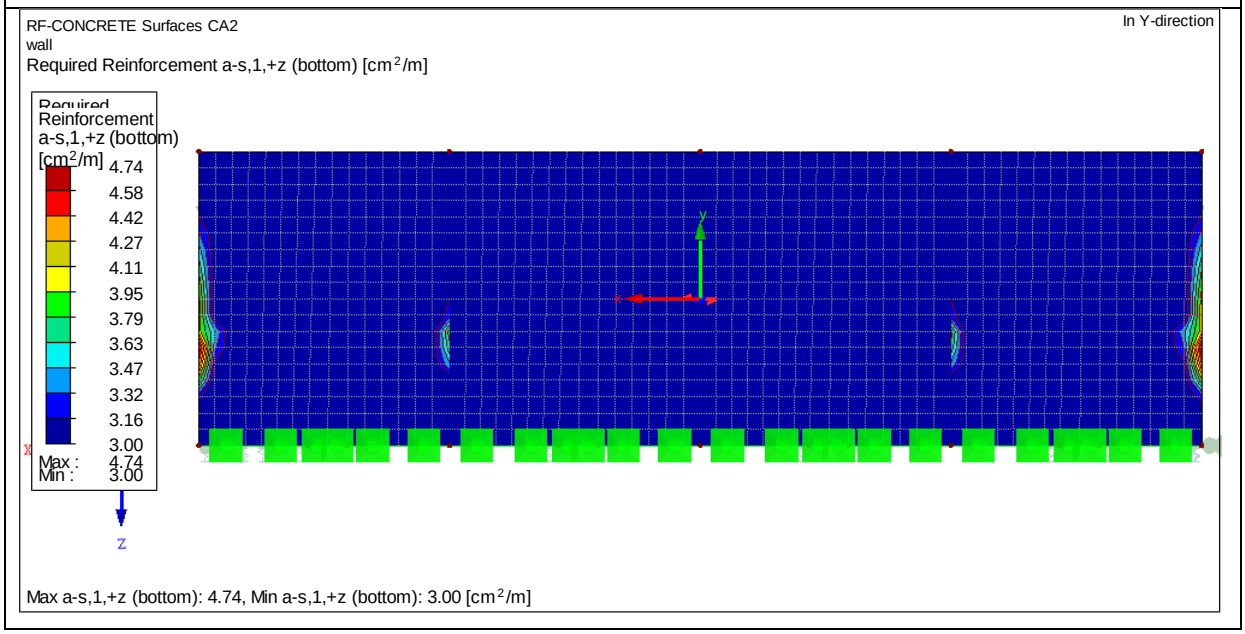


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.





PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

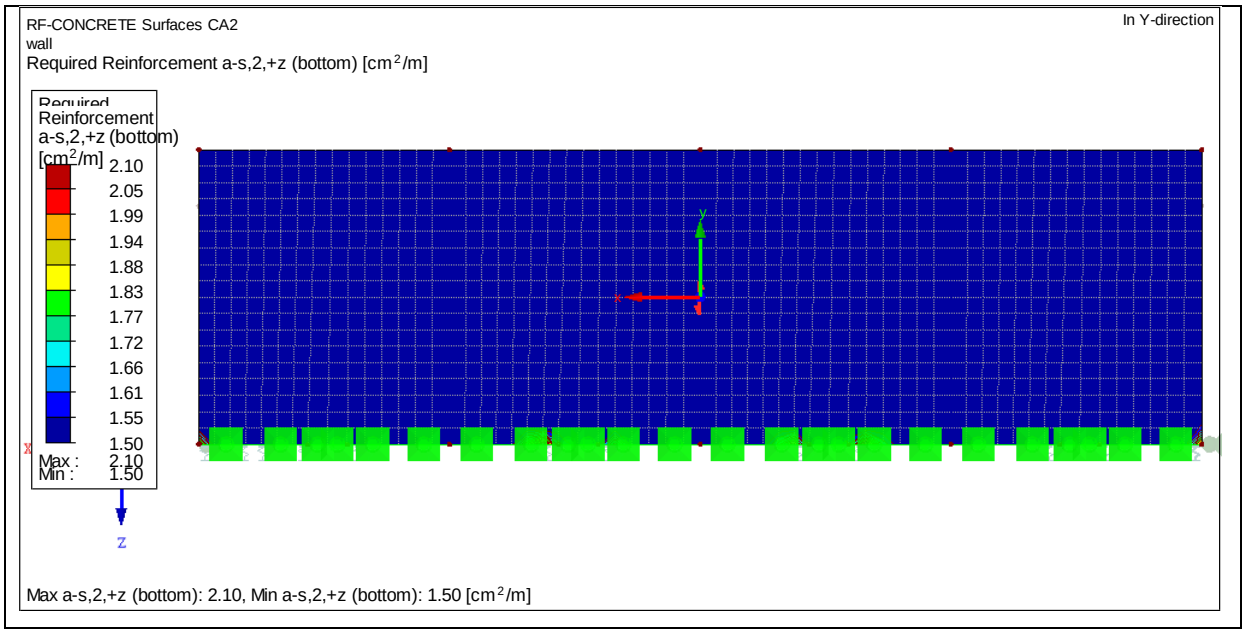


Figure 26 Required reinforcement area for long side wall

According to SBC 304, Ch. 14.3 and ACI 318 Ch. 11,

Provide vertical reinforcement T12@150mm, $A_{v,p}$ = 7,54 cm^2/m.

ρ_v = 0.0025%> $\rho_{v,min}$ = 0.0012, for $d_{bar} \leq 16mm$ and $f_{yk} \geq 420MPa$,

ρ_v = 0.0025%< $\rho_{v,max}$ = 0.08,

s_v = 150mm< $s_{v,max}$ = min (3h,450mm), ok

Provide horizontal reinforcement T16@150mm, $A_{h,p}$ = 13,04 cm^2/m to resist shrinkage.

ρ_h = 0.0043%> $\rho_{h,min}$ = 0.0020, for $d_{bar} \leq 16mm$ and $f_{yk} \geq 420MPa$,

ρ_h = 0.0025%< $\rho_{h,max}$ = 0.08,

s_h = 150mm< $_{sh,max}$ = min (3h,450mm), ok

A02

Below is an additional check regarding minimum reinforcement required to ensure controlled cracking for constraint according to DIN 1045-1.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

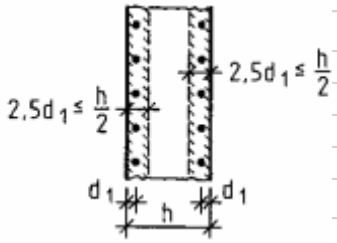
Shrinkage in concrete, Peter Gebhard, Germany — Versio 1.0

1. Structure information			2. Material property		
concrete f_{ck}	**35**	N/mm²	f_{ctm}	3.21	N/mm²
concrete is:	**young**		$f_{ct,eff}$	1.60	N/mm²
Slastic modulus of steel E_s	**200000**	N/mm²			
shrinkage reinforcement ϕ d_s	**16**	mm		design factor k_c	
cross section height h_1	**300**	mm	**tension**	k_c	1
cross section width b_1	**1000**	mm			
concrete cover	**50**	mm		design factor k	
shear reinforcement ϕ	**0**	mm	thin structure	k	0.8
crack width limit w_k	**0.3**	mm	thick structure	k	0.5
			value used in calculation	***k***	**0.8**

effective cross section

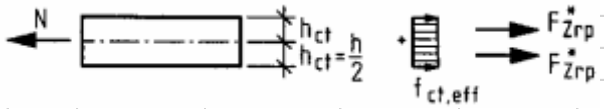
effective depth d_1 58 mm

$$A_{c,eff} = \begin{Bmatrix} 2{,}5 d_1 b \\ \leq \frac{h}{2} b \end{Bmatrix}$$



eff. concrete tensile zone $A_{c,eff}$ 145000 mm² /face

concrete tensile zone A_{ct} = 150000 mm² /pinta $A_{ct}=b*h_1/2$



SHRINKAGE FORCE IN STEEL

shrinkage force in steel $F_{Zrp} = k_c k A_{ct} f_{ct,eff}$ 192597.7 N /face

CRACKING FORCE

cracking force $F_{cr} = A_{c,eff} f_{ct,eff}$ 232722.3 N /face

SHRINKAGE REINFORCEMENT Fcr>FZrp -> As2

reinforcement 1 $A_{s1} = \sqrt{\frac{d_s A_{c,eff}}{3{,}6 W_k E_s}\left(F_{Zrp} - 0{,}4 F_{cr}\right)}$ 1034 mm² /face

reinforcement 2 $A_{s2} = \sqrt{\frac{d_s 0{,}6 F_{ZRp}^2}{3{,}6 W_k E_s f_{ct.eff}}}$ 1014 mm² /face

Required steel area per surface	**$A_{s.vaad}$=**	**1014**	**mm²**	**/m/face**
	ϕ [mm]	‹k_{vaad} [mm]		
Bar size and required spacing	16	198	/face	

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

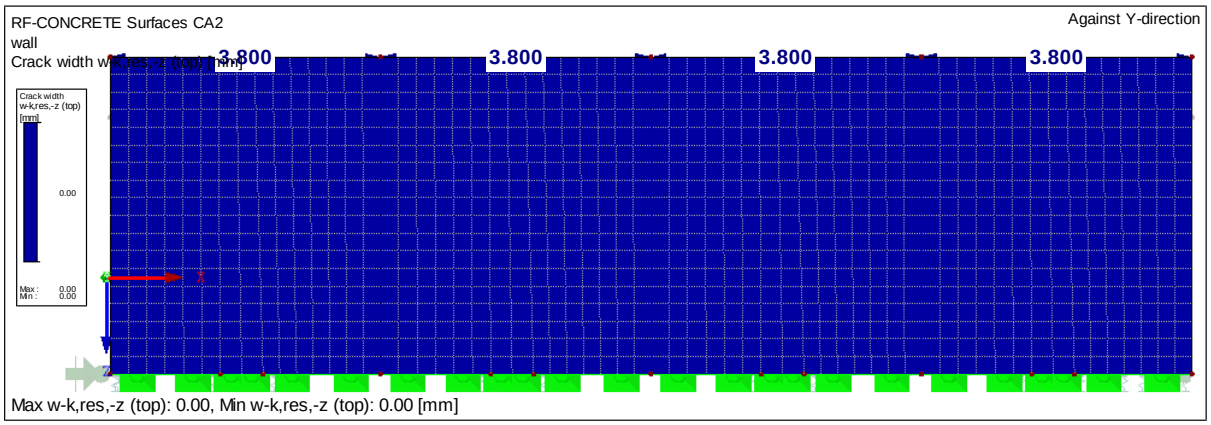


Figure 27 Crack at top $w_{k,top}$

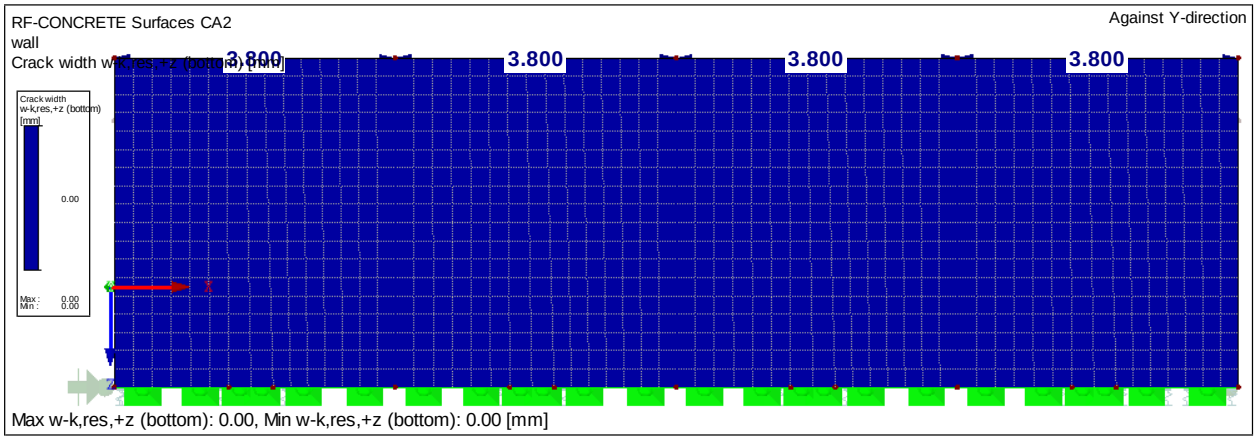


Figure 28 Crack at bottom $w_{k,bottom}$

$w_k < w_{k,max} = 0.20mm$, ok!

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

7.5. Analysis for intermediate walls

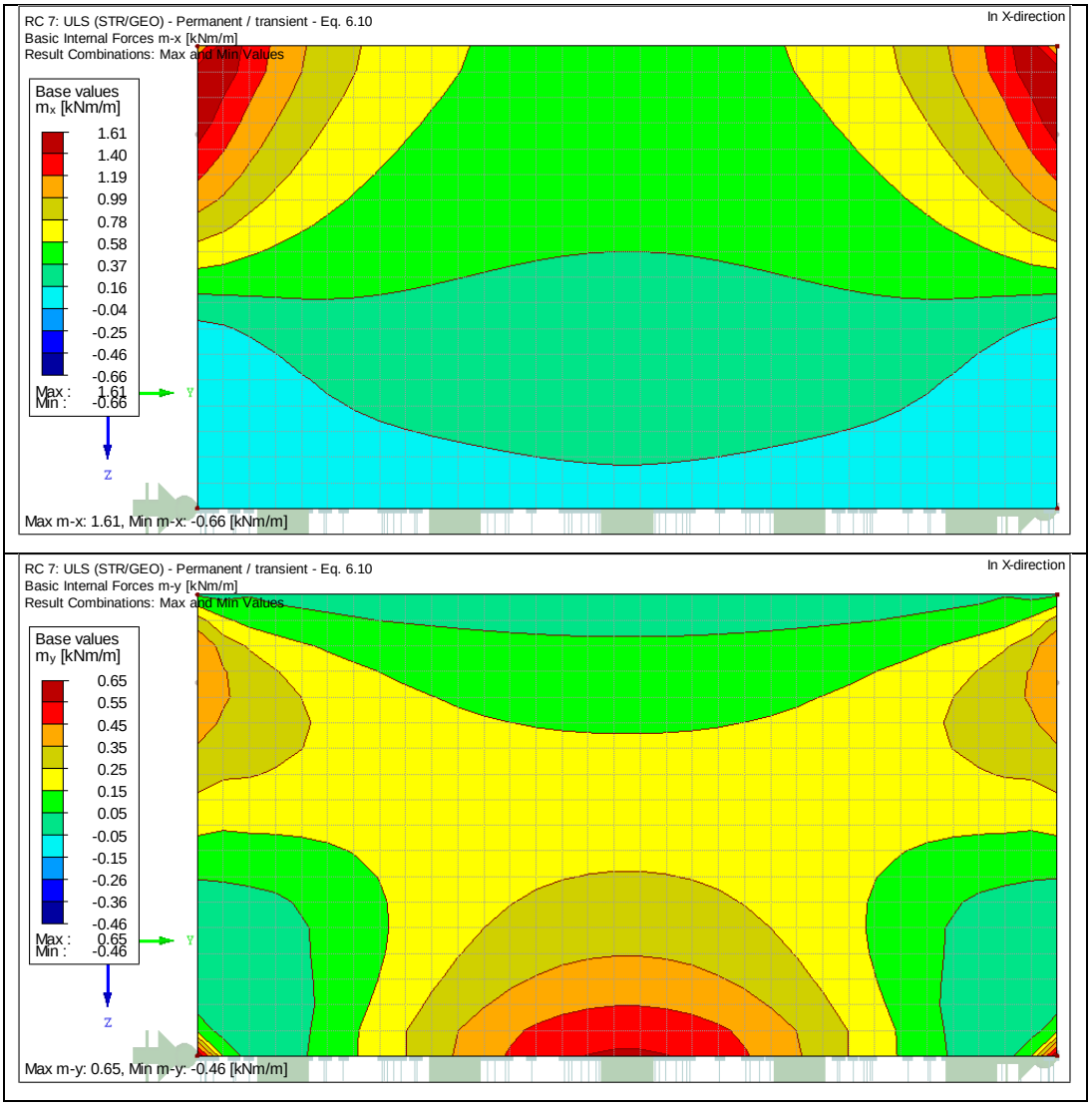


Figure 29 Bending moment results for intermediate walls

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

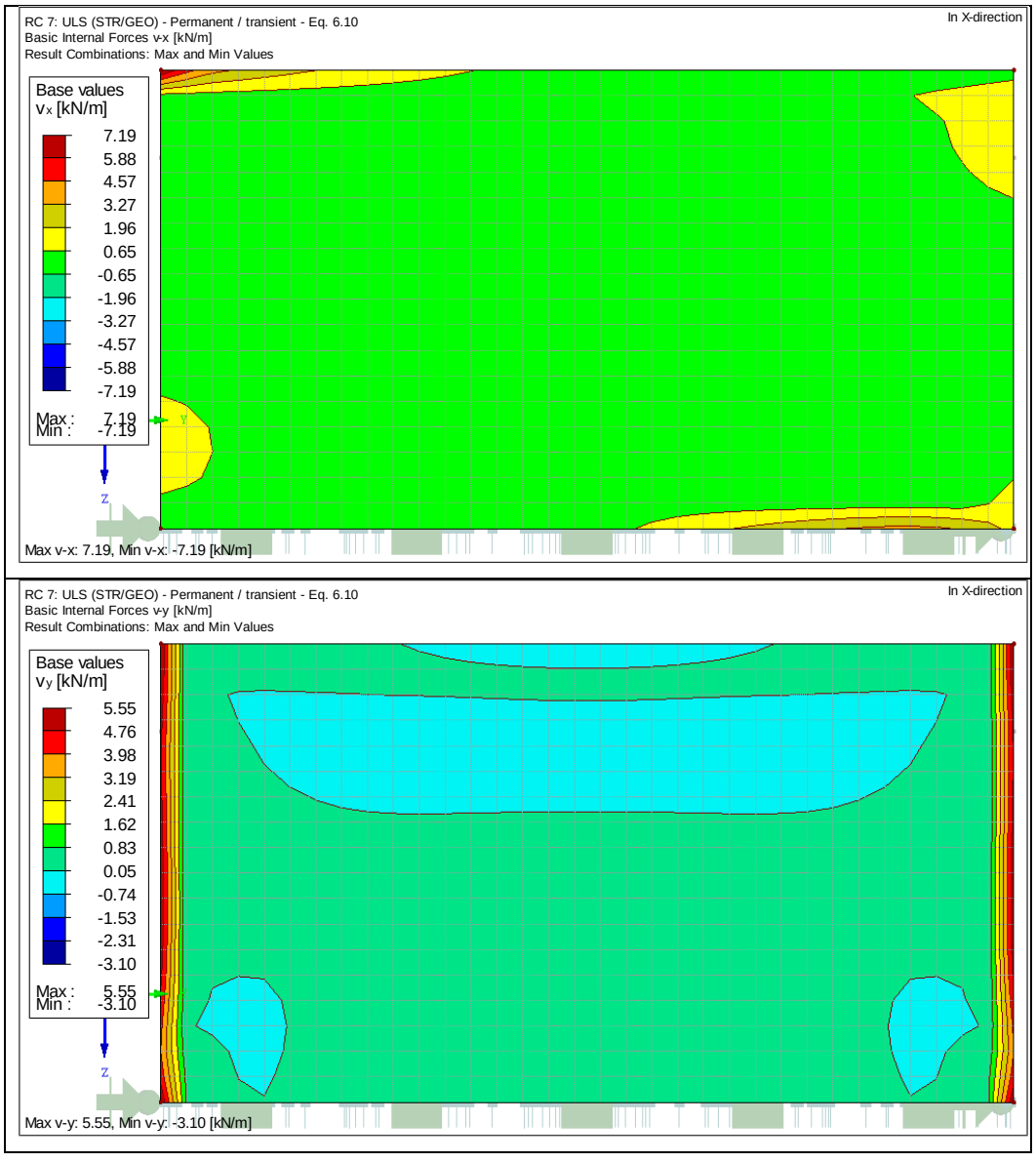


Figure 30 Shear force results for intermediate walls

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

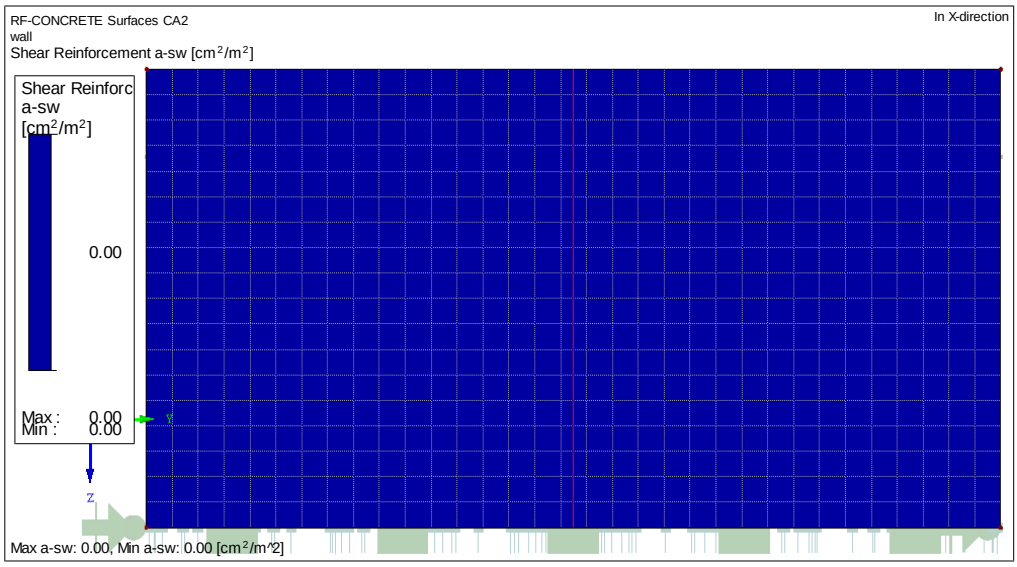


Figure 31 Required shear reinforcement area

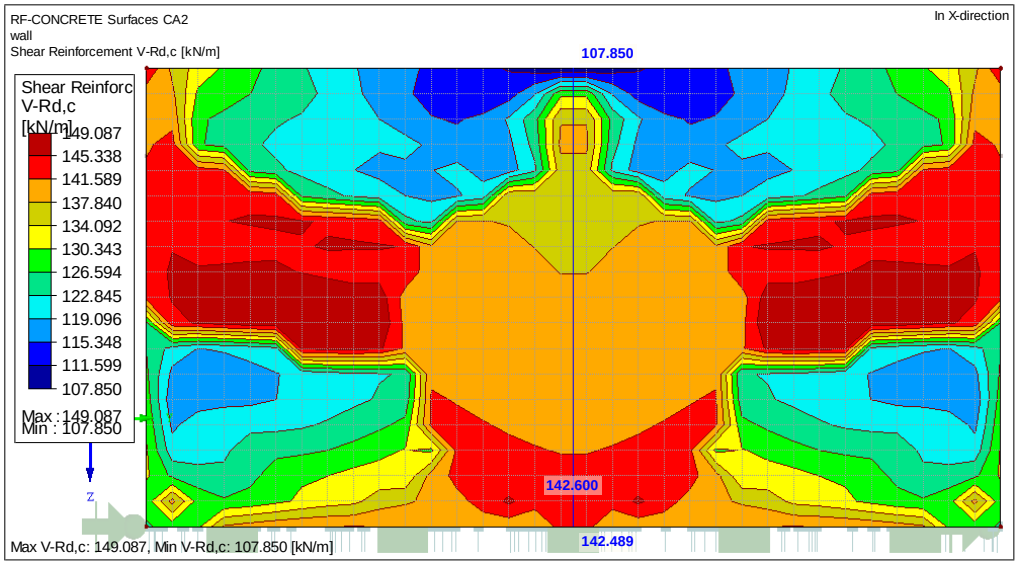


Figure 32 $V_{Rd,c}$ design shear resistance without shear reinforcement

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

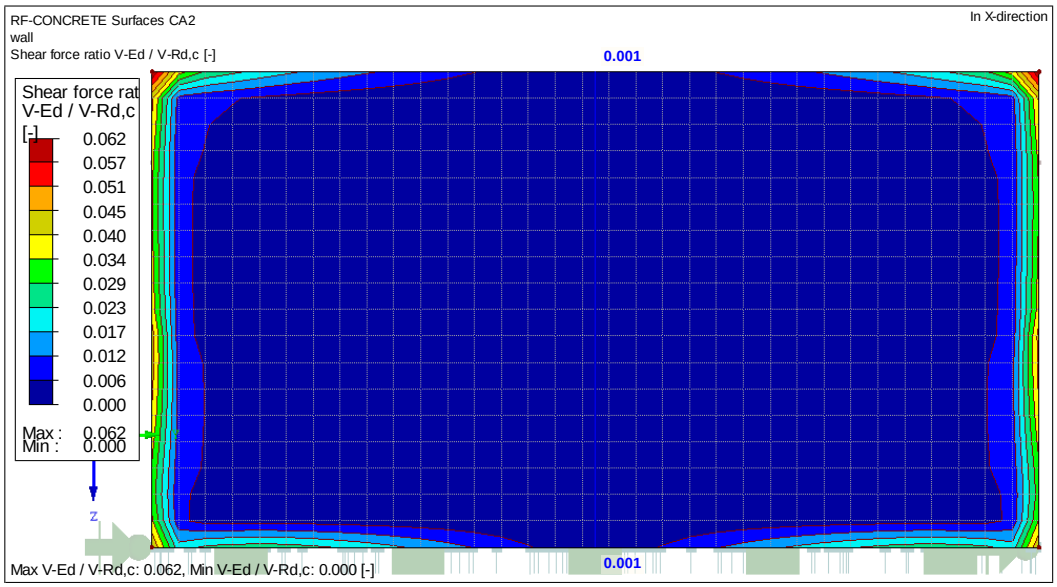


Figure 33 utility ratio $V_{Ed}/V_{Rd,c}$

utility ratio= applied shear force/ shear resistance= $V_{Ed}/V_{Rd,c}$< 1.0, ok!

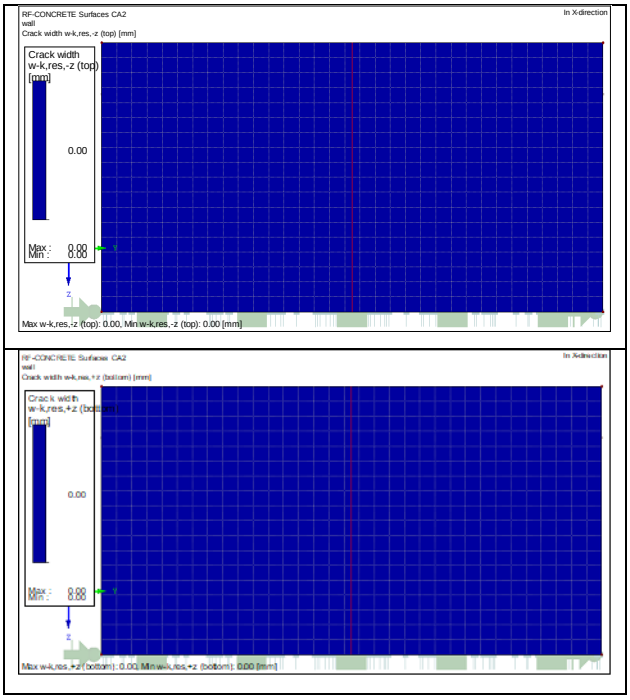
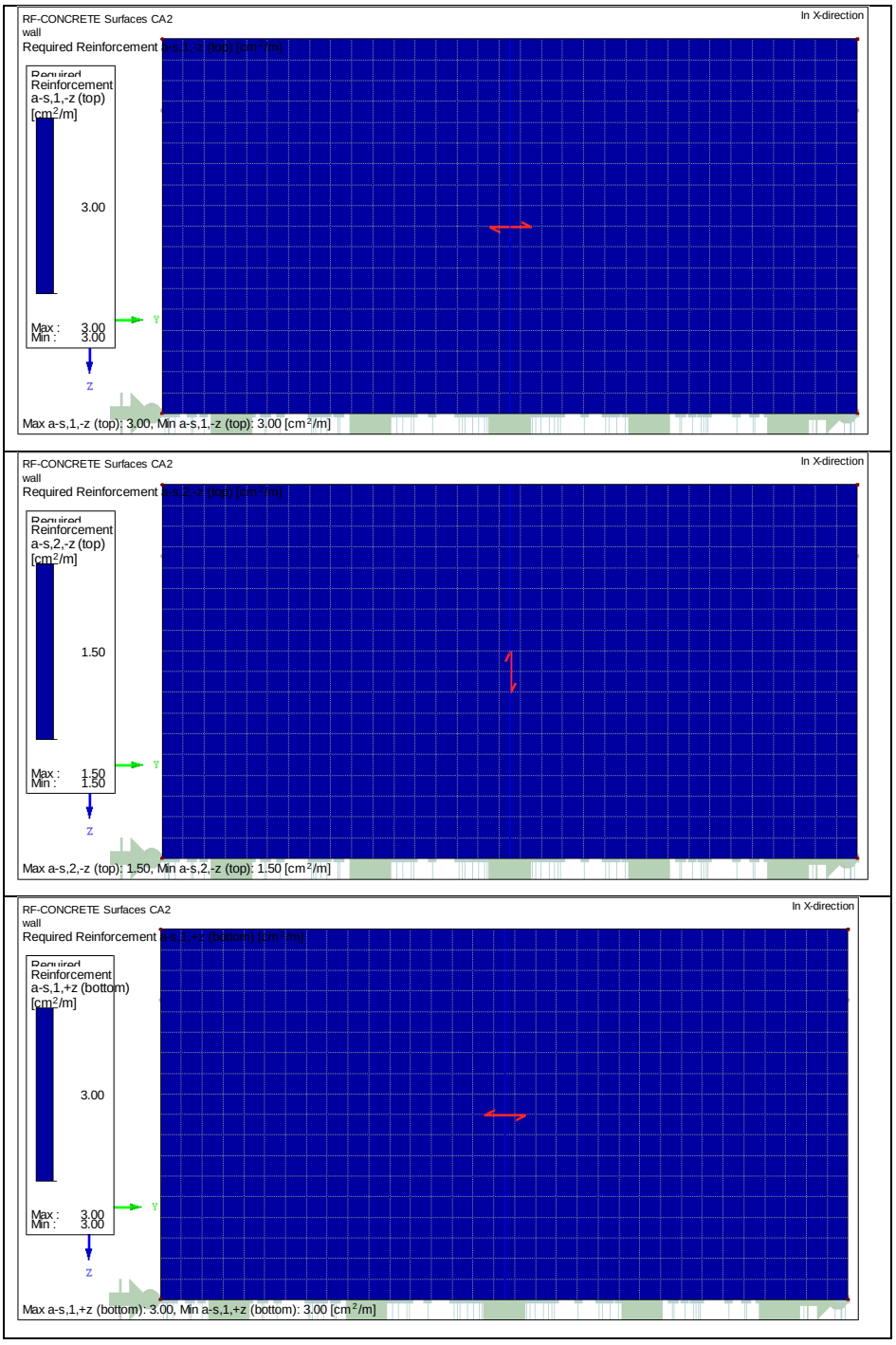


Figure 34 Cracking check, $w_k < w_{k,max}$ = 0.20mm, ok!

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

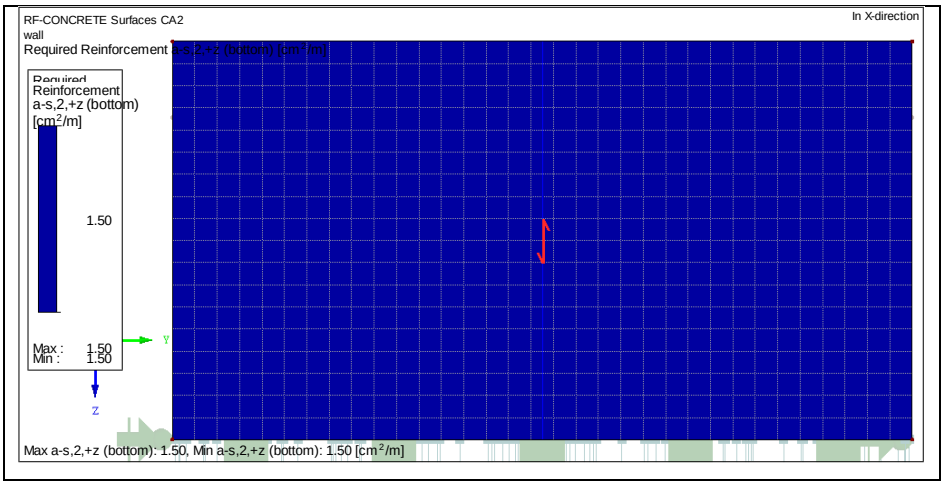


Figure 35 Required reinforcement area for intermediate walls

Provide basic reinforcement T12@150, A_{st} = 7.54cm^2/m.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

7.6. Analysis for slab

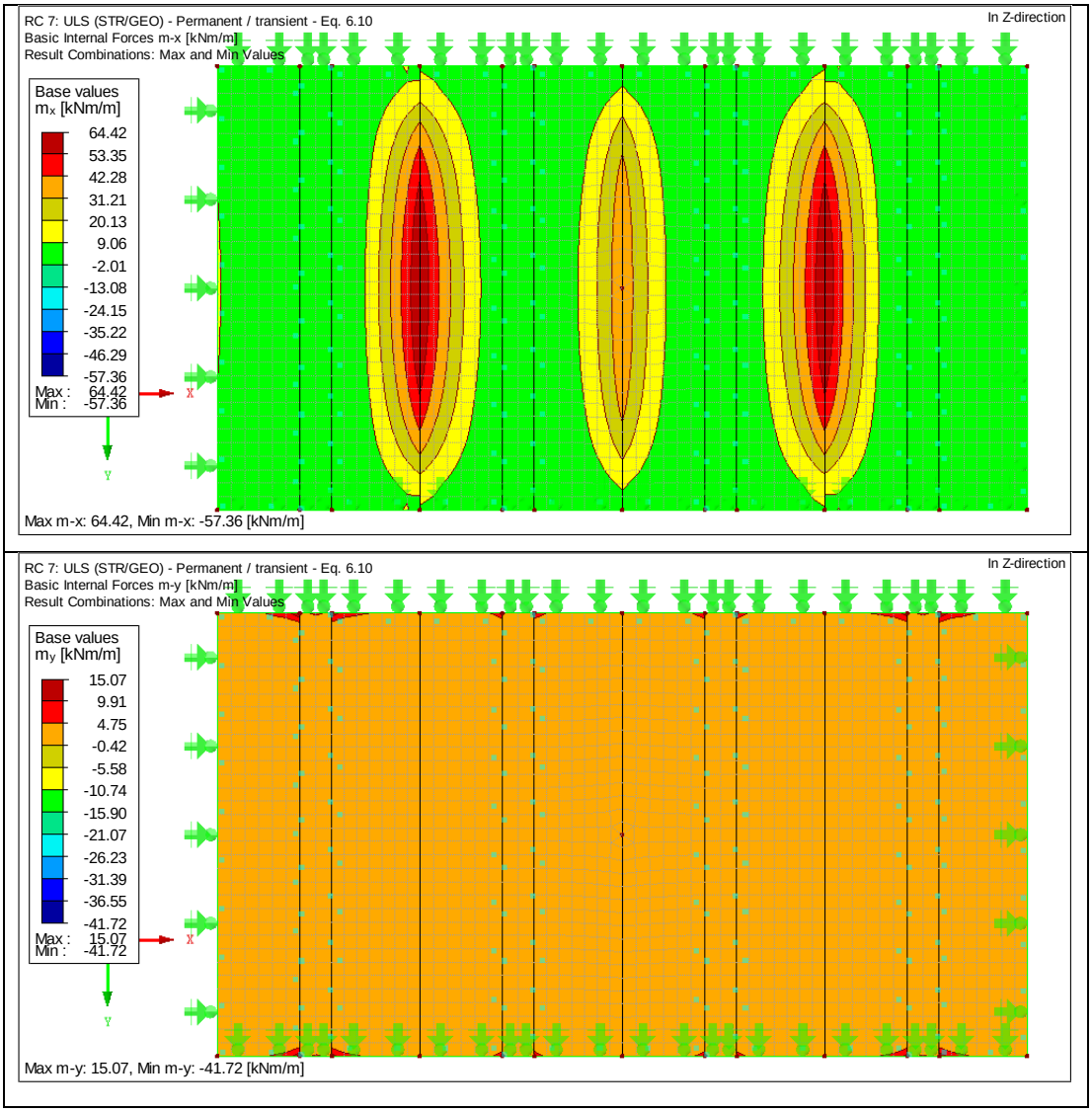


Figure 36 Bending moment results for base slab

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

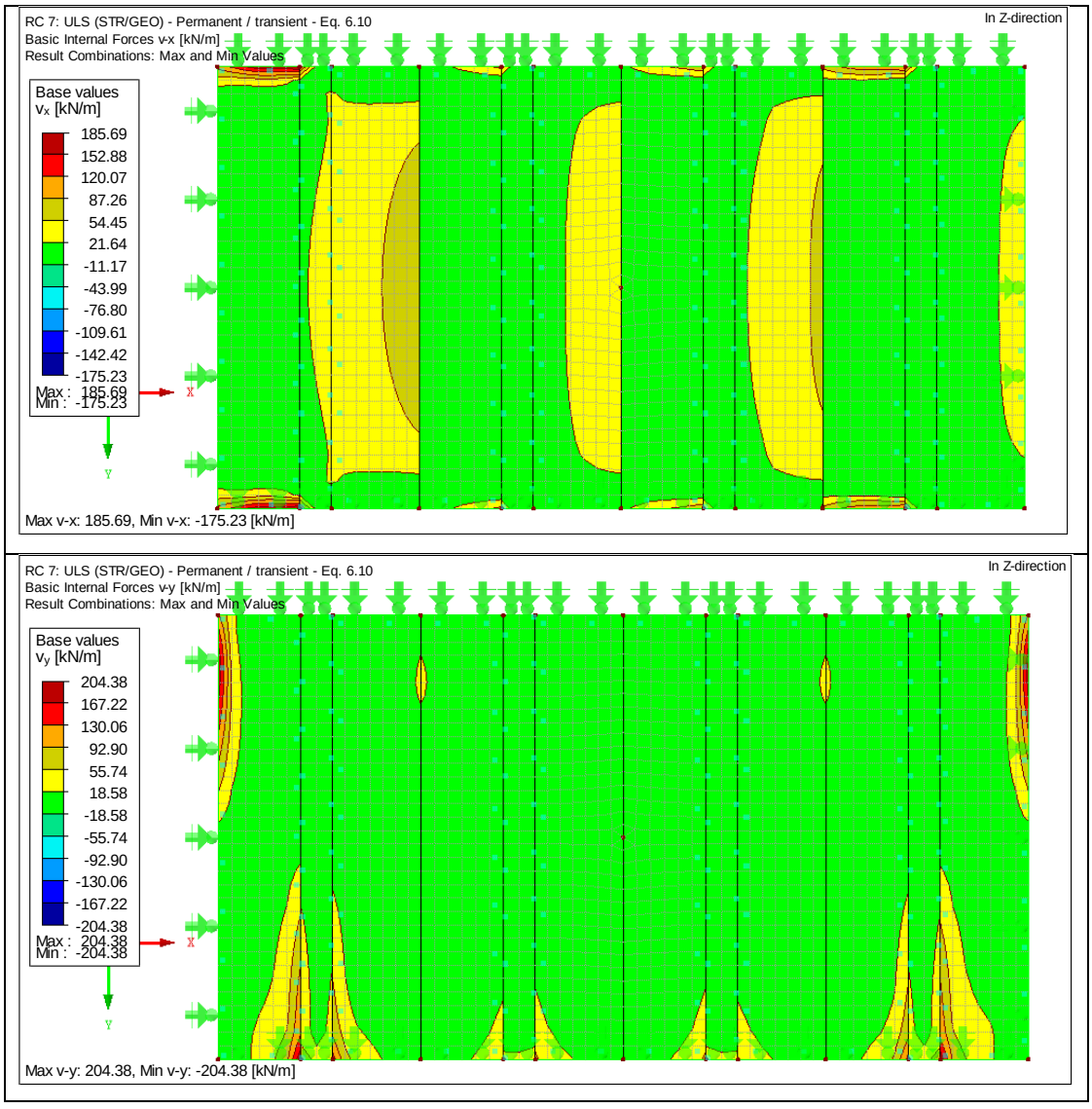


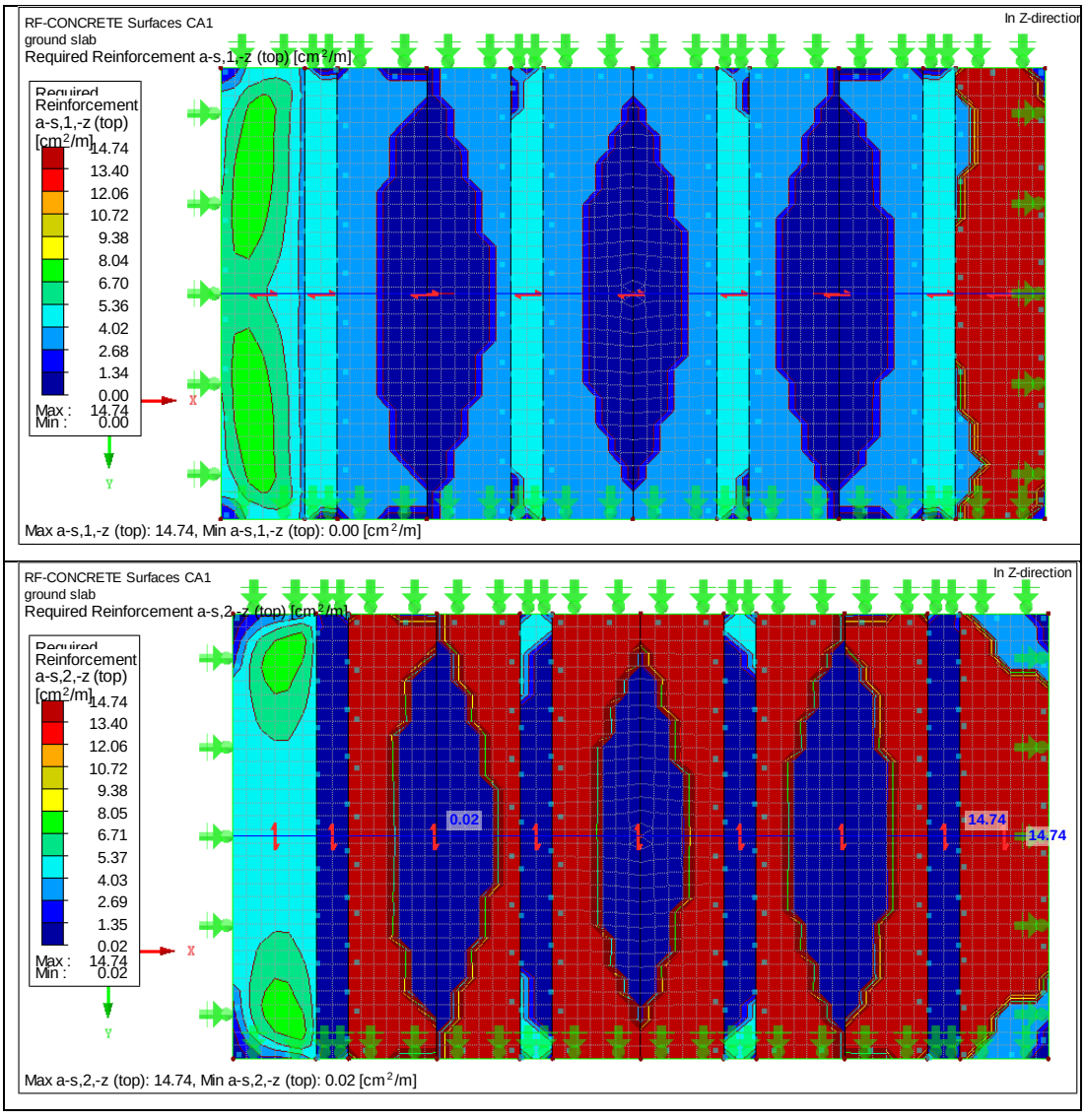
Figure 37 Shear force results for base slab

The internal force v_x corresponds to the shear force distribution in the direction of the local axis x of the surface, and the internal force v_y to the shear force distribution in the direction of the local axis y. The resulting shear force, which is also used in the shear design, is then calculated as $v_{Ed} = \sqrt{(v_x^2 + v_y^2)}$

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

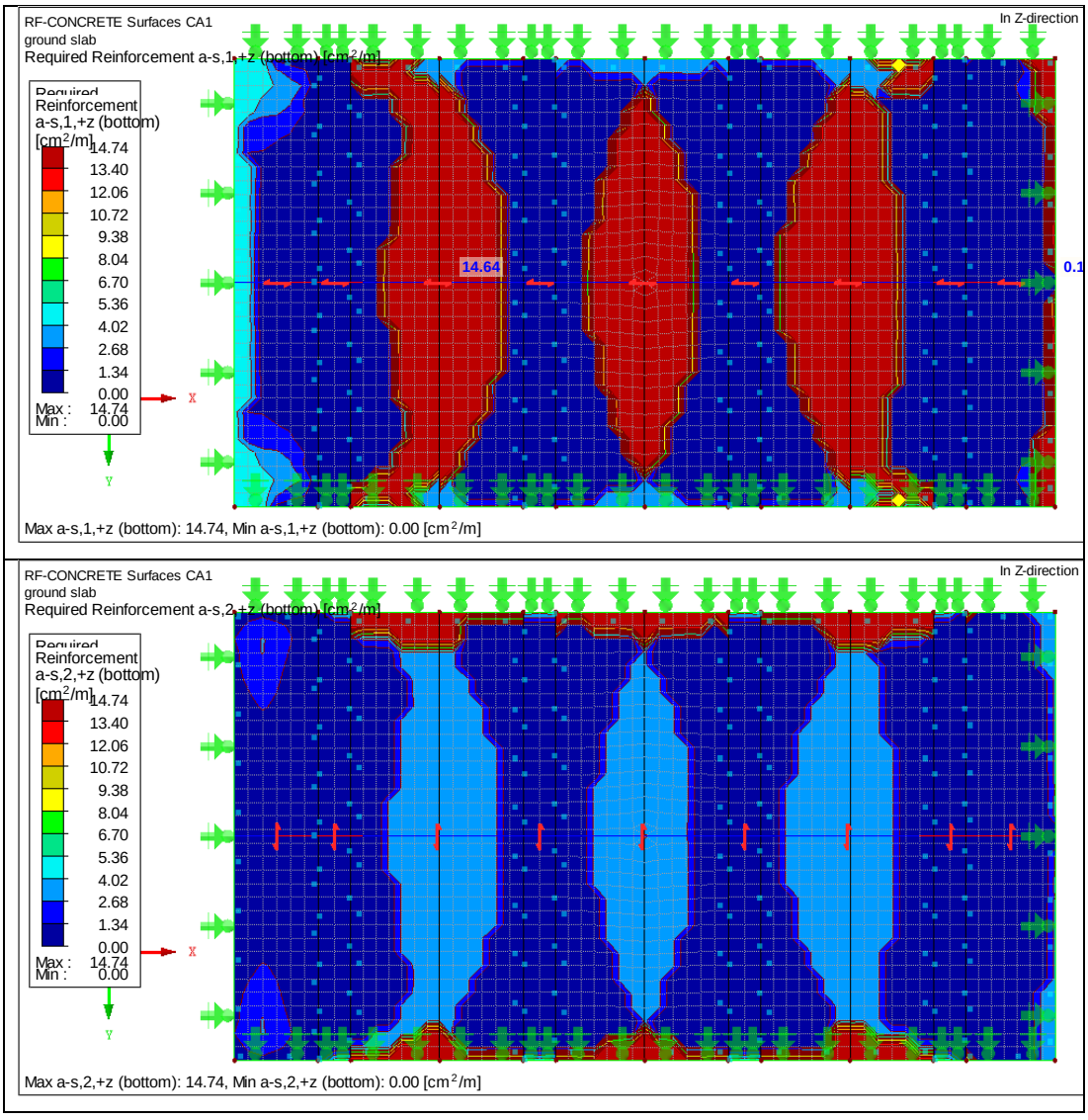


Figure 38 Required reinforcement area for base slab

For slab thickness t = 300mm, Provide T12@150, A_{sp}= 7.54 cm^2/m at top and T16@125, A_{sp}= 16.08 cm^2/m at bottom.

For slab thickness t = 800mm, Provide T16@125, A_{sp}= 16.08 cm^2/m at top and bottom.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

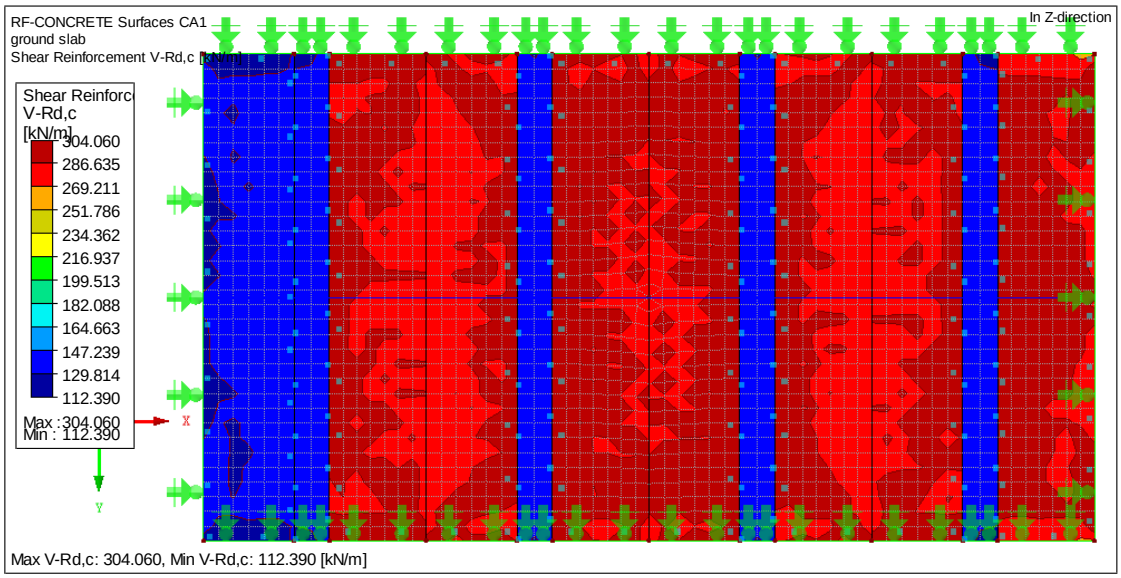


Figure 39 $V_{Rd,c}$ design shear resistance without shear reinforcement

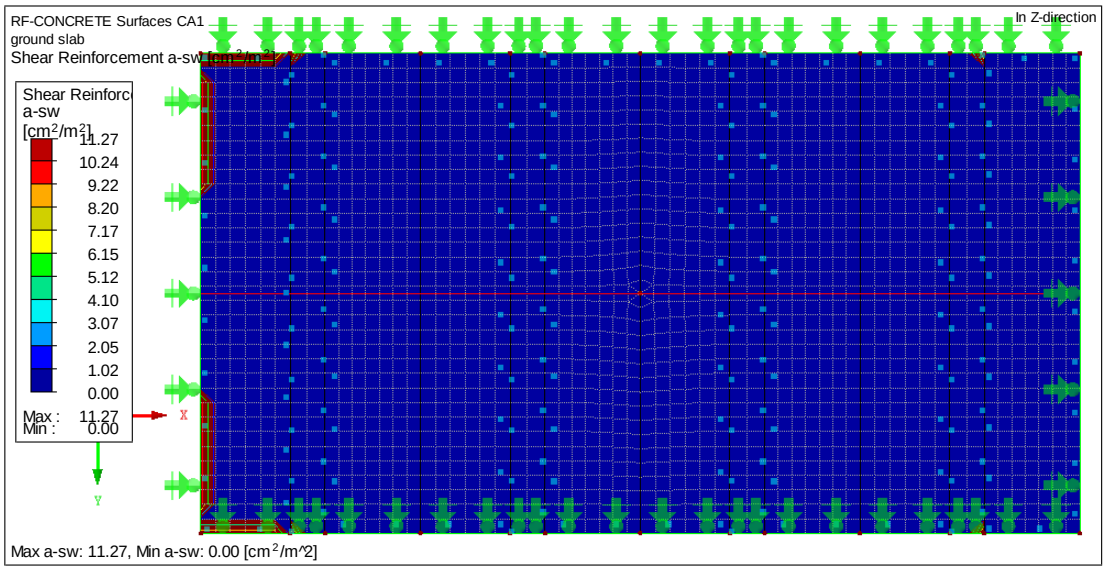


Figure 40 Required shear reinforcement

starter bars at both faces (T12/125mm, Asv = 18.09 cm^2/m) improves shear resistance.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

8. ANALYSIS FOR DISTRIBUTION BOX

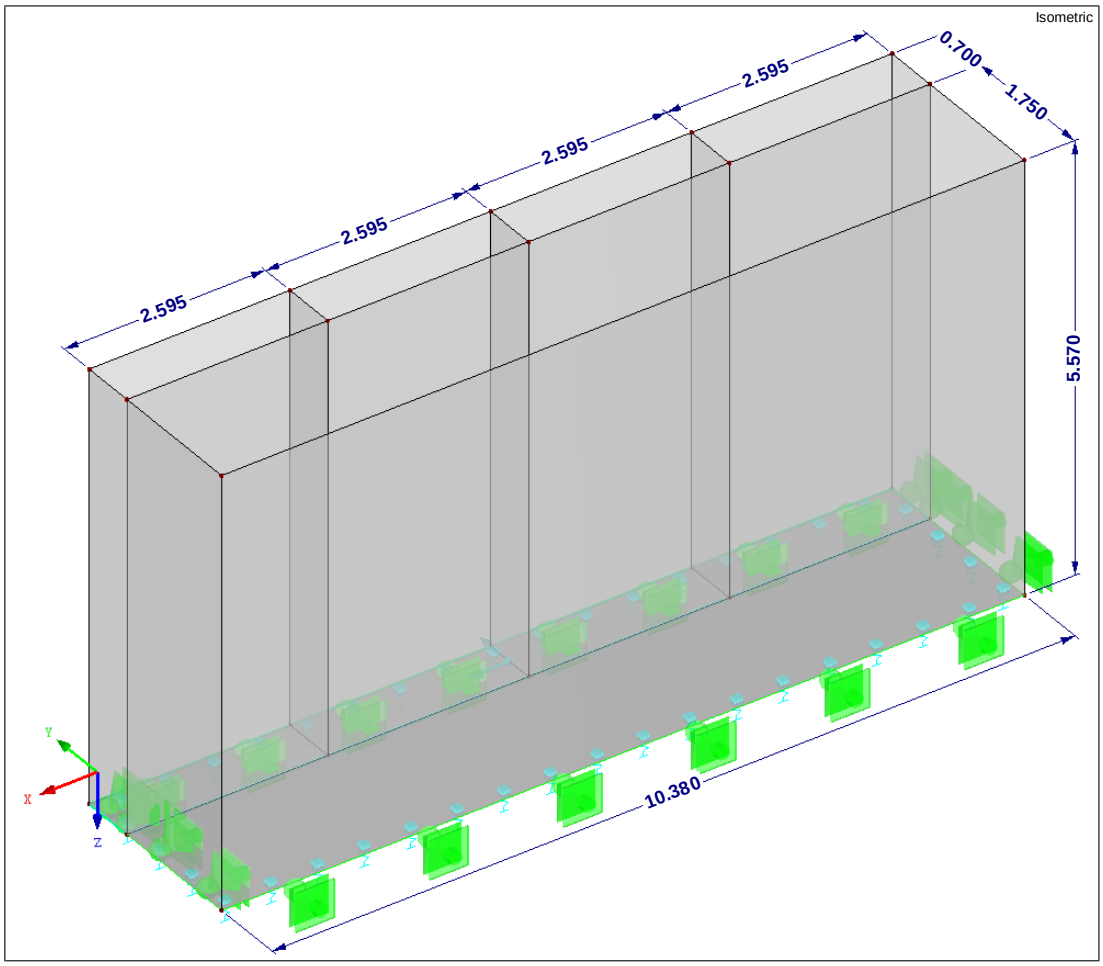


Figure 41 Structure dimension

For distribution box, the external walls have a thickness of 300mm, intermediate wall thickness 200mm/300mm and foundation.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

8.1. Load case in RFEM

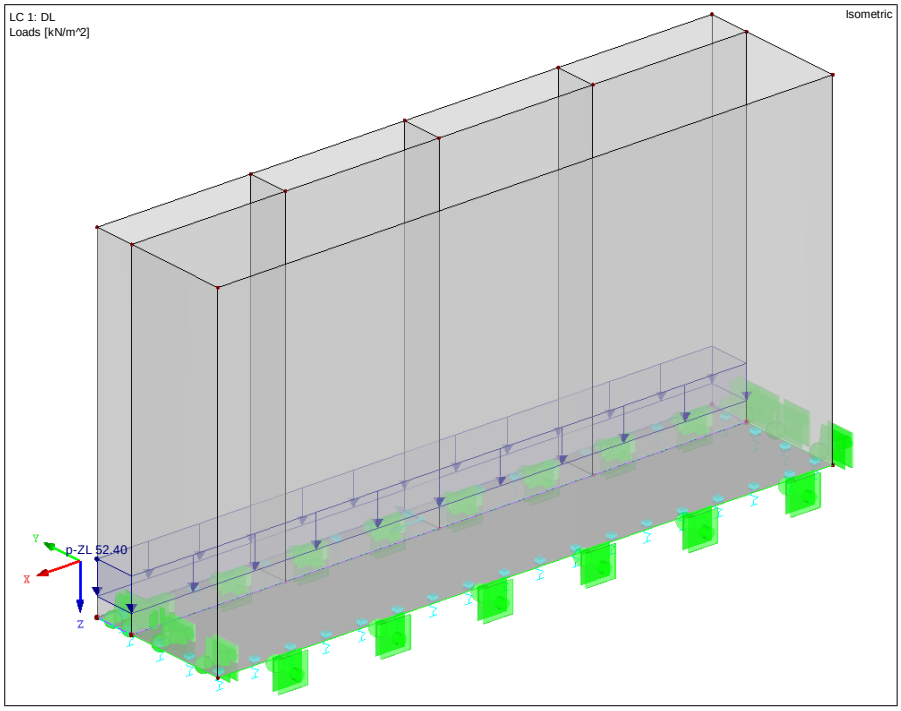
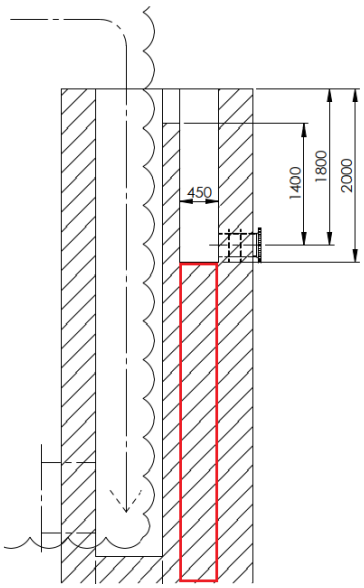


Figure 42 Self-weight



concrete weight is applied as surface load on slab

25kN/m^3* (5.57m-2m)*450mm*(10380mm-300mm*3) /(0.70m*10.38m) = 52.4kN/m^2

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

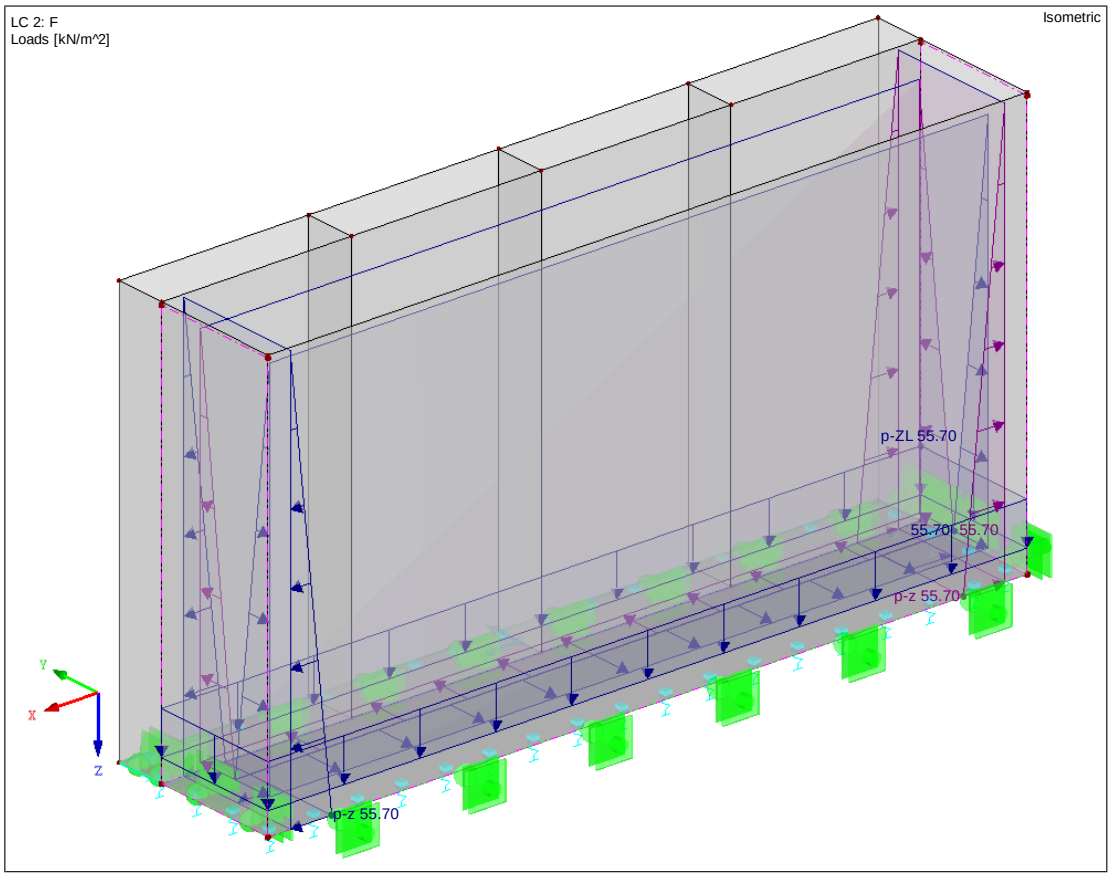


Figure 43 Fluid load

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

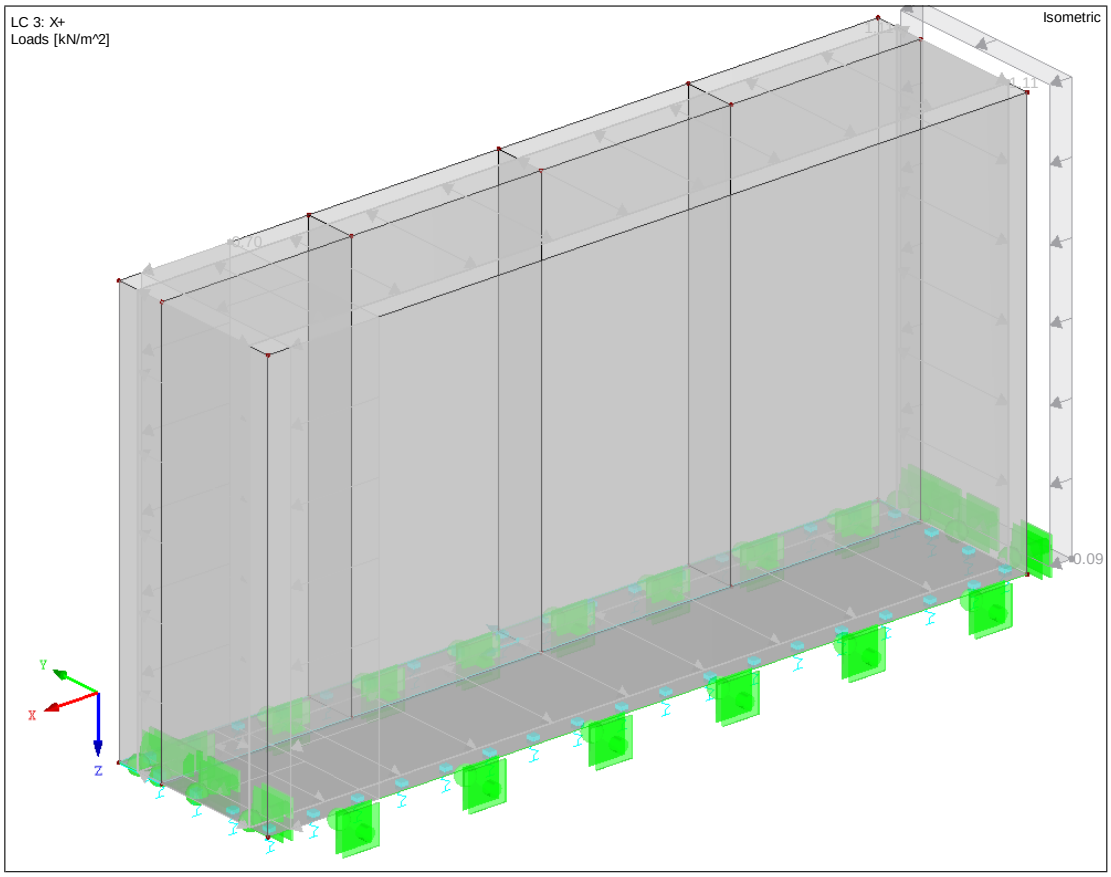


Figure 44 Wind in X (pos)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

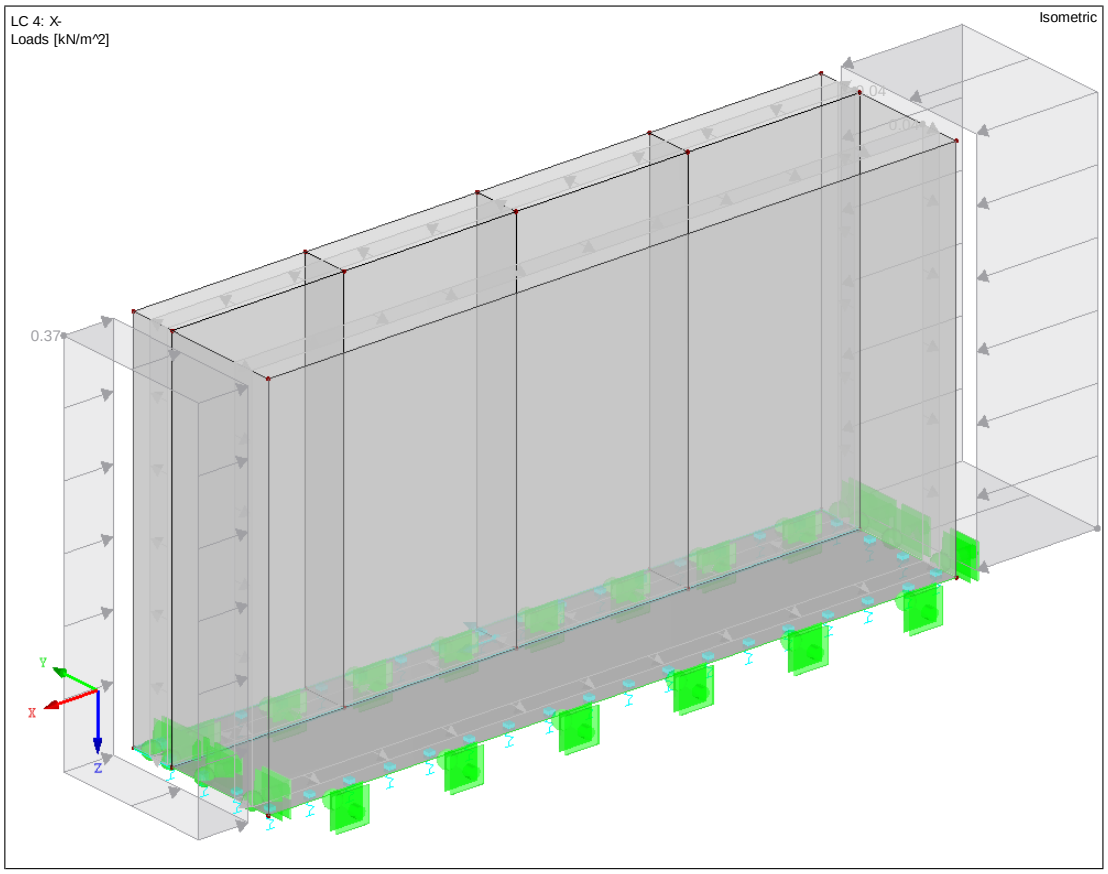


Figure 45 Wind in X (neg)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

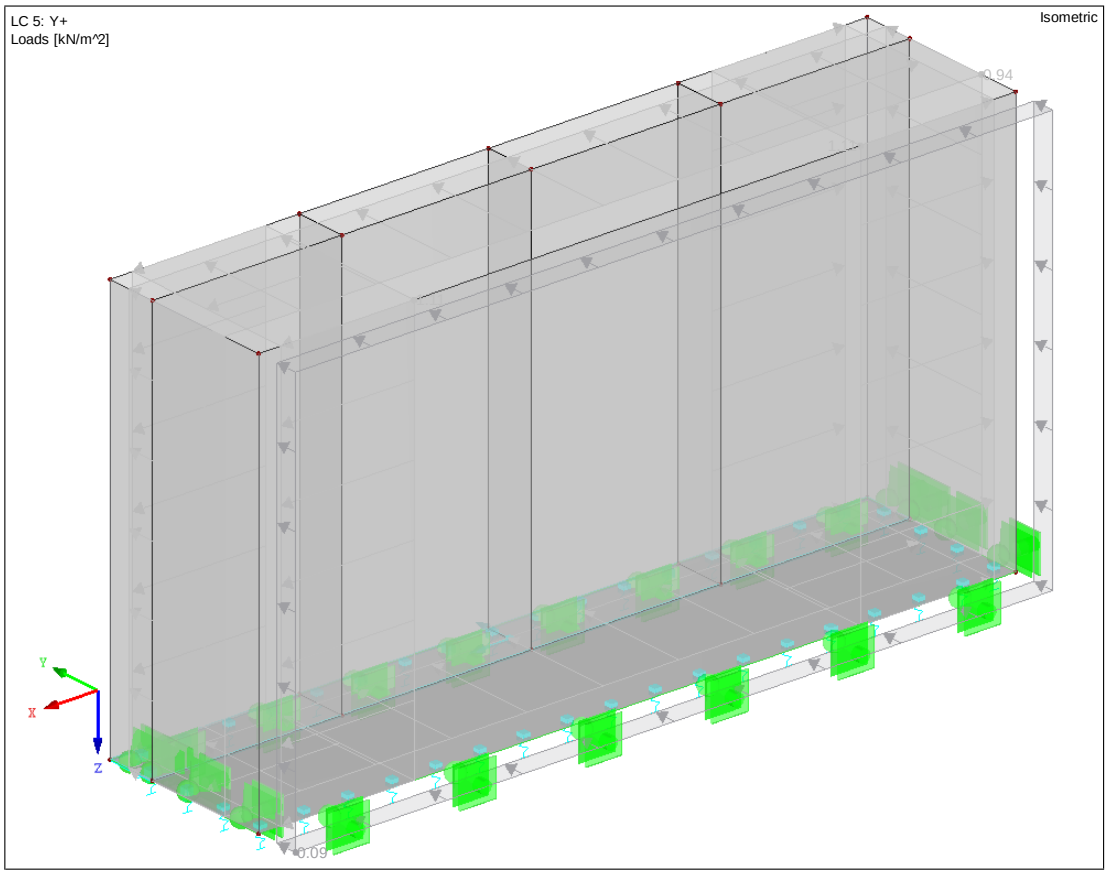


Figure 46 Wind in Y (pos)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

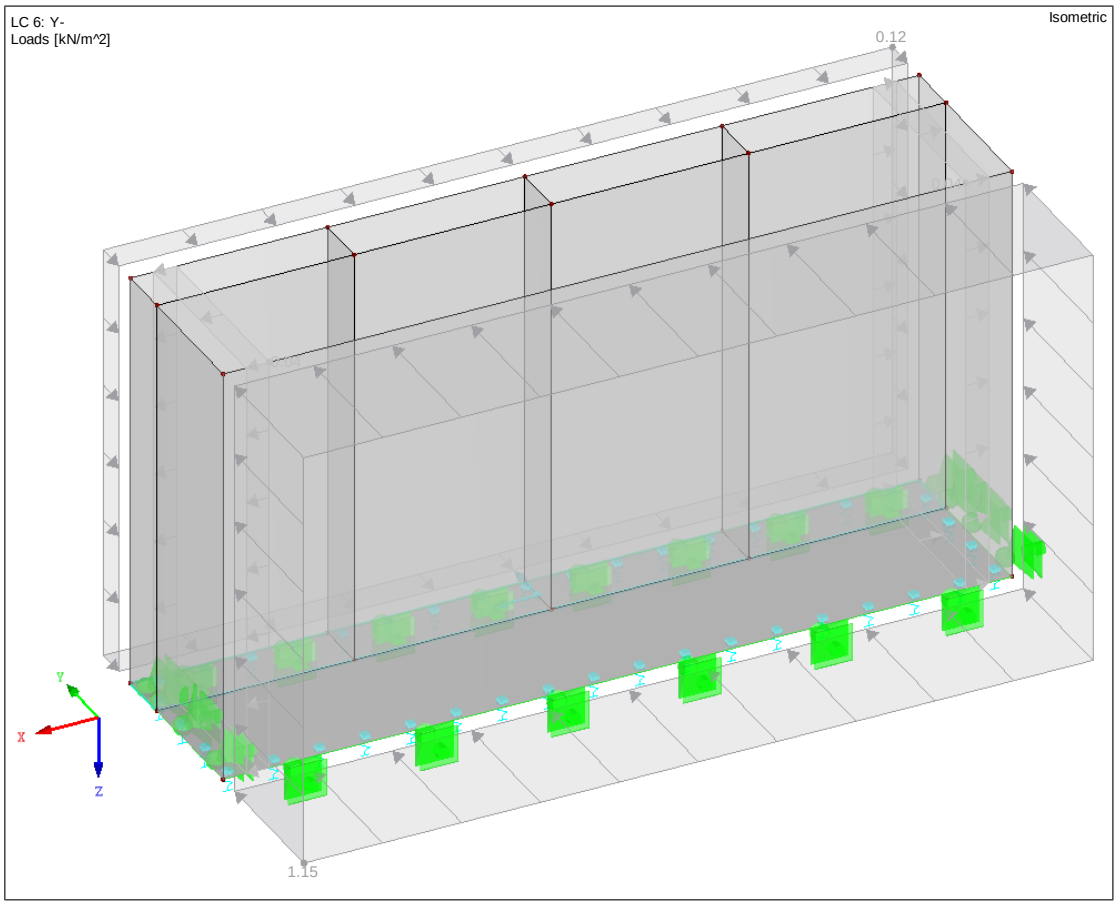


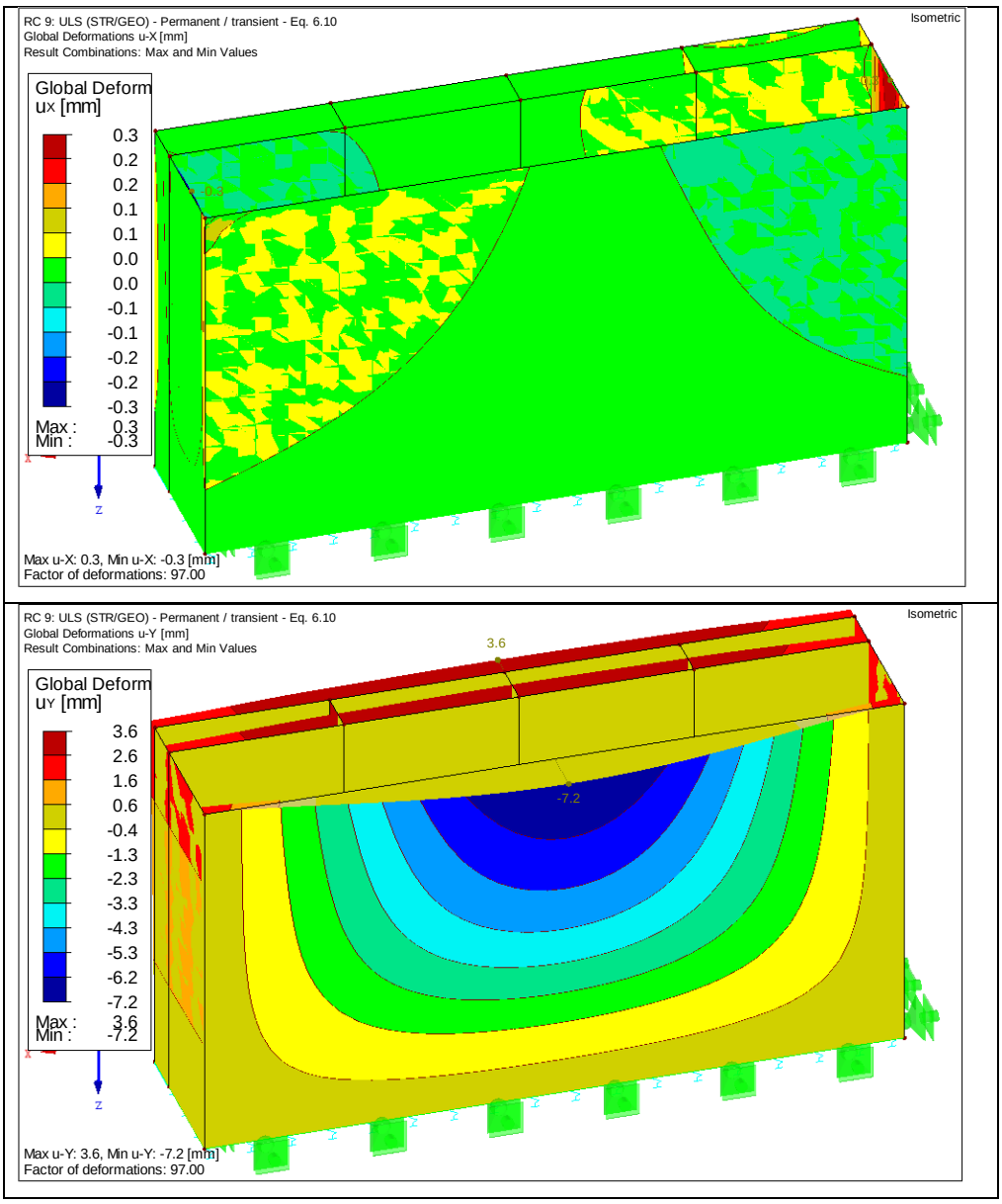
Figure 47 Wind in Y (neg)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

8.2. Global deformation results



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

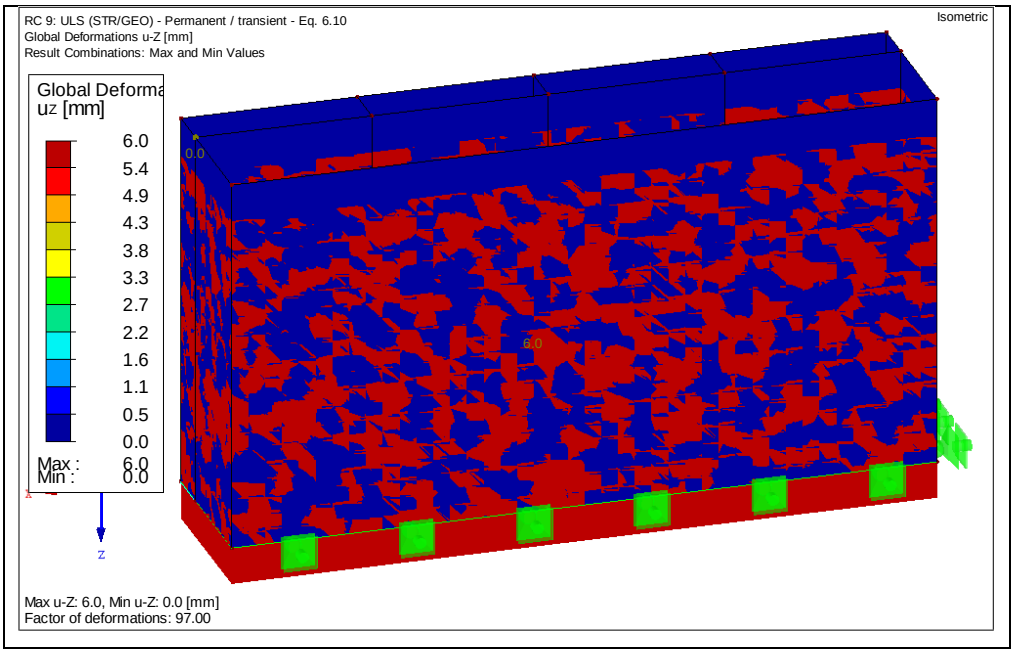


Figure 48 Global deformations

$u < u_{max}$ =50mm, ok!

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

8.3. Analysis for wall

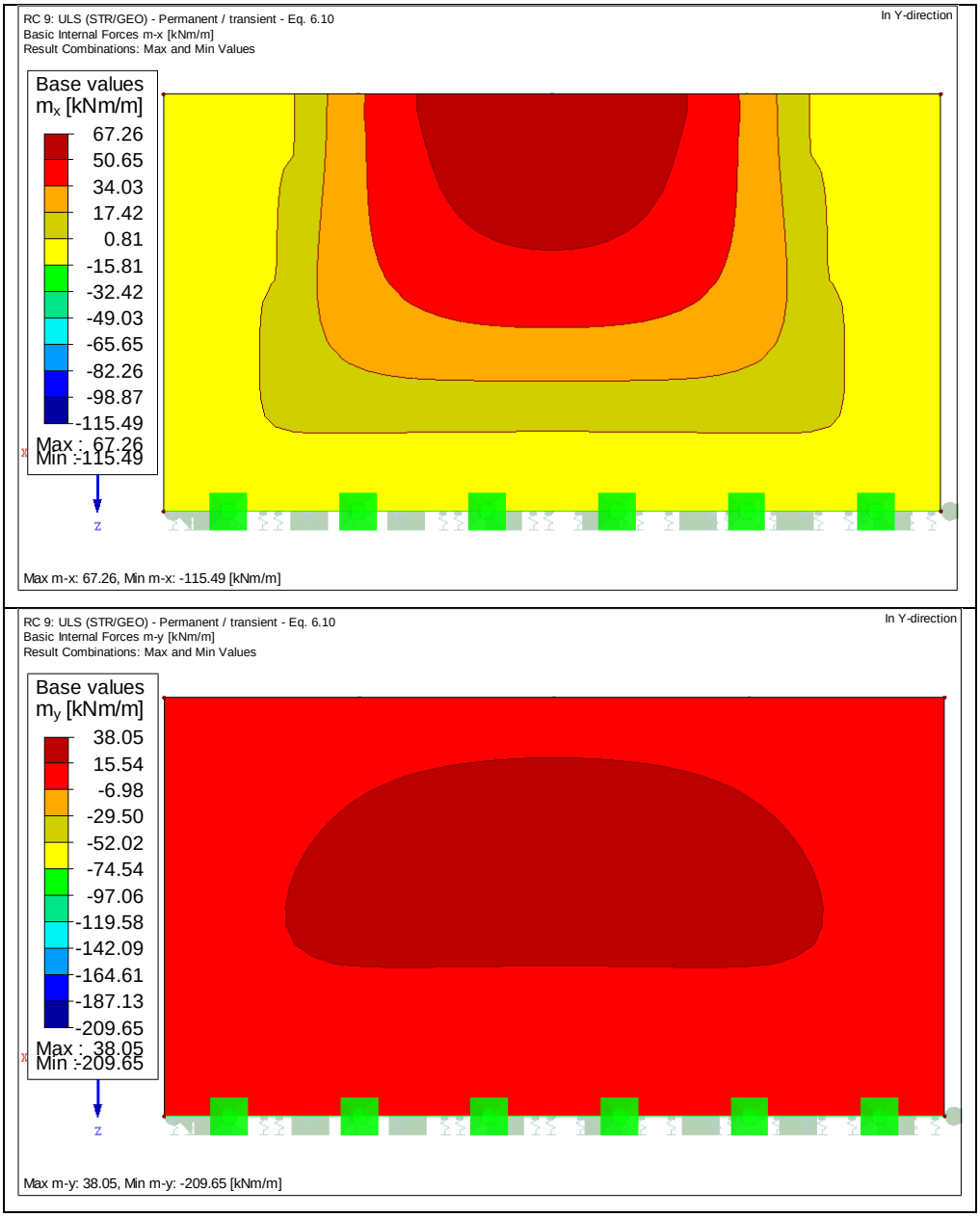
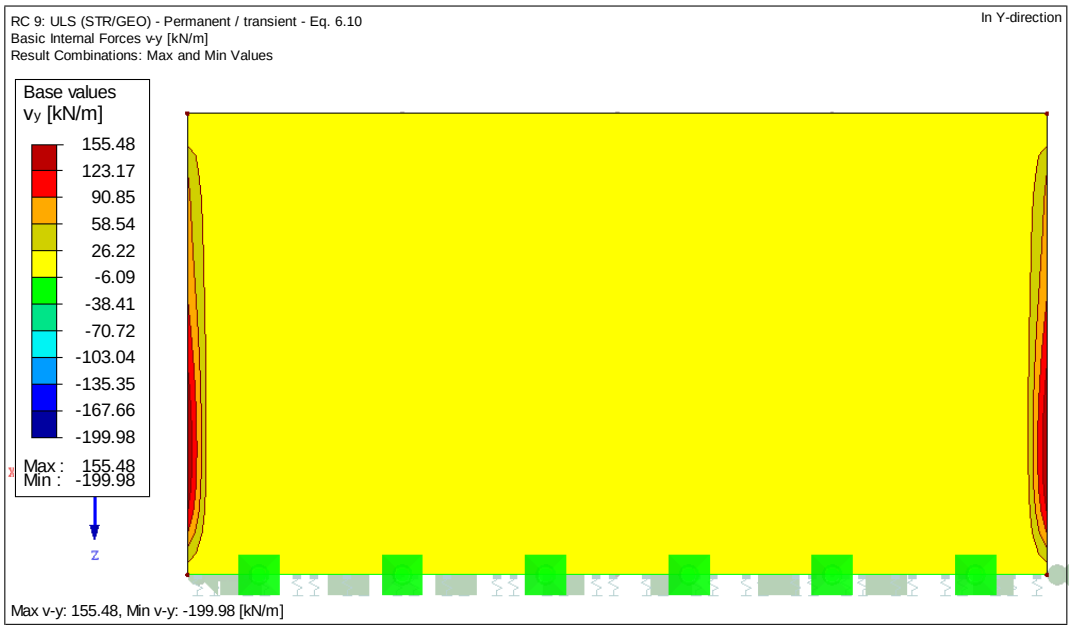


Figure 49 Max bending moment results for wall

PROPRIETARY & CONFIDENTIAL

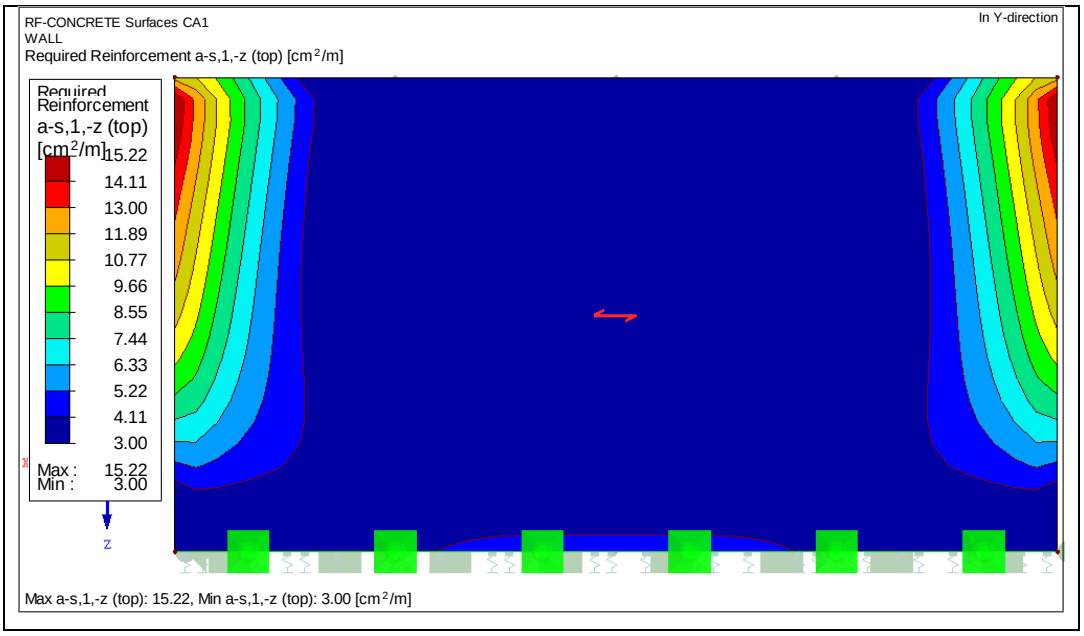
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



A02

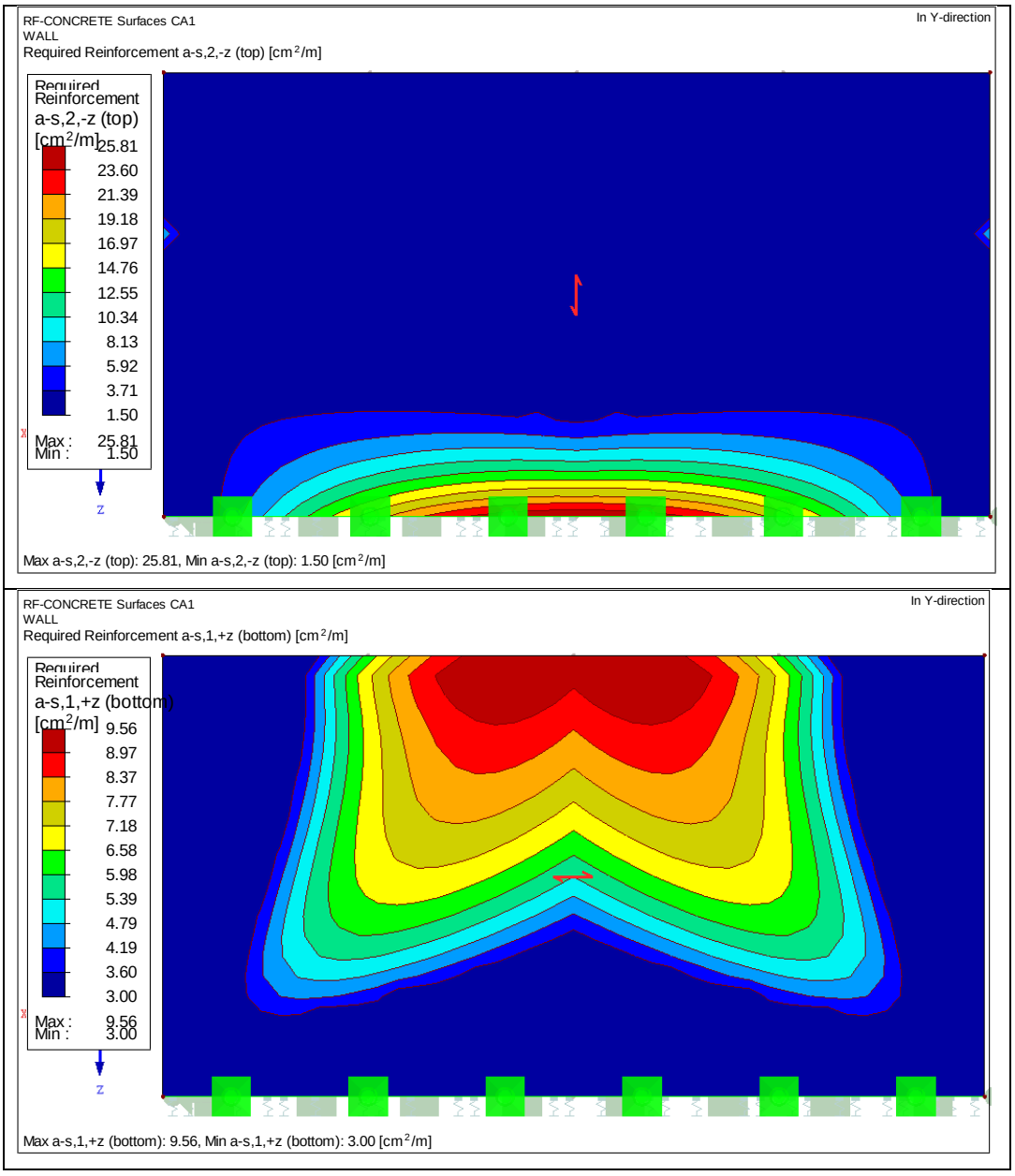
Figure 50 Shear force for wall



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

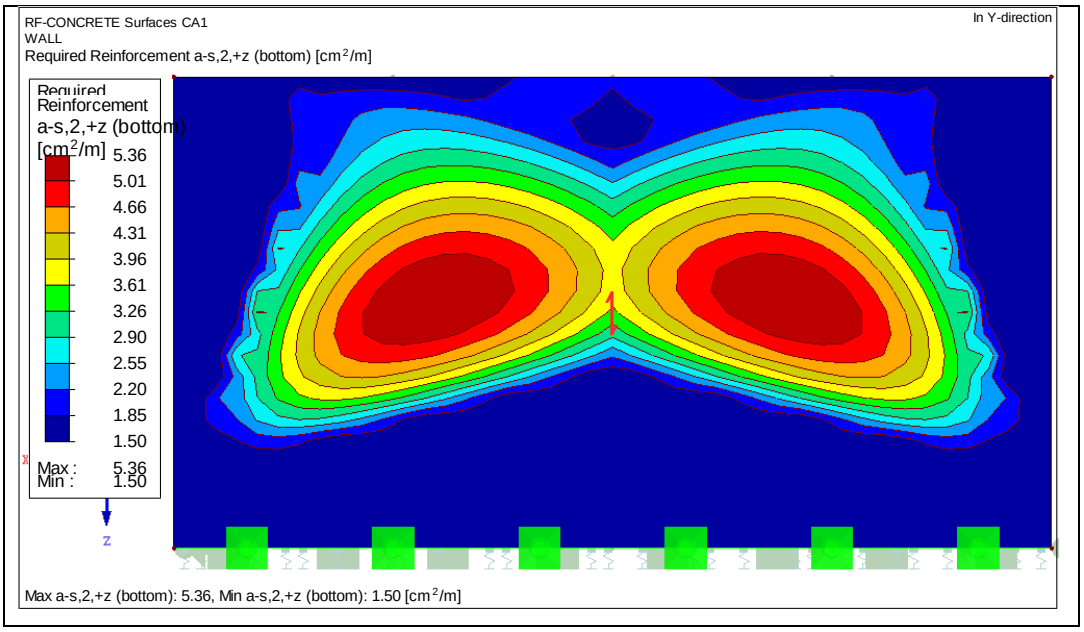
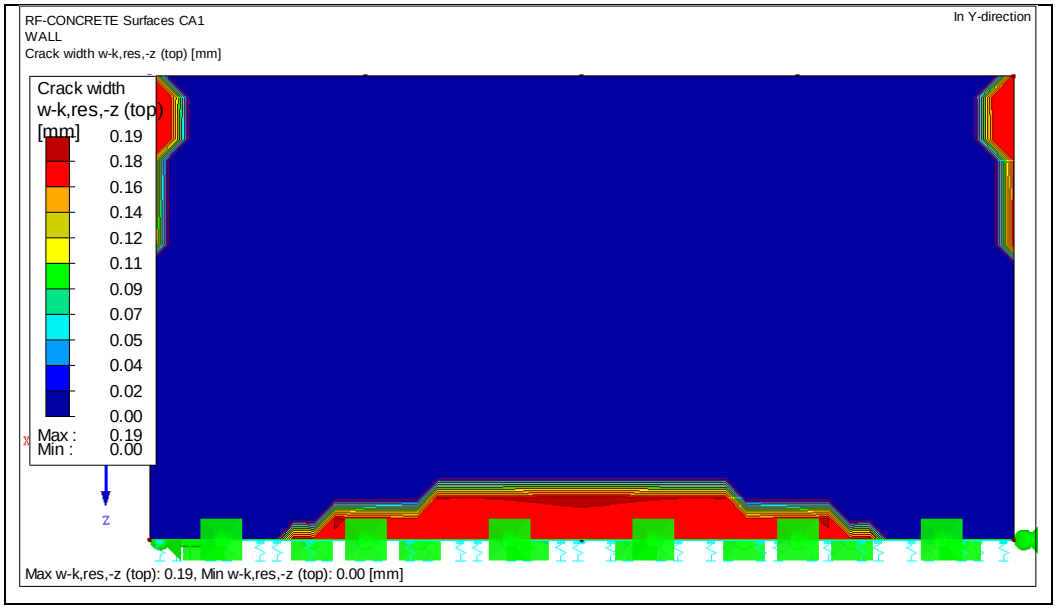


Figure 51 Required reinforcement for wall

Provide basic reinforcement T12@150, A_{st} = 7.54 cm^2/m. Starter bars at bottom (T16/150mm) and stirrups at edges (T12/150mm) increases provided reinforcement area.



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

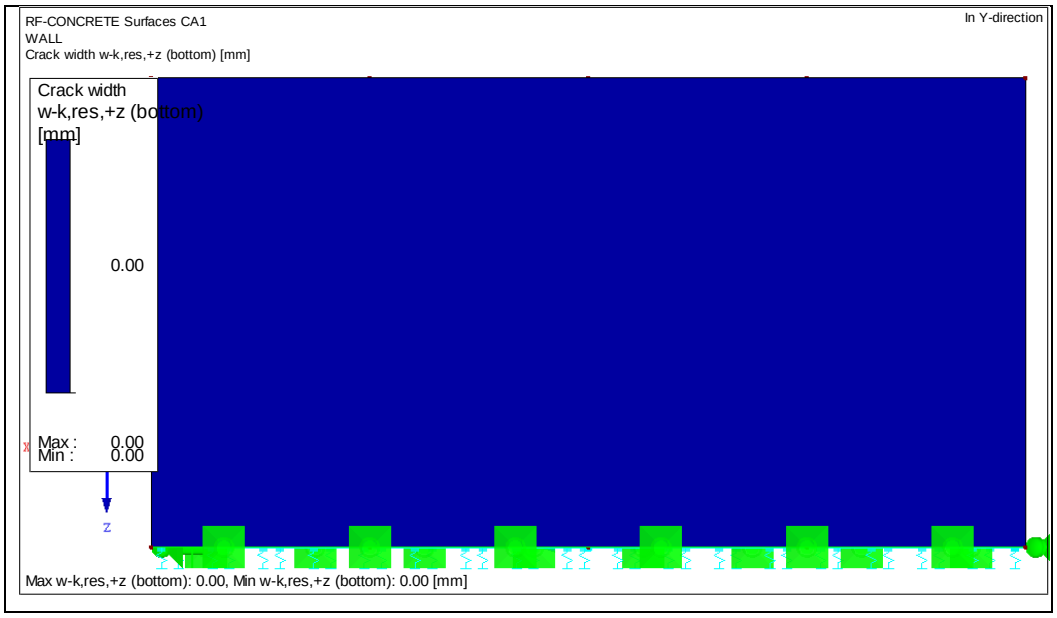


Figure 52 Cracking check

$w_k < w_{k,max}$=0.2mm, ok! In addition, starter bars at bottom (T16/150mm) and stirrups at edges (T12/150mm) decreases cracking.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

8.4. Analysis for slab

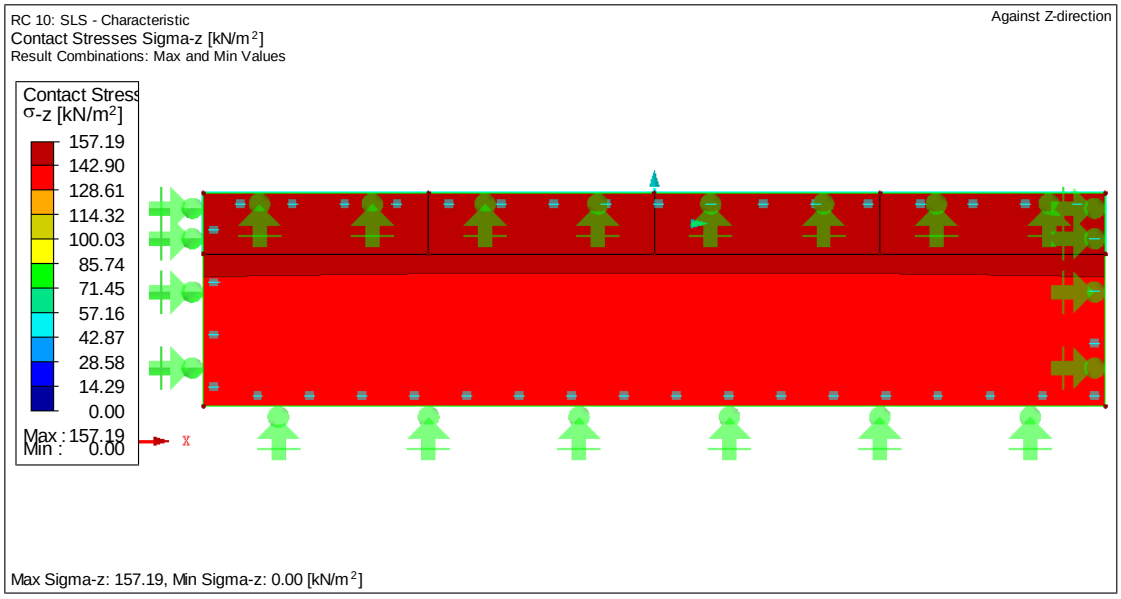
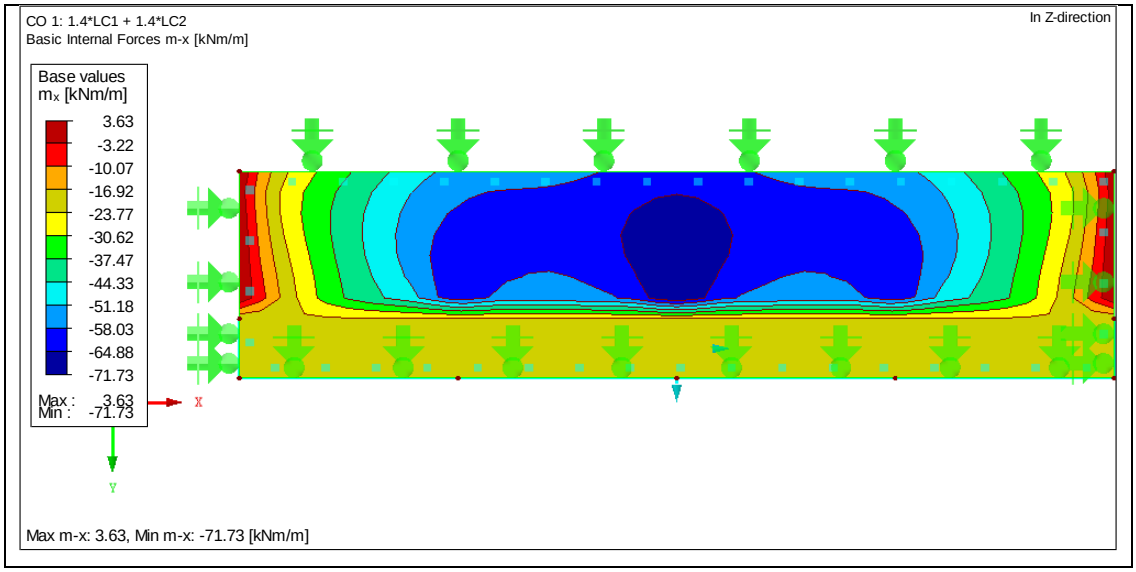


Figure 53 Soil contact stress on slab



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

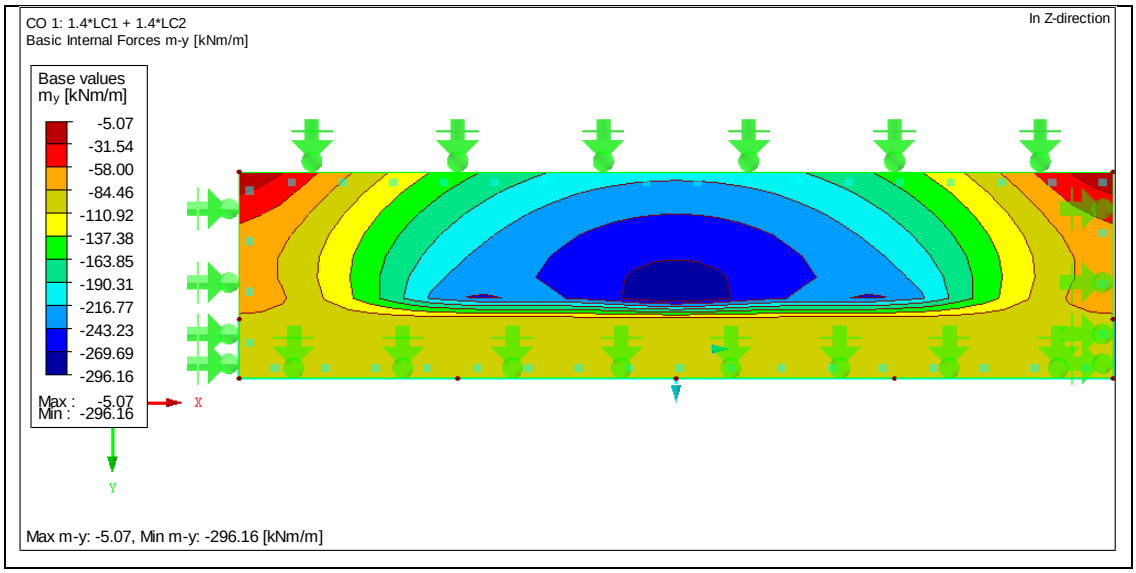
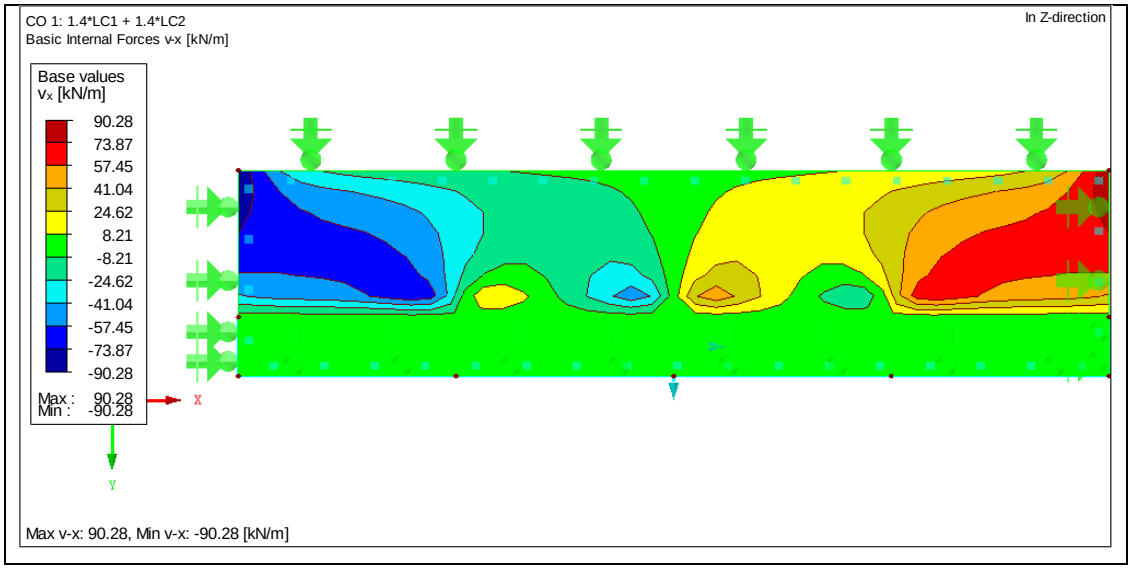


Figure 54 Max bending moment results for slab



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

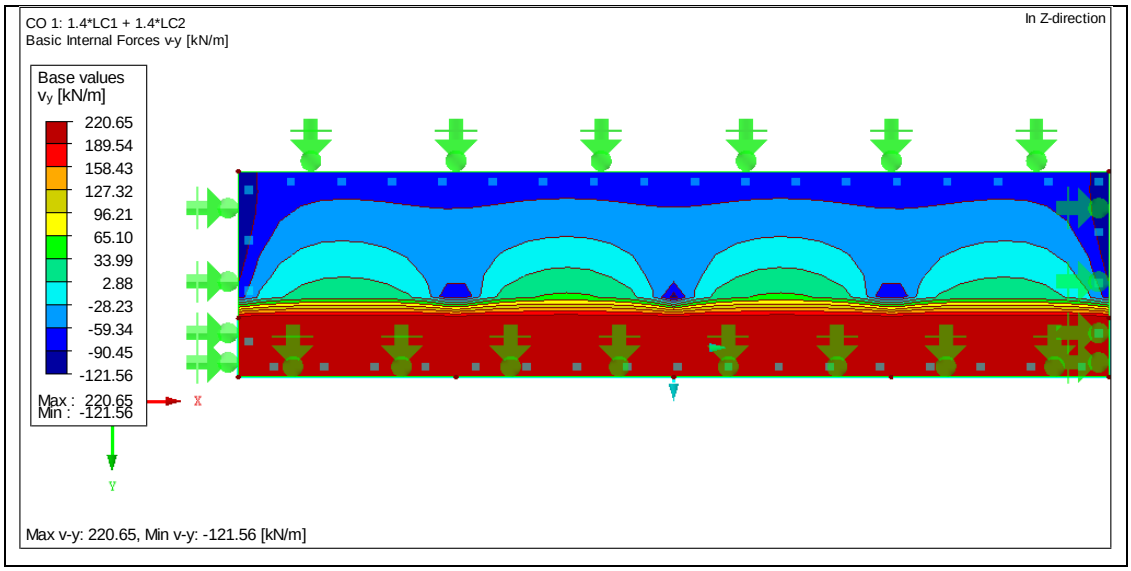


Figure 55 Shear force results for slab

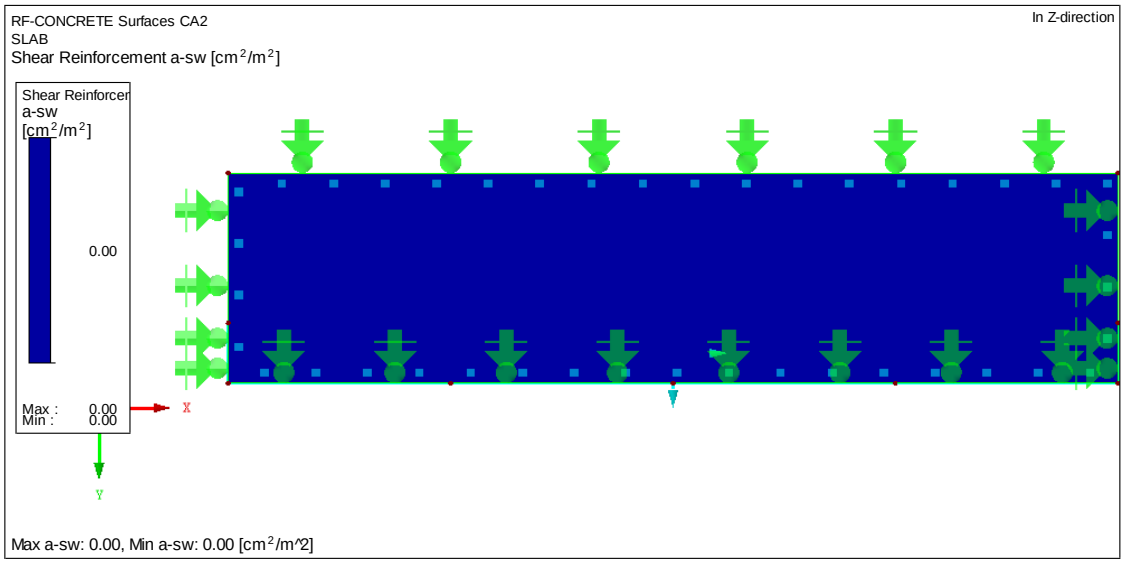


Figure 56 Required shear reinforcement

Provide T16/150mm as shear reinforcement.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

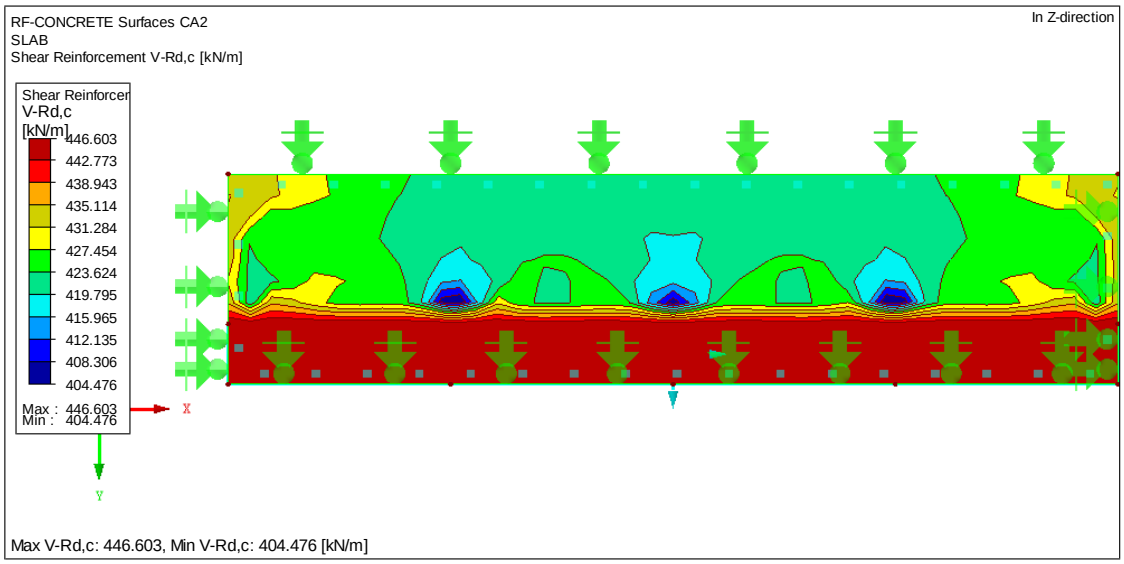


Figure 57 Shear resistance without shear reinforcement

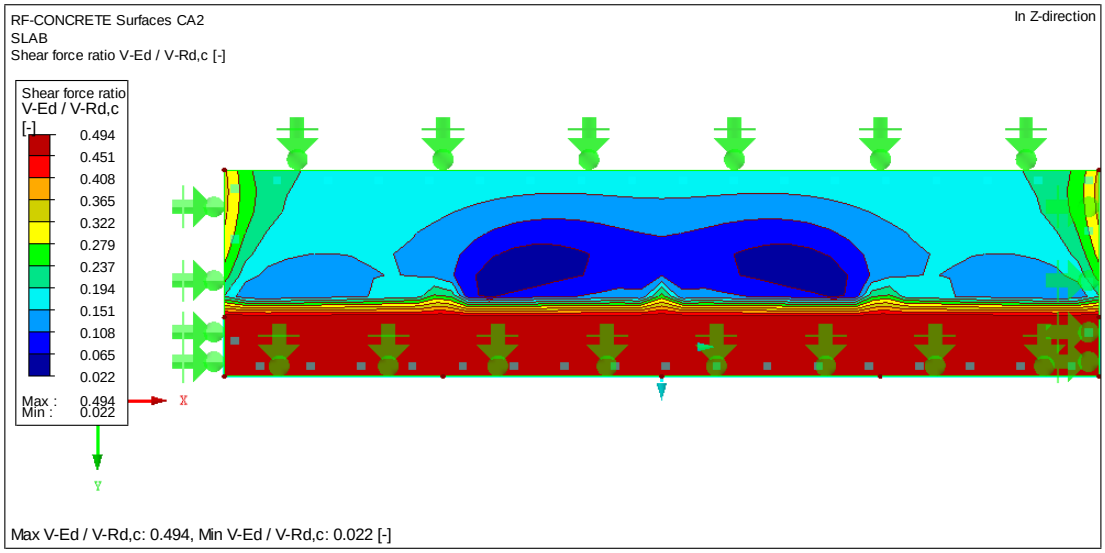


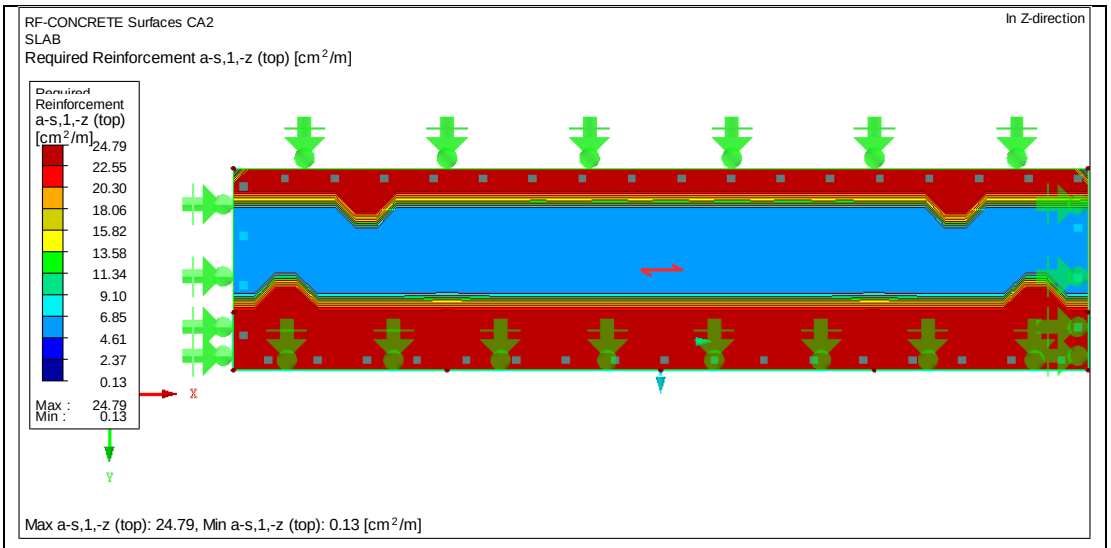
Figure 58 utility ratio $V_{Ed}/V_{Rd,c}$

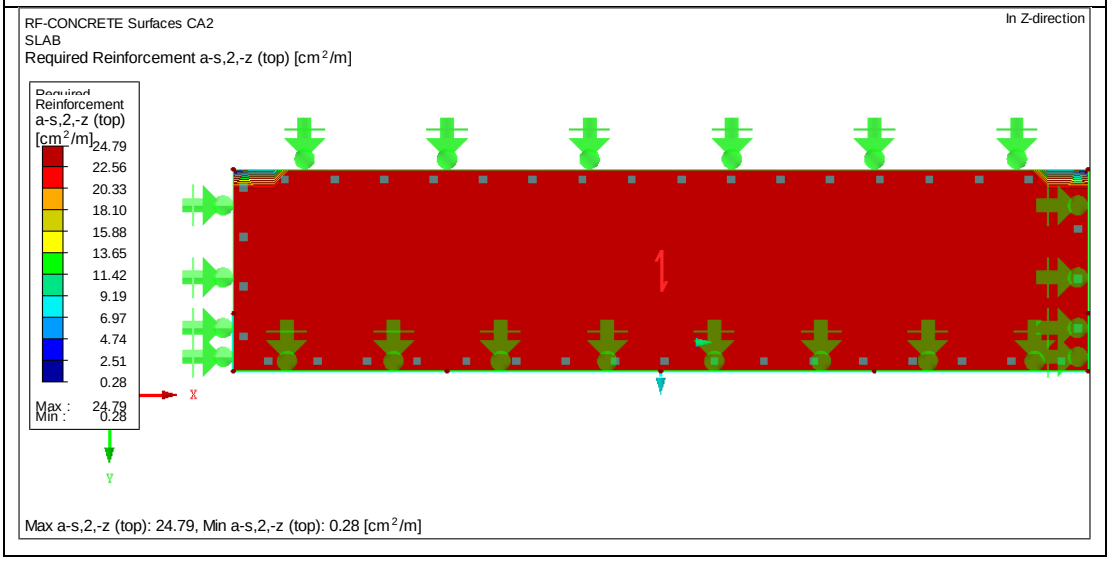
utility ratio= applied shear force/ shear resistance= $V_{Ed}/V_{Rd,c}$< 1.0, ok!

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.





PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

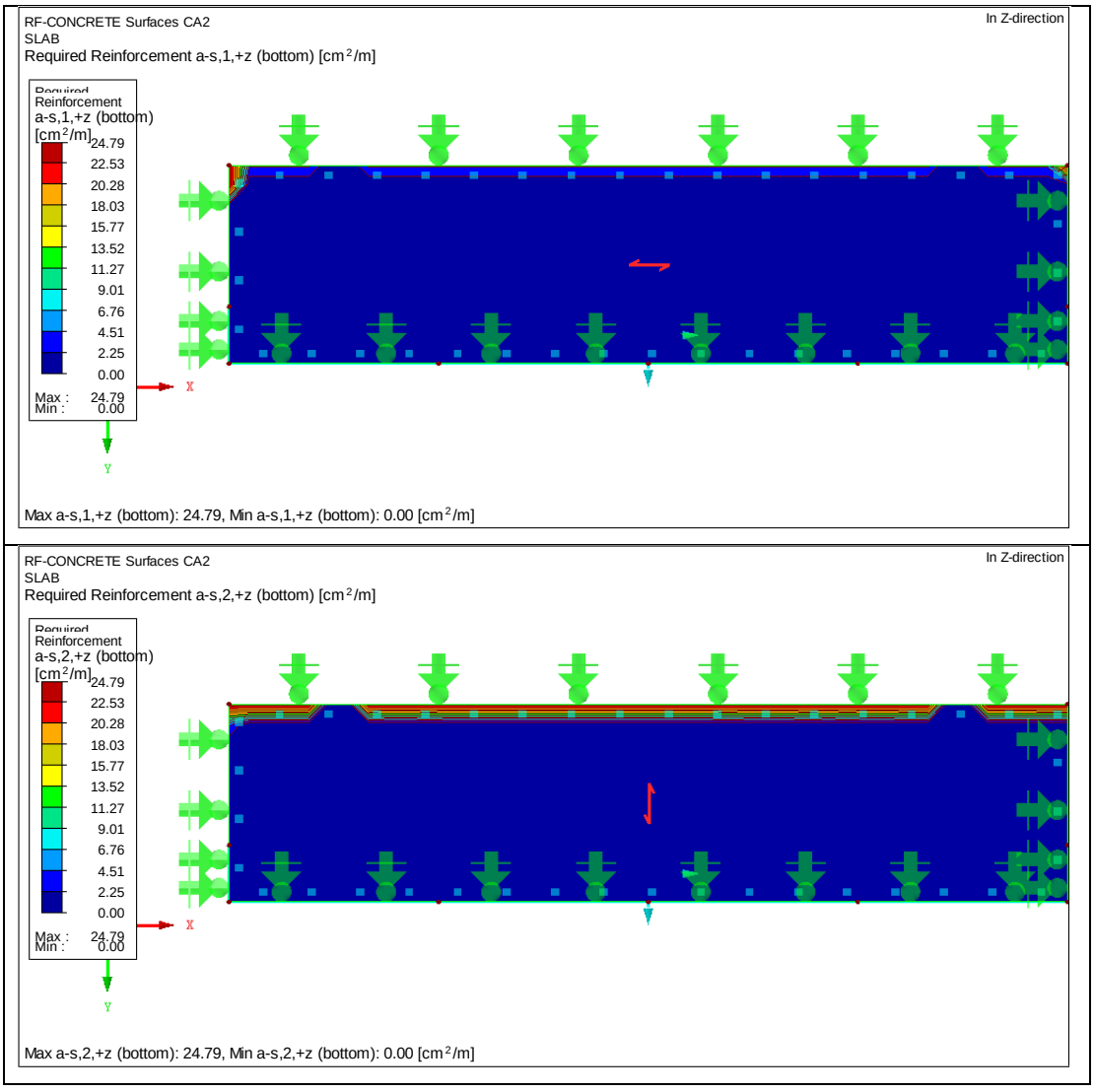


Figure 59 Required reinforcement for foundation

Provide T25@150mm, Asp = 29,45cm^2/m

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

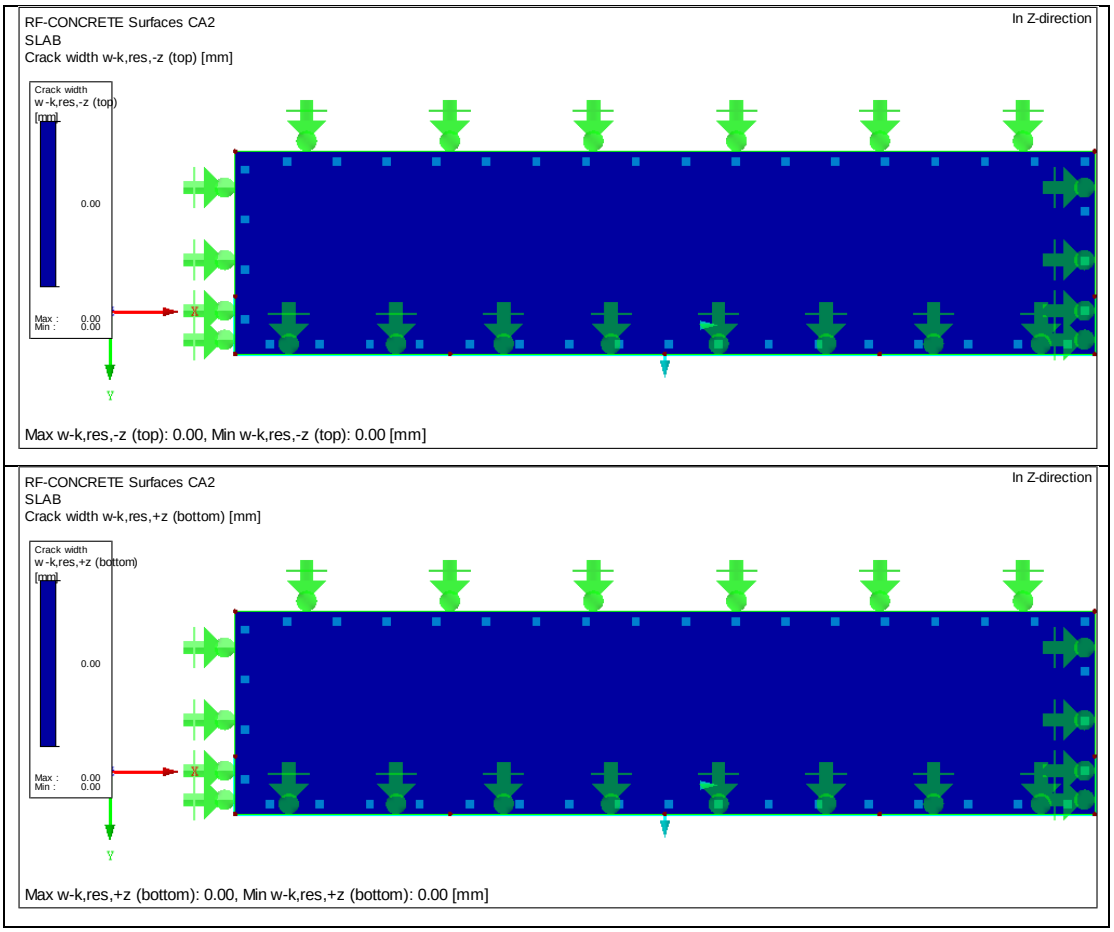


Figure 60 Crack width

crack width at top and bottom $w_k < w_{k,max}$ = 0.20mm, ok!

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

REFERENCES

[1] Saudi Building Code (SBC) 301, Structural - Loading and Forces.

[2] Saudi Building Code (SBC) 304, Structural – Concrete Structures.

[3] Project Design Criteria, MD-216-1000-EG-ST-CRT-1001-A03.

[4] Overall geotechnical report, MD-216-3000-EG-CV-RPT-1008-C01.

[5] General arrangement drawing, raw water treatment plan, 1-8/E-J, MD-216-49W1-EG-CV-GAD-1001

[6] General arrangement drawing, raw water treatment plan, 7-12/E-J, MD-216-49W1-EG-CV-GAD-1002

[7] P06204M002 Sand filter layout

[8] P06204M015 Sand filter model 1

[9] P06204M016 Sand filter model 2

[10] P06204M021 Sand filter distribution box

[11] Detail drawing, Raw water treatment plant, Dimension drawing, sand filter and distribution box, MD-216-49W1-EG-CV-DED-1003

[12] Detail drawing, Raw water treatment plant, Reinforcement drawing, Sand filter bottom slab, MD-216-49W1-EG-CV-DED-1004

[13] Detail drawing, Raw water treatment plant, Reinforcement drawing, Sand filter bottom walls, MD-216-49W1-EG-CV-DED-1005

[14] ACI318, Building Code Requirements for Structural Concrete

[15] ACI350R, Environmental Engineering Concrete Structures

[16] DIN 1045-1. Plain, reinforced and prestressed concrete structures - Part 1: Design and construction

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

ATTACHMENT 1 INPUT DRAWING FROM FLOOTECH

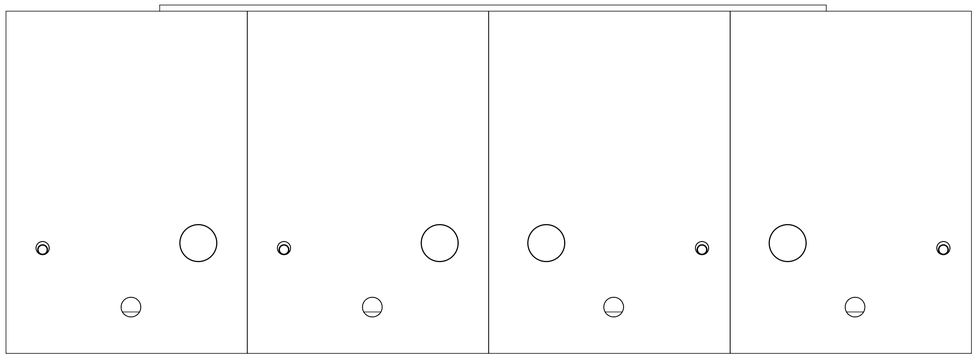
PROPRIETARY & CONFIDENTIAL

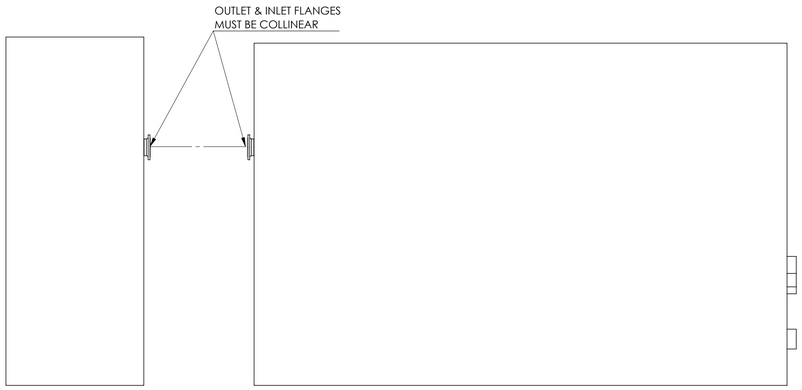
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

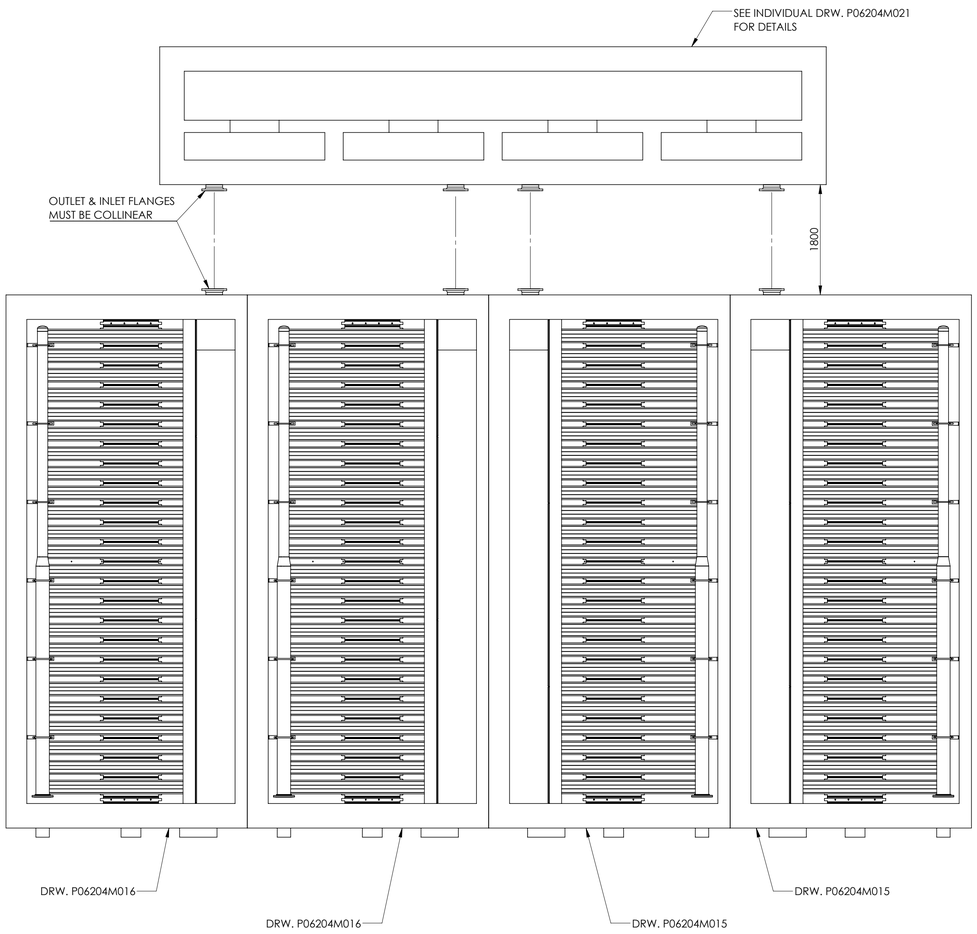
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

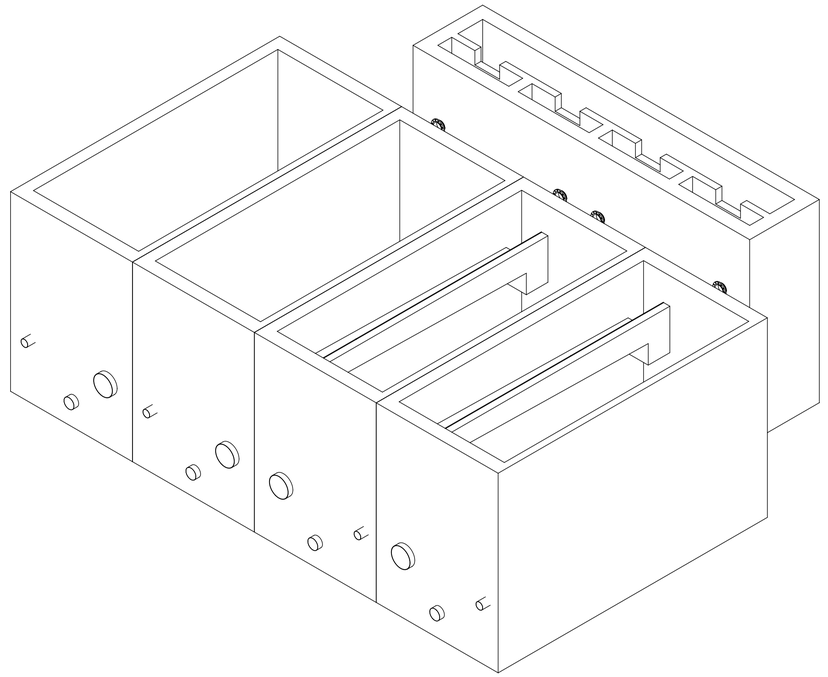
DRAWING NUMBER: –

REV: –









PRE-ENGINEERING DRAWINGS
- NOT FINAL -

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		01	12-08-2021	REVISED AS PER CLIENT COMMENT	MVa	NL		SL	

Flootech

LARSEN & TOUBRO

This drawing is the property of *LARSEN & TOUBRO* and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Outotec

The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

FLOOTECH DRAWING NUMBER: P06204M002.00 — REVISION: –

LARSEN & TOUBRO DRAWING NUMBER: – — REVISION: –

OUTOTEC DRAWING NUMBER: – — DOCUMENT ID: – — REVISION: –

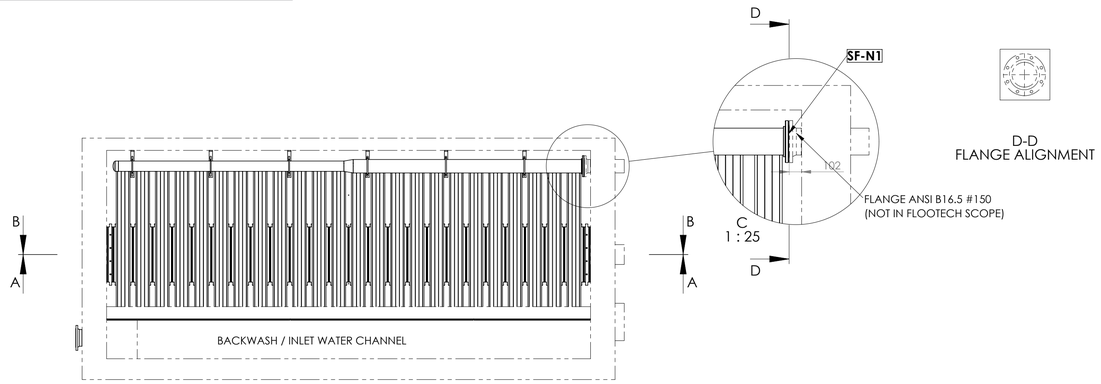
DRAWN	MVa	2.6.2021
CHECKED	MVa	2.6.2021
ENGINEER	NL	2.6.2021
APPROVED	SL	2.6.2021
EMPLOYER		
SCALE	1:50	A1

MANSOURAH - MASSARAH GOLD PROJECT

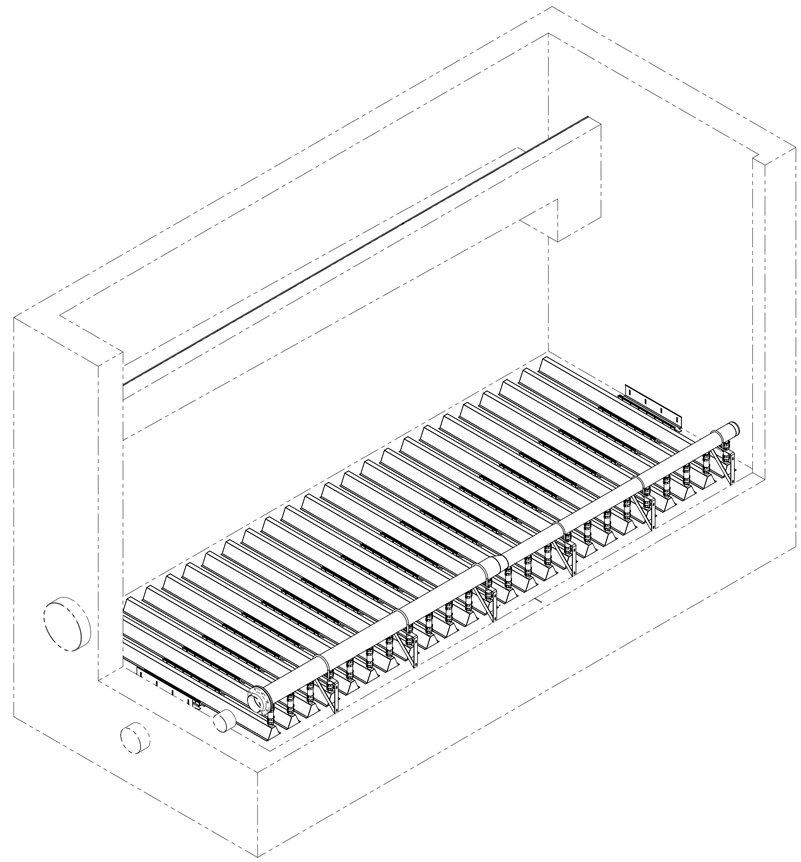
Sand filter + distribution box
Layout

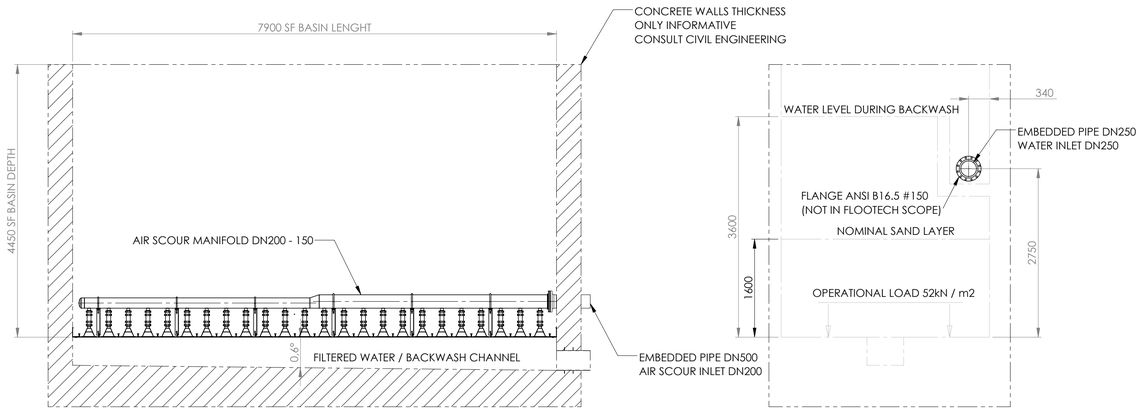
DRAWING NUMBER: – – – – – – — REV: –

DRAWING NUMBER – – – – – – REV

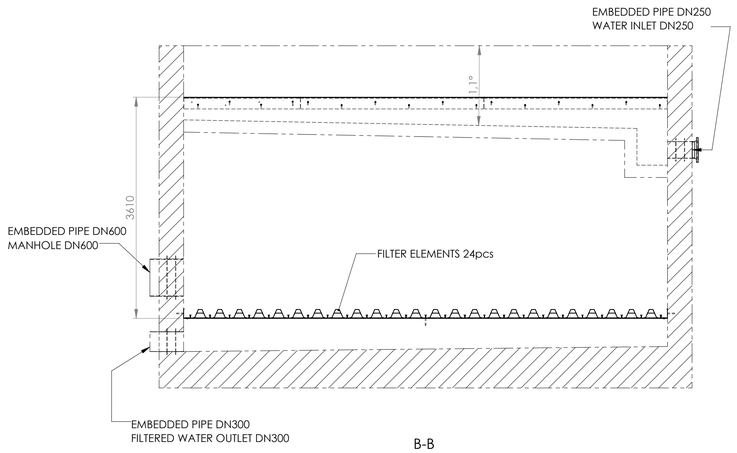


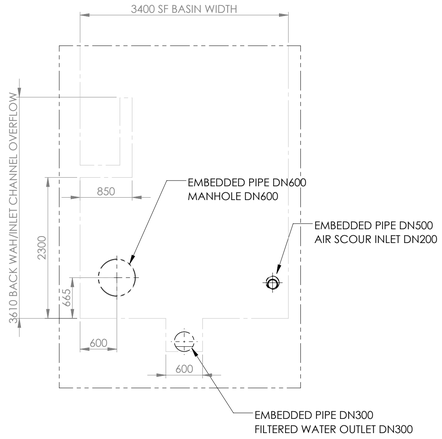
D-D
FLANGE ALIGNMENT





A-A





B-B

EMBEDDED PIPES AND CIVIL CONSTRUCTION
IN BUYER SCOPE OF SUPPLY

PRE-ENGINEERING DRAWINGS
- NOT FINAL -

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		01	12-08-2021	REVISED AS PER CLIENT COMMENT	MVa	NL		SL	

Flootech

FLOOTECH DRAWING NUMBER: P06204M015.00 REVISION: –

LARSEN & TOUBRO DRAWING NUMBER: – REVISION: –

OUTOTEC DRAWING NUMBER: – – – – – DOCUMENT ID: REVISION: –

This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

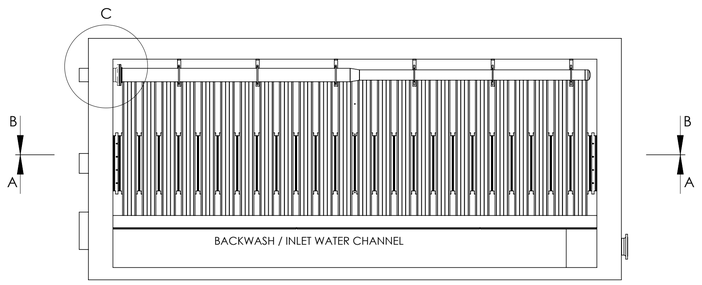
This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

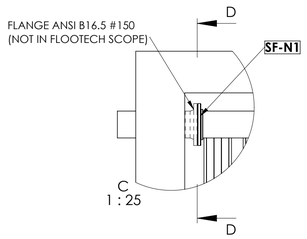
DRAWN	MVa	2.6.2021
CHECKED	MVa	2.6.2021
ENGINEER	NL	2.6.2021
APPROVED	SL	2.6.2021
EMPLOYER		
SCALE	1:50	A1

MANSOURAH - MASSARAH GOLD PROJECT
Sand filtration
FlooSand G
General arrangement drawing

DRAWING NUMBER: – – – – – – REV: –

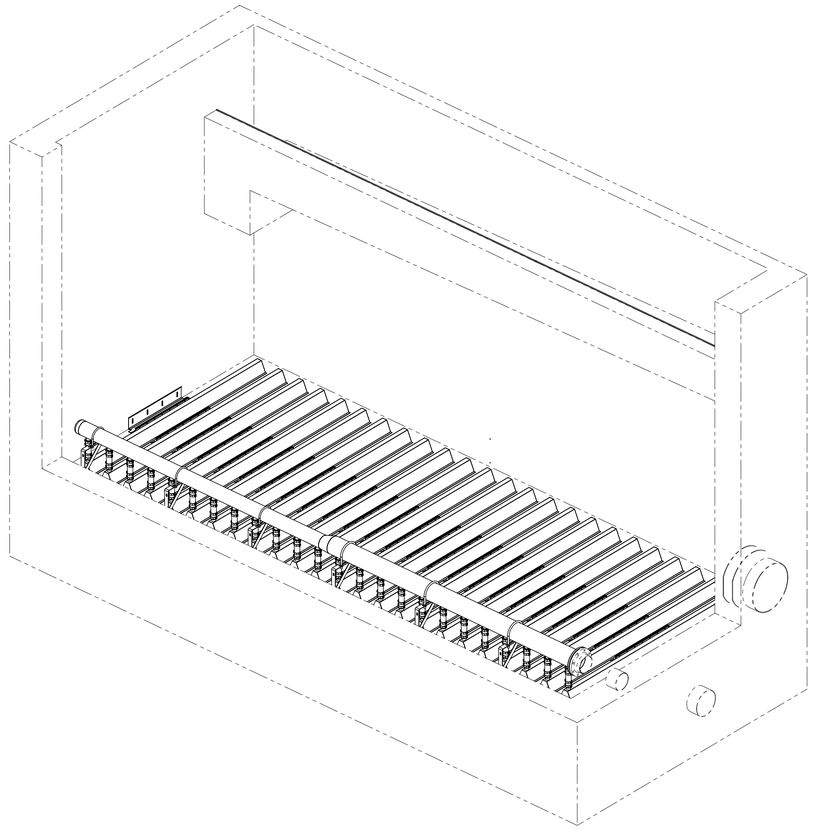
DRAWING NUMBER – – – – – – – REV

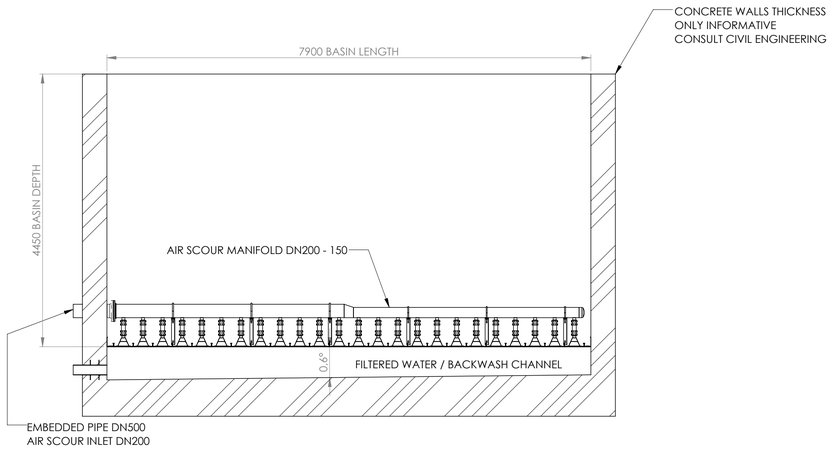




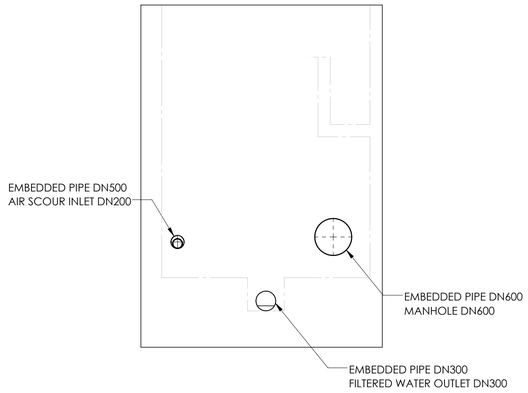


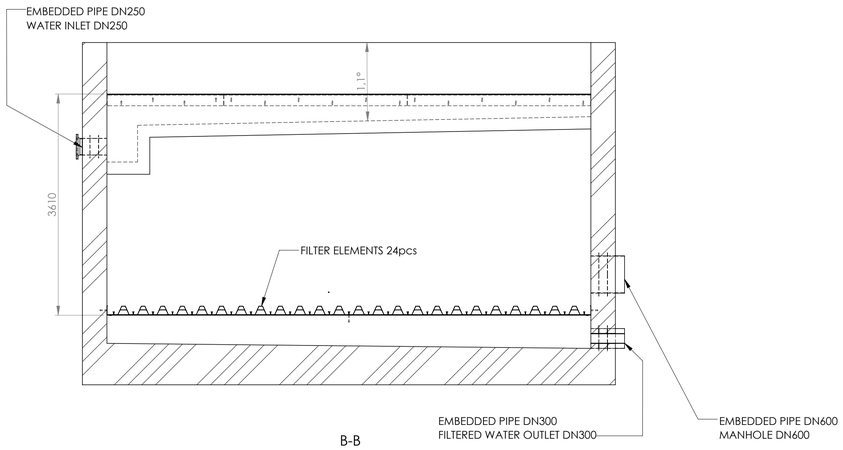
D-D
FLANGE ALIGNMENT



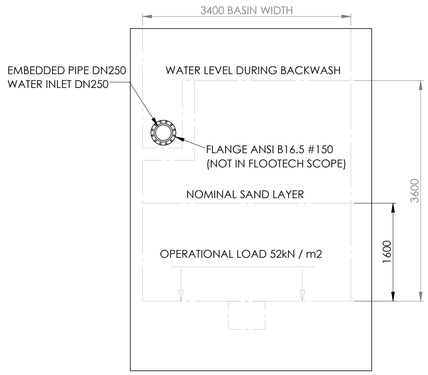


A-A





B-B



EMBEDDED PIPES AND CIVIL CONSTRUCTION
IN BUYER SCOPE OF SUPPLY

PRE-ENGINEERING DRAWINGS
- NOT FINAL -

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		01	12-08-2021	REVISED AS PER CLIENT COMMENT	MVa	NL		SL	

Flootech

LARSEN & TOUBRO

This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

FLOOTECH DRAWING NUMBER: P06204M016.00 — REVISION: –

LARSEN & TOUBRO DRAWING NUMBER: – — REVISION: –

OUTOTEC DRAWING NUMBER: — — — — — DOCUMENT ID: — REVISION: –

Outotec

The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

MA'ADEN Saudi Arabian Mining Company شركة التعدين العربية السعودية

This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

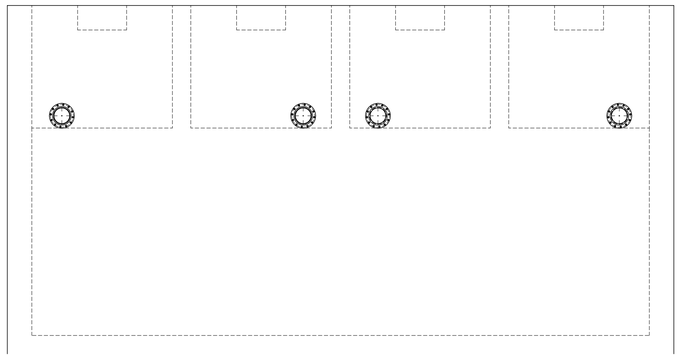
DRAWN	MVa	2.6.2021
CHECKED	MVa	2.6.2021
ENGINEER	NL	2.6.2021
APPROVED	SL	2.6.2021
EMPLOYER		
SCALE	1:50	A1

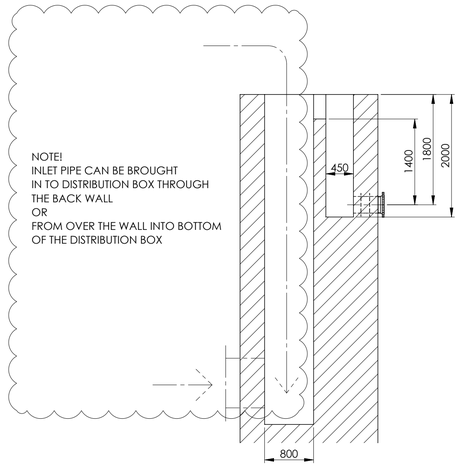
MANSOURAH - MASSARAH GOLD PROJECT
Flotation + sand filtration
Assembly drawing
Main assembly

DRAWING NUMBER: – – – – – – – REV: –

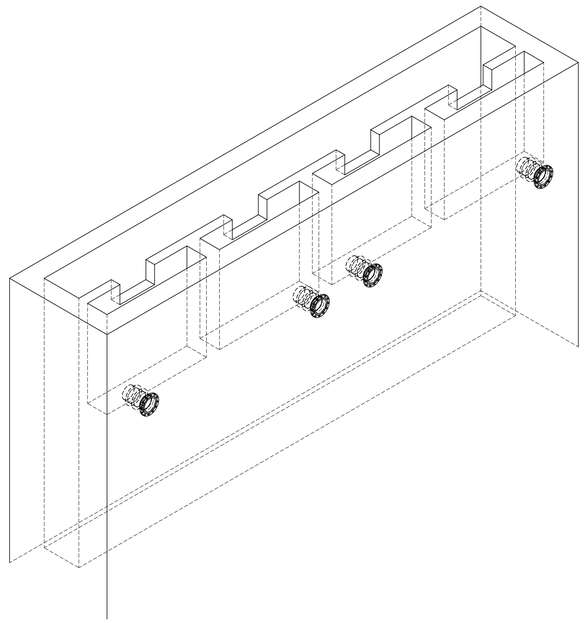
DRAWING NUMBER – – – – – –

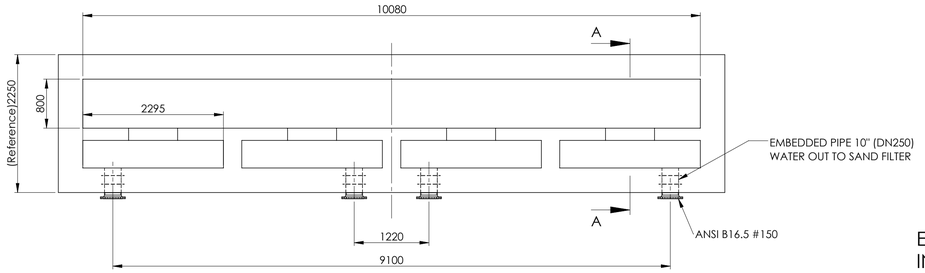
REV –





A-A
PRINCIPLE CROSS SECTION
OF EACH CHAMBER





EMBEDDED PIPES AND CIVIL CONSTRUCTION
IN BUYER SCOPE OF SUPPLY

WALL THICKNESSES TO BE DEFINED BY CIVIL

PRE-ENGINEERING DRAWINGS
- NOT FINAL -

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		01	12-08-2021	REVISED AS PER CLIENT COMMENT	MVa	NL		SL	

Flootech

FLOOTECH DRAWING NUMBER: P06204M021.00 REVISION: –

LARSEN & TOUBRO

This drawing is the property of *LARSEN & TOUBRO* and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Outotec

The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

MA'ADEN Saudi Arabian Mining Company

This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be regroduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

LARSEN & TOUBRO DRAWING NUMBER: – REVISION: –

OUTOTEC DRAWING NUMBER: – – – – DOCUMENT ID: REVISION: –

DRAWN	MVa	2.6.2021
CHECKED	MVa	2.6.2021
ENGINEER	NL	2.6.2021
APPROVED	SL	2.6.2021
EMPLOYER		
SCALE	1:50	A1

MANSOURAH - MASSARAH GOLD PROJECT
Distribution box
Main dimension drawing
49W1-DBX-001

DRAWING NUMBER: – – – – – – REV: –

ATTACHMENT 2 COMMENT RESOLUTION SHEET

When comments are received from Employer, they need to be placed in a comment resolution table, and the filled table is always attached to the end of the document.

COMMENT TABLE	Document Name:	Document Number:	Revision No.	Comment Date:
General Comment:				

Sr. No	Chapter	Regarding	Comment	Resolution
1	4	design basis	exposure class, crack, deformation, soil pressure	added in p.3
2	4	design codes	design codes	included in Ch. 2 Scope
3	4	load pattern &maintenance	needed	added in p.4
4	6	design basis	Exposure categories and classes	added as Ch. 4
5	7.1	pattern load	Please check if patter load is possible per process design criteria	see point 3
6	7.3	shear force results	shear force results	shear force results added. shear is not critical, bending and tension is in control
5	7.3	crack width check	conclusion	added
6	7.4	shear force results	shear force results	shear force results added.
7	7.4	calculation sheet source	Please show the reference of this calculation sheet and indicate it in reference list, too	added. it is an extra check.
8	7.4	shear force results	shear force results	shear force results added.
9	7.4	soil bearing capacity	soil bearing capacity	soil bearing pressure = 'contact stress' in RFEM, see Figure 13
10	7.6.2	shear force results	shear force results	shear force results added.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

11	8.4	soil bearing capacity	soil bearing capacity	soil bearing pressure = 'contact stress' in RFEM, see Figure 39

1 Remember also to always mark the made changes with a triangle to the left side of the document.

You can also reply to the comments on an excel template, then it will be coded as a separate document and submitted to Employer together with the revised document itself. You can find the instruction and the excel template from "Document Codes and Numbering Procedure" [1].

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
Load Case LC1 - DL				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	2511.46	kN	
	Sum of support reactions in Z	2511.46	kN	Deviation 0.00%
	Resultant of reactions about X	323.950	kNm	At center of gravity of model (X:0.000, Y:-0.501, Z:-1.705 m)
	Resultant of reactions about Y	0.000	kNm	At center of gravity of model
	Resultant of reactions about Z	0.000	kNm	At center of gravity of model
	Max. displacement in X	-0.0	mm	FE Mesh Node No. 729 (X: -4.671, Y: 0.000, Z: -5.570 m)
	Max. displacement in Y	5.0	mm	FE Mesh Node No. 391 (X: 0.000, Y: 0.700, Z: -5.570 m)
	Max. displacement in Z	4.0	mm	FE Mesh Node No. 319 (X: 5.190, Y: 0.700, Z: -5.570 m)
	Max. vector displacement	6.4	mm	FE Mesh Node No. 391 (X: 0.000, Y: 0.700, Z: -5.570 m)
	Max. rotation about X	0.9	mrad	FE Mesh Node No. 1415 (X: 0.260, Y: 0.700, Z: -0.253 m)
	Max. rotation about Y	-0.0	mrad	FE Mesh Node No. 3469 (X: 4.943, Y: -0.750, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 955 (X: 3.893, Y: 0.700, Z: -5.570 m)
	Method of analysis	Linear		Geometrically linear analysis
	Reduction of stiffness			Cross-sections, Members, Surfaces
	Number of load increments	1		
	Number of iterations	1		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.404E+201470		
	Infinity Norm	2.628E+11		
Load Case LC2 - F				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	1011.79	kN	
	Sum of support reactions in Z	1011.79	kN	Deviation 0.00%
	Resultant of reactions about X	-378.566	kNm	At center of gravity of model (X:0.000, Y:-0.501, Z:-1.705 m)
	Resultant of reactions about Y	0.000	kNm	At center of gravity of model
	Resultant of reactions about Z	0.000	kNm	At center of gravity of model
	Max. displacement in X	-0.2	mm	FE Mesh Node No. 3113 (X: 5.190, Y: -1.000, Z: -5.570 m)
	Max. displacement in Y	-10.1	mm	FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
	Max. displacement in Z	2.2	mm	FE Mesh Node No. 2016 (X: 5.190, Y: -1.750, Z: -4.810 m)
	Max. vector displacement	10.4	mm	FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
	Max. rotation about X	-2.2	mrad	FE Mesh Node No. 2553 (X: 0.000, Y: -1.750, Z: -1.772 m)
	Max. rotation about Y	-0.1	mrad	FE Mesh Node No. 2899 (X: -5.190, Y: -1.250, Z: -5.570 m)
	Max. rotation about Z	-1.5	mrad	FE Mesh Node No. 1958 (X: -3.460, Y: -1.750, Z: -5.570 m)
	Method of analysis	Linear		Geometrically linear analysis
	Reduction of stiffness			Cross-sections, Members, Surfaces
	Number of load increments	1		
	Number of iterations	1		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.404E+201470		
	Infinity Norm	2.628E+11		
Load Case LC3 - X+				
	Sum of loads in X	10.73	kN	
	Sum of support reactions in X	10.73	kN	Deviation 0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	0.00	kN	
	Sum of support reactions in Z	0.00	kN	
	Resultant of reactions about X	0.000	kNm	At center of gravity of model (X:0.000, Y:-0.501, Z:-1.705 m)
	Resultant of reactions about Y	-11.589	kNm	At center of gravity of model
	Resultant of reactions about Z	0.259	kNm	At center of gravity of model
	Max. displacement in X	0.0	mm	FE Mesh Node No. 2922 (X: -5.190, Y: -1.000, Z: -5.570 m)
	Max. displacement in Y	-0.4	mm	FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
	Max. displacement in Z	0.0	mm	FE Mesh Node No. 1973 (X: 5.190, Y: -1.750, Z: -5.064 m)
	Max. vector displacement	0.4	mm	FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
	Max. rotation about X	-0.1	mrad	FE Mesh Node No. 2424 (X: 0.000, Y: -1.750, Z: -2.532 m)
	Max. rotation about Y	-0.0	mrad	FE Mesh Node No. 2899 (X: -5.190, Y: -1.250, Z: -5.570 m)
	Max. rotation about Z	-0.1	mrad	FE Mesh Node No. 1958 (X: -3.460, Y: -1.750, Z: -5.570 m)
	Method of analysis	Linear		Geometrically linear analysis
	Reduction of stiffness			Cross-sections, Members, Surfaces
	Number of load increments	1		
	Number of iterations	1		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.404E+201470		
	Infinity Norm	2.628E+11		

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
Load Case LC4 - X-				
	Sum of loads in X	10.73	kN	
	Sum of support reactions in X	10.73	kN	Deviation 0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	0.00	kN	
	Sum of support reactions in Z	0.00	kN	
	Resultant of reactions about X	0.000	kNm	At center of gravity of model (X:0.000, Y:-0.501, Z:-1.705 m)
	Resultant of reactions about Y	-11.589	kNm	At center of gravity of model
	Resultant of reactions about Z	0.259	kNm	At center of gravity of model
	Max. displacement in X	0.0	mm	FE Mesh Node No. 2945 (X: -5.190, Y: -0.750, Z: -5.570 m)
	Max. displacement in Y	-0.0	mm	FE Mesh Node No. 1932 (X: -0.247, Y: -1.750, Z: -5.570 m)
	Max. displacement in Z	-0.0	mm	FE Mesh Node No. 386 (X: -5.190, Y: 0.700, Z: -5.570 m)
	Max. vector displacement	0.0	mm	FE Mesh Node No. 2945 (X: -5.190, Y: -0.750, Z: -5.570 m)
	Max. rotation about X	-0.0	mrad	FE Mesh Node No. 2382 (X: -0.247, Y: -1.750, Z: -2.785 m)
	Max. rotation about Y	-0.0	mrad	FE Mesh Node No. 2965 (X: -5.190, Y: -0.750, Z: -0.506 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 1964 (X: -4.201, Y: -1.750, Z: -5.570 m)
	Method of analysis	Linear		Geometrically linear analysis
	Reduction of stiffness			Cross-sections, Members, Surfaces
	Number of load increments	1		
	Number of iterations	1		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.404E+201470		
	Infinity Norm	2.628E+11		
Load Case LC5 - Y+				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	59.72	kN	
	Sum of support reactions in Y	59.72	kN	Deviation 0.00%
	Sum of loads in Z	0.00	kN	
	Sum of support reactions in Z	0.00	kN	
	Resultant of reactions about X	64.514	kNm	At center of gravity of model (X:0.000, Y:-0.501, Z:-1.705 m)
	Resultant of reactions about Y	0.000	kNm	At center of gravity of model
	Resultant of reactions about Z	0.000	kNm	At center of gravity of model
	Max. displacement in X	-0.0	mm	FE Mesh Node No. 2968 (X: -5.190, Y: -0.500, Z: -5.570 m)
	Max. displacement in Y	2.2	mm	FE Mesh Node No. 1210 (X: 0.779, Y: 0.700, Z: -5.570 m)
	Max. displacement in Z	0.5	mm	FE Mesh Node No. 1256 (X: 0.000, Y: 0.700, Z: -4.304 m)
	Max. vector displacement	2.3	mm	FE Mesh Node No. 1210 (X: 0.779, Y: 0.700, Z: -5.570 m)
	Max. rotation about X	0.4	mrad	FE Mesh Node No. 1371 (X: 1.298, Y: 0.700, Z: -1.266 m)
	Max. rotation about Y	-0.0	mrad	FE Mesh Node No. 12 (X: 3.893, Y: 0.000, Z: -5.570 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 951 (X: 4.412, Y: 0.700, Z: -5.570 m)
	Method of analysis	Linear		Geometrically linear analysis
	Reduction of stiffness			Cross-sections, Members, Surfaces
	Number of load increments	1		
	Number of iterations	1		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.404E+201470		
	Infinity Norm	2.628E+11		
Load Case LC6 - Y-				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	59.72	kN	
	Sum of support reactions in Y	59.72	kN	Deviation 0.00%
	Sum of loads in Z	0.00	kN	
	Sum of support reactions in Z	0.00	kN	
	Resultant of reactions about X	64.514	kNm	At center of gravity of model (X:0.000, Y:-0.501, Z:-1.705 m)
	Resultant of reactions about Y	0.000	kNm	At center of gravity of model
	Resultant of reactions about Z	0.000	kNm	At center of gravity of model
	Max. displacement in X	-0.0	mm	FE Mesh Node No. 2922 (X: -5.190, Y: -1.000, Z: -5.570 m)
	Max. displacement in Y	2.6	mm	FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
	Max. displacement in Z	-0.5	mm	FE Mesh Node No. 1973 (X: 5.190, Y: -1.750, Z: -5.064 m)
	Max. vector displacement	2.6	mm	FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
	Max. rotation about X	0.5	mrad	FE Mesh Node No. 2381 (X: 0.000, Y: -1.750, Z: -2.785 m)
	Max. rotation about Y	-0.0	mrad	FE Mesh Node No. 3115 (X: 5.190, Y: -1.250, Z: -5.570 m)
	Max. rotation about Z	-0.1	mrad	FE Mesh Node No. 1902 (X: 3.460, Y: -1.750, Z: -5.570 m)
	Method of analysis	Linear		Geometrically linear analysis
	Reduction of stiffness			Cross-sections, Members, Surfaces
	Number of load increments	1		
	Number of iterations	1		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.404E+201470		
	Infinity Norm	2.628E+11		

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
Load Combination CO1 - 1.4*LC1 + 1.4*LC2				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	4932.55	kN	
	Sum of support reactions in Z	4932.55	kN	Deviation 0.00%
	Resultant of reactions about X	-76.9	kNm	At center of gravity of model (X:0.0, Y:-0.5, Z:-1.7 m)
	Resultant of reactions about Y	0.0	kNm	At center of gravity of model
	Resultant of reactions about Z	0.0	kNm	At center of gravity of model
	Max. displacement in X	-0.3	mm	FE Mesh Node No. 3113 (X: 5.190, Y: -1.000, Z: -5.570 m)
	Max. displacement in Y	-7.2	mm	FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
	Max. displacement in Z	6.0	mm	FE Mesh Node No. 1336 (X: 0.000, Y: 0.700, Z: -2.279 m)
	Max. vector displacement	9.2	mm	FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
	Max. rotation about X	-1.8	mrad	FE Mesh Node No. 2553 (X: 0.000, Y: -1.750, Z: -1.772 m)
	Max. rotation about Y	-0.1	mrad	FE Mesh Node No. 2899 (X: -5.190, Y: -1.250, Z: -5.570 m)
	Max. rotation about Z	-2.1	mrad	FE Mesh Node No. 1958 (X: -3.460, Y: -1.750, Z: -5.570 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	10		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.366E+201470		
	Infinity Norm	2.628E+11		
Load Combination CO2 - 1.2*LC1 + 1.2*LC2 + 0.8*LC3				
	Sum of loads in X	8.58	kN	
	Sum of support reactions in X	8.58	kN	Deviation -0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	4227.90	kN	
	Sum of support reactions in Z	4227.90	kN	Deviation 0.00%
	Resultant of reactions about X	-65.9	kNm	At center of gravity of model (X:0.0, Y:-0.5, Z:-1.7 m)
	Resultant of reactions about Y	-9.3	kNm	At center of gravity of model
	Resultant of reactions about Z	0.2	kNm	At center of gravity of model
	Max. displacement in X	0.2	mm	FE Mesh Node No. 2922 (X: -5.190, Y: -1.000, Z: -5.570 m)
	Max. displacement in Y	-6.5	mm	FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
	Max. displacement in Z	5.1	mm	FE Mesh Node No. 1336 (X: 0.000, Y: 0.700, Z: -2.279 m)
	Max. vector displacement	8.1	mm	FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
	Max. rotation about X	-1.6	mrad	FE Mesh Node No. 2553 (X: 0.000, Y: -1.750, Z: -1.772 m)
	Max. rotation about Y	-0.1	mrad	FE Mesh Node No. 2899 (X: -5.190, Y: -1.250, Z: -5.570 m)
	Max. rotation about Z	-1.9	mrad	FE Mesh Node No. 1958 (X: -3.460, Y: -1.750, Z: -5.570 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	10		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.372E+201470		
	Infinity Norm	2.628E+11		
Load Combination CO3 - 1.2*LC1 + 1.2*LC2 + 0.8*LC4				
	Sum of loads in X	8.58	kN	
	Sum of support reactions in X	8.58	kN	Deviation 0.00%
	Sum of loads in Y	-0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	4227.90	kN	
	Sum of support reactions in Z	4227.90	kN	Deviation 0.00%
	Resultant of reactions about X	-65.8	kNm	At center of gravity of model (X:0.0, Y:-0.5, Z:-1.7 m)
	Resultant of reactions about Y	-9.3	kNm	At center of gravity of model
	Resultant of reactions about Z	0.2	kNm	At center of gravity of model
	Max. displacement in X	0.2	mm	FE Mesh Node No. 2922 (X: -5.190, Y: -1.000, Z: -5.570 m)
	Max. displacement in Y	-6.2	mm	FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
	Max. displacement in Z	5.1	mm	FE Mesh Node No. 1336 (X: 0.000, Y: 0.700, Z: -2.279 m)
	Max. vector displacement	7.9	mm	FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
	Max. rotation about X	-1.5	mrad	FE Mesh Node No. 2553 (X: 0.000, Y: -1.750, Z: -1.772 m)
	Max. rotation about Y	-0.1	mrad	FE Mesh Node No. 2899 (X: -5.190, Y: -1.250, Z: -5.570 m)
	Max. rotation about Z	-1.8	mrad	FE Mesh Node No. 1958 (X: -3.460, Y: -1.750, Z: -5.570 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	10		

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.371E+201470		
	Infinity Norm	2.628E+11		
Load Combination CO4 - 1.2*LC1 + 1.2*LC2 + 0.8*LC5				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	47.78	kN	
	Sum of support reactions in Y	47.78	kN	Deviation -0.00%
	Sum of loads in Z	4227.90	kN	
	Sum of support reactions in Z	4227.90	kN	Deviation 0.00%
	Resultant of reactions about X	-13.1	kNm	At center of gravity of model (X:0.0, Y:-0.5, Z:-1.7 m)
	Resultant of reactions about Y	0.0	kNm	At center of gravity of model
	Resultant of reactions about Z	0.0	kNm	At center of gravity of model
	Max. displacement in X	-0.2	mm	FE Mesh Node No. 3113 (X: 5.190, Y: -1.000, Z: -5.570 m)
	Max. displacement in Y	-4.4	mm	FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
	Max. displacement in Z	5.5	mm	FE Mesh Node No. 1336 (X: 0.000, Y: 0.700, Z: -2.279 m)
	Max. vector displacement	6.6	mm	FE Mesh Node No. 391 (X: 0.000, Y: 0.700, Z: -5.570 m)
	Max. rotation about X	-1.2	mrad	FE Mesh Node No. 2553 (X: 0.000, Y: -1.750, Z: -1.772 m)
	Max. rotation about Y	-0.1	mrad	FE Mesh Node No. 2899 (X: -5.190, Y: -1.250, Z: -5.570 m)
	Max. rotation about Z	-1.8	mrad	FE Mesh Node No. 1958 (X: -3.460, Y: -1.750, Z: -5.570 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	10		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.374E+201470		
	Infinity Norm	2.628E+11		
Load Combination CO5 - 1.2*LC1 + 1.2*LC2 + 0.8*LC6				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	47.78	kN	
	Sum of support reactions in Y	47.78	kN	Deviation -0.00%
	Sum of loads in Z	4227.90	kN	
	Sum of support reactions in Z	4227.90	kN	Deviation 0.00%
	Resultant of reactions about X	-13.1	kNm	At center of gravity of model (X:0.0, Y:-0.5, Z:-1.7 m)
	Resultant of reactions about Y	0.0	kNm	At center of gravity of model
	Resultant of reactions about Z	0.0	kNm	At center of gravity of model
	Max. displacement in X	-0.2	mm	FE Mesh Node No. 3113 (X: 5.190, Y: -1.000, Z: -5.570 m)
	Max. displacement in Y	-4.1	mm	FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
	Max. displacement in Z	5.5	mm	FE Mesh Node No. 1336 (X: 0.000, Y: 0.700, Z: -2.279 m)
	Max. vector displacement	6.5	mm	FE Mesh Node No. 391 (X: 0.000, Y: 0.700, Z: -5.570 m)
	Max. rotation about X	-1.1	mrad	FE Mesh Node No. 2553 (X: 0.000, Y: -1.750, Z: -1.772 m)
	Max. rotation about Y	-0.1	mrad	FE Mesh Node No. 2899 (X: -5.190, Y: -1.250, Z: -5.570 m)
	Max. rotation about Z	-1.7	mrad	FE Mesh Node No. 1958 (X: -3.460, Y: -1.750, Z: -5.570 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	10		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.372E+201470		
	Infinity Norm	2.628E+11		
Load Combination CO6 - 0.9*LC1 + 1.6*LC3				
	Sum of loads in X	17.16	kN	
	Sum of support reactions in X	17.16	kN	Deviation 0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	2260.32	kN	
	Sum of support reactions in Z	2260.32	kN	Deviation 0.00%
	Resultant of reactions about X	294.1	kNm	At center of gravity of model (X:0.0, Y:-0.5, Z:-1.7 m)
	Resultant of reactions about Y	-18.6	kNm	At center of gravity of model
	Resultant of reactions about Z	0.4	kNm	At center of gravity of model
	Max. displacement in X	0.1	mm	FE Mesh Node No. 2922 (X: -5.190, Y: -1.000, Z: -5.570 m)
	Max. displacement in Y	4.7	mm	FE Mesh Node No. 1212 (X: 0.519, Y: 0.700, Z: -5.570 m)
	Max. displacement in Z	3.7	mm	FE Mesh Node No. 945 (X: 5.190, Y: 0.700, Z: -5.317 m)
	Max. vector displacement	5.9	mm	FE Mesh Node No. 1210 (X: 0.779, Y: 0.700, Z: -5.570 m)

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Max. rotation about X	0.8	mrad	FE Mesh Node No. 1372 (X: 1.038, Y: 0.700, Z: -1.266 m)
	Max. rotation about Y	-0.0	mrad	FE Mesh Node No. 2899 (X: -5.190, Y: -1.250, Z: -5.570 m)
	Max. rotation about Z	-0.2	mrad	FE Mesh Node No. 1958 (X: -3.460, Y: -1.750, Z: -5.570 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	10		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.364E+201470		
	Infinity Norm	2.628E+11		
Load Combination CO7 - 0.9*LC1 + 1.6*LC4				
	Sum of loads in X	17.16	kN	
	Sum of support reactions in X	17.16	kN	Deviation 0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	2260.32	kN	
	Sum of support reactions in Z	2260.32	kN	Deviation 0.00%
	Resultant of reactions about X	294.2	kNm	At center of gravity of model (X:0.0, Y:-0.5, Z:-1.7 m)
	Resultant of reactions about Y	-18.6	kNm	At center of gravity of model
	Resultant of reactions about Z	0.4	kNm	At center of gravity of model
	Max. displacement in X	0.0	mm	FE Mesh Node No. 319 (X: 5.190, Y: 0.700, Z: -5.570 m)
	Max. displacement in Y	4.5	mm	FE Mesh Node No. 391 (X: 0.000, Y: 0.700, Z: -5.570 m)
	Max. displacement in Z	3.7	mm	FE Mesh Node No. 319 (X: 5.190, Y: 0.700, Z: -5.570 m)
	Max. vector displacement	5.8	mm	FE Mesh Node No. 961 (X: 3.114, Y: 0.700, Z: -5.570 m)
	Max. rotation about X	0.8	mrad	FE Mesh Node No. 1393 (X: 0.779, Y: 0.700, Z: -0.760 m)
	Max. rotation about Y	-0.0	mrad	FE Mesh Node No. 3469 (X: 4.943, Y: -0.750, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 1964 (X: -4.201, Y: -1.750, Z: -5.570 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	10		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.361E+201470		
	Infinity Norm	2.628E+11		
Load Combination CO8 - 0.9*LC1 + 1.6*LC5				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	95.55	kN	
	Sum of support reactions in Y	95.55	kN	Deviation 0.00%
	Sum of loads in Z	2260.32	kN	
	Sum of support reactions in Z	2260.32	kN	Deviation 0.00%
	Resultant of reactions about X	399.4	kNm	At center of gravity of model (X:0.0, Y:-0.5, Z:-1.7 m)
	Resultant of reactions about Y	0.0	kNm	At center of gravity of model
	Resultant of reactions about Z	0.0	kNm	At center of gravity of model
	Max. displacement in X	-0.0	mm	FE Mesh Node No. 729 (X: -4.671, Y: 0.000, Z: -5.570 m)
	Max. displacement in Y	8.1	mm	FE Mesh Node No. 1210 (X: 0.779, Y: 0.700, Z: -5.570 m)
	Max. displacement in Z	4.4	mm	FE Mesh Node No. 945 (X: 5.190, Y: 0.700, Z: -5.317 m)
	Max. vector displacement	9.2	mm	FE Mesh Node No. 1210 (X: 0.779, Y: 0.700, Z: -5.570 m)
	Max. rotation about X	1.5	mrad	FE Mesh Node No. 1372 (X: 1.038, Y: 0.700, Z: -1.266 m)
	Max. rotation about Y	-0.0	mrad	FE Mesh Node No. 317 (X: 5.190, Y: 0.700, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 953 (X: 4.152, Y: 0.700, Z: -5.570 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	10		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.366E+201470		
	Infinity Norm	2.628E+11		
Load Combination CO9 - 0.9*LC1 + 1.6*LC6				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	95.55	kN	
	Sum of support reactions in Y	95.55	kN	Deviation -0.00%

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Sum of loads in Z	2260.32	kN	
	Sum of support reactions in Z	2260.32	kN	Deviation 0.00%
	Resultant of reactions about X	399.7	kNm	At center of gravity of model (X:0.0, Y:-0.5, Z:-1.7 m)
	Resultant of reactions about Y	0.0	kNm	At center of gravity of model
	Resultant of reactions about Z	0.0	kNm	At center of gravity of model
	Max. displacement in X	-0.0	mm	FE Mesh Node No. 2922 (X: -5.190, Y: -1.000, Z: -5.570 m)
	Max. displacement in Y	8.7	mm	FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
	Max. displacement in Z	4.4	mm	FE Mesh Node No. 319 (X: 5.190, Y: 0.700, Z: -5.570 m)
	Max. vector displacement	9.2	mm	FE Mesh Node No. 319 (X: 5.190, Y: 0.700, Z: -5.570 m)
	Max. rotation about X	1.6	mrad	FE Mesh Node No. 2381 (X: 0.000, Y: -1.750, Z: -2.785 m)
	Max. rotation about Y	-0.0	mrad	FE Mesh Node No. 3115 (X: 5.190, Y: -1.250, Z: -5.570 m)
	Max. rotation about Z	-0.2	mrad	FE Mesh Node No. 1902 (X: 3.460, Y: -1.750, Z: -5.570 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	10		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.363E+201470		
	Infinity Norm	2.628E+11		
Load Combination CO10 - LC1 + LC2				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	3523.25	kN	
	Sum of support reactions in Z	3523.25	kN	Deviation 0.00%
	Resultant of reactions about X	-54.8	kNm	At center of gravity of model (X:0.0, Y:-0.5, Z:-1.7 m)
	Resultant of reactions about Y	0.0	kNm	At center of gravity of model
	Resultant of reactions about Z	0.0	kNm	At center of gravity of model
	Max. displacement in X	-0.2	mm	FE Mesh Node No. 3113 (X: 5.190, Y: -1.000, Z: -5.570 m)
	Max. displacement in Y	-5.2	mm	FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
	Max. displacement in Z	4.3	mm	FE Mesh Node No. 1336 (X: 0.000, Y: 0.700, Z: -2.279 m)
	Max. vector displacement	6.6	mm	FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
	Max. rotation about X	-1.3	mrad	FE Mesh Node No. 2553 (X: 0.000, Y: -1.750, Z: -1.772 m)
	Max. rotation about Y	-0.1	mrad	FE Mesh Node No. 2899 (X: -5.190, Y: -1.250, Z: -5.570 m)
	Max. rotation about Z	-1.5	mrad	FE Mesh Node No. 1958 (X: -3.460, Y: -1.750, Z: -5.570 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	10		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.377E+201470		
	Infinity Norm	2.628E+11		
Load Combination CO11 - LC1 + LC2 + LC3				
	Sum of loads in X	10.73	kN	
	Sum of support reactions in X	10.73	kN	Deviation -0.00%
	Sum of loads in Y	-0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	3523.25	kN	
	Sum of support reactions in Z	3523.25	kN	Deviation 0.00%
	Resultant of reactions about X	-54.9	kNm	At center of gravity of model (X:0.0, Y:-0.5, Z:-1.7 m)
	Resultant of reactions about Y	-11.6	kNm	At center of gravity of model
	Resultant of reactions about Z	0.3	kNm	At center of gravity of model
	Max. displacement in X	0.2	mm	FE Mesh Node No. 2922 (X: -5.190, Y: -1.000, Z: -5.570 m)
	Max. displacement in Y	-5.6	mm	FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
	Max. displacement in Z	4.3	mm	FE Mesh Node No. 1336 (X: 0.000, Y: 0.700, Z: -2.279 m)
	Max. vector displacement	6.9	mm	FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
	Max. rotation about X	-1.4	mrad	FE Mesh Node No. 2553 (X: 0.000, Y: -1.750, Z: -1.772 m)
	Max. rotation about Y	-0.1	mrad	FE Mesh Node No. 2899 (X: -5.190, Y: -1.250, Z: -5.570 m)
	Max. rotation about Z	-1.6	mrad	FE Mesh Node No. 1958 (X: -3.460, Y: -1.750, Z: -5.570 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	10		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Stiffness matrix determinant	1.378E+201470		
	Infinity Norm	2.628E+11		
Load Combination CO12 - LC1 + LC2 + LC4				
	Sum of loads in X	10.73	kN	
	Sum of support reactions in X	10.73	kN	Deviation 0.00%
	Sum of loads in Y	-0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	3523.25	kN	
	Sum of support reactions in Z	3523.25	kN	Deviation 0.00%
	Resultant of reactions about X	-54.8	kNm	At center of gravity of model (X:0.0, Y:-0.5, Z:-1.7 m)
	Resultant of reactions about Y	-11.6	kNm	At center of gravity of model
	Resultant of reactions about Z	0.3	kNm	At center of gravity of model
	Max. displacement in X	0.2	mm	FE Mesh Node No. 2922 (X: -5.190, Y: -1.000, Z: -5.570 m)
	Max. displacement in Y	-5.2	mm	FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
	Max. displacement in Z	4.3	mm	FE Mesh Node No. 1336 (X: 0.000, Y: 0.700, Z: -2.279 m)
	Max. vector displacement	6.6	mm	FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
	Max. rotation about X	-1.3	mrad	FE Mesh Node No. 2553 (X: 0.000, Y: -1.750, Z: -1.772 m)
	Max. rotation about Y	-0.1	mrad	FE Mesh Node No. 2899 (X: -5.190, Y: -1.250, Z: -5.570 m)
	Max. rotation about Z	-1.5	mrad	FE Mesh Node No. 1958 (X: -3.460, Y: -1.750, Z: -5.570 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		$N, V_y, V_z, M_y, M_z, M_T$
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	10		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.376E+201470		
	Infinity Norm	2.628E+11		
Load Combination CO13 - LC1 + LC2 + LC5				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	59.72	kN	
	Sum of support reactions in Y	59.72	kN	Deviation 0.00%
	Sum of loads in Z	3523.25	kN	
	Sum of support reactions in Z	3523.25	kN	Deviation 0.00%
	Resultant of reactions about X	10.8	kNm	At center of gravity of model (X:0.0, Y:-0.5, Z:-1.7 m)
	Resultant of reactions about Y	0.0	kNm	At center of gravity of model
	Resultant of reactions about Z	0.0	kNm	At center of gravity of model
	Max. displacement in X	-0.2	mm	FE Mesh Node No. 3113 (X: 5.190, Y: -1.000, Z: -5.570 m)
	Max. displacement in Y	3.7	mm	FE Mesh Node No. 391 (X: 0.000, Y: 0.700, Z: -5.570 m)
	Max. displacement in Z	4.8	mm	FE Mesh Node No. 1336 (X: 0.000, Y: 0.700, Z: -2.279 m)
	Max. vector displacement	6.0	mm	FE Mesh Node No. 391 (X: 0.000, Y: 0.700, Z: -5.570 m)
	Max. rotation about X	-0.9	mrad	FE Mesh Node No. 2553 (X: 0.000, Y: -1.750, Z: -1.772 m)
	Max. rotation about Y	-0.1	mrad	FE Mesh Node No. 2899 (X: -5.190, Y: -1.250, Z: -5.570 m)
	Max. rotation about Z	-1.5	mrad	FE Mesh Node No. 1958 (X: -3.460, Y: -1.750, Z: -5.570 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		$N, V_y, V_z, M_y, M_z, M_T$
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	10		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.379E+201470		
	Infinity Norm	2.628E+11		
Load Combination CO14 - LC1 + LC2 + LC6				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	59.72	kN	
	Sum of support reactions in Y	59.72	kN	Deviation -0.00%
	Sum of loads in Z	3523.25	kN	
	Sum of support reactions in Z	3523.25	kN	Deviation 0.00%
	Resultant of reactions about X	10.9	kNm	At center of gravity of model (X:0.0, Y:-0.5, Z:-1.7 m)
	Resultant of reactions about Y	0.0	kNm	At center of gravity of model
	Resultant of reactions about Z	0.0	kNm	At center of gravity of model
	Max. displacement in X	-0.2	mm	FE Mesh Node No. 3113 (X: 5.190, Y: -1.000, Z: -5.570 m)
	Max. displacement in Y	3.6	mm	FE Mesh Node No. 391 (X: 0.000, Y: 0.700, Z: -5.570 m)
	Max. displacement in Z	4.7	mm	FE Mesh Node No. 1336 (X: 0.000, Y: 0.700, Z: -2.279 m)
	Max. vector displacement	6.0	mm	FE Mesh Node No. 391 (X: 0.000, Y: 0.700, Z: -5.570 m)
	Max. rotation about X	0.8	mrad	FE Mesh Node No. 448 (X: 1.298, Y: 0.000, Z: -0.760 m)
	Max. rotation about Y	-0.1	mrad	FE Mesh Node No. 2899 (X: -5.190, Y: -1.250, Z: -5.570 m)
	Max. rotation about Z	-1.4	mrad	FE Mesh Node No. 1958 (X: -3.460, Y: -1.750, Z: -5.570 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		$N, V_y, V_z, M_y, M_z, M_T$

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	10		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.377E+201470		
	Infinity Norm	2.628E+11		
Load Combination CO15 - 0.6*LC1 + LC3				
	Sum of loads in X	10.73	kN	
	Sum of support reactions in X	10.73	kN	Deviation -0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	1506.88	kN	
	Sum of support reactions in Z	1506.88	kN	Deviation 0.00%
	Resultant of reactions about X	195.5	kNm	At center of gravity of model (X:0.0, Y:-0.5, Z:-1.7 m)
	Resultant of reactions about Y	-11.6	kNm	At center of gravity of model
	Resultant of reactions about Z	0.2	kNm	At center of gravity of model
	Max. displacement in X	0.0	mm	FE Mesh Node No. 2922 (X: -5.190, Y: -1.000, Z: -5.570 m)
	Max. displacement in Y	3.1	mm	FE Mesh Node No. 1212 (X: 0.519, Y: 0.700, Z: -5.570 m)
	Max. displacement in Z	2.4	mm	FE Mesh Node No. 945 (X: 5.190, Y: 0.700, Z: -5.317 m)
	Max. vector displacement	3.9	mm	FE Mesh Node No. 1210 (X: 0.779, Y: 0.700, Z: -5.570 m)
	Max. rotation about X	0.6	mrad	FE Mesh Node No. 1372 (X: 1.038, Y: 0.700, Z: -1.266 m)
	Max. rotation about Y	-0.0	mrad	FE Mesh Node No. 2899 (X: -5.190, Y: -1.250, Z: -5.570 m)
	Max. rotation about Z	-0.1	mrad	FE Mesh Node No. 1958 (X: -3.460, Y: -1.750, Z: -5.570 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	10		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.377E+201470		
	Infinity Norm	2.628E+11		
Load Combination CO16 - 0.6*LC1 + LC4				
	Sum of loads in X	10.73	kN	
	Sum of support reactions in X	10.73	kN	Deviation 0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	1506.88	kN	
	Sum of support reactions in Z	1506.88	kN	Deviation 0.00%
	Resultant of reactions about X	195.5	kNm	At center of gravity of model (X:0.0, Y:-0.5, Z:-1.7 m)
	Resultant of reactions about Y	-11.6	kNm	At center of gravity of model
	Resultant of reactions about Z	0.2	kNm	At center of gravity of model
	Max. displacement in X	0.0	mm	FE Mesh Node No. 319 (X: 5.190, Y: 0.700, Z: -5.570 m)
	Max. displacement in Y	3.0	mm	FE Mesh Node No. 391 (X: 0.000, Y: 0.700, Z: -5.570 m)
	Max. displacement in Z	2.4	mm	FE Mesh Node No. 319 (X: 5.190, Y: 0.700, Z: -5.570 m)
	Max. vector displacement	3.9	mm	FE Mesh Node No. 961 (X: 3.114, Y: 0.700, Z: -5.570 m)
	Max. rotation about X	0.5	mrad	FE Mesh Node No. 1404 (X: 0.519, Y: 0.700, Z: -0.506 m)
	Max. rotation about Y	-0.0	mrad	FE Mesh Node No. 3469 (X: 4.943, Y: -0.750, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 1962 (X: -3.954, Y: -1.750, Z: -5.570 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	10		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.375E+201470		
	Infinity Norm	2.628E+11		
Load Combination CO17 - 0.6*LC1 + LC5				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	59.72	kN	
	Sum of support reactions in Y	59.72	kN	Deviation 0.00%
	Sum of loads in Z	1506.88	kN	
	Sum of support reactions in Z	1506.88	kN	Deviation 0.00%
	Resultant of reactions about X	260.9	kNm	At center of gravity of model (X:0.0, Y:-0.5, Z:-1.7 m)
	Resultant of reactions about Y	0.0	kNm	At center of gravity of model
	Resultant of reactions about Z	0.0	kNm	At center of gravity of model

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Max. displacement in X	-0.0	mm	FE Mesh Node No. 729 (X: -4.671, Y: 0.000, Z: -5.570 m)
	Max. displacement in Y	5.3	mm	FE Mesh Node No. 1210 (X: 0.779, Y: 0.700, Z: -5.570 m)
	Max. displacement in Z	2.9	mm	FE Mesh Node No. 945 (X: 5.190, Y: 0.700, Z: -5.317 m)
	Max. vector displacement	6.0	mm	FE Mesh Node No. 1210 (X: 0.779, Y: 0.700, Z: -5.570 m)
	Max. rotation about X	0.9	mrad	FE Mesh Node No. 1372 (X: 1.038, Y: 0.700, Z: -1.266 m)
	Max. rotation about Y	-0.0	mrad	FE Mesh Node No. 317 (X: 5.190, Y: 0.700, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 953 (X: 4.152, Y: 0.700, Z: -5.570 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	10		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.379E+201470		
	Infinity Norm	2.628E+11		
Load Combination CO18 - 0.6*LC1 + LC6				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	59.72	kN	
	Sum of support reactions in Y	59.72	kN	Deviation 0.00%
	Sum of loads in Z	1506.88	kN	
	Sum of support reactions in Z	1506.88	kN	Deviation 0.00%
	Resultant of reactions about X	261.0	kNm	At center of gravity of model (X:0.0, Y:-0.5, Z:-1.7 m)
	Resultant of reactions about Y	0.0	kNm	At center of gravity of model
	Resultant of reactions about Z	0.0	kNm	At center of gravity of model
	Max. displacement in X	-0.0	mm	FE Mesh Node No. 2922 (X: -5.190, Y: -1.000, Z: -5.570 m)
	Max. displacement in Y	5.6	mm	FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
	Max. displacement in Z	2.9	mm	FE Mesh Node No. 319 (X: 5.190, Y: 0.700, Z: -5.570 m)
	Max. vector displacement	6.0	mm	FE Mesh Node No. 319 (X: 5.190, Y: 0.700, Z: -5.570 m)
	Max. rotation about X	1.0	mrad	FE Mesh Node No. 2381 (X: 0.000, Y: -1.750, Z: -2.785 m)
	Max. rotation about Y	-0.0	mrad	FE Mesh Node No. 3115 (X: 5.190, Y: -1.250, Z: -5.570 m)
	Max. rotation about Z	-0.1	mrad	FE Mesh Node No. 1902 (X: 3.460, Y: -1.750, Z: -5.570 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	10		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.377E+201470		
	Infinity Norm	2.628E+11		
Load Combination CO19 - SLS - Quasi-permanent				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	3219.72	kN	
	Sum of support reactions in Z	3219.72	kN	Deviation 0.00%
	Resultant of reactions about X	59.7	kNm	At center of gravity of model (X:0.0, Y:-0.5, Z:-1.7 m)
	Resultant of reactions about Y	0.0	kNm	At center of gravity of model
	Resultant of reactions about Z	0.0	kNm	At center of gravity of model
	Max. displacement in X	-0.1	mm	FE Mesh Node No. 3113 (X: 5.190, Y: -1.000, Z: -5.570 m)
	Max. displacement in Y	2.5	mm	FE Mesh Node No. 391 (X: 0.000, Y: 0.700, Z: -5.570 m)
	Max. displacement in Z	4.2	mm	FE Mesh Node No. 1326 (X: 0.000, Y: 0.700, Z: -2.532 m)
	Max. vector displacement	4.9	mm	FE Mesh Node No. 391 (X: 0.000, Y: 0.700, Z: -5.570 m)
	Max. rotation about X	-0.6	mrad	FE Mesh Node No. 2553 (X: 0.000, Y: -1.750, Z: -1.772 m)
	Max. rotation about Y	-0.0	mrad	FE Mesh Node No. 2899 (X: -5.190, Y: -1.250, Z: -5.570 m)
	Max. rotation about Z	-1.0	mrad	FE Mesh Node No. 1958 (X: -3.460, Y: -1.750, Z: -5.570 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	10		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	1.040E+11		
	Minimum value of element of stiffness matrix on diagonal	2.587E+07		
	Stiffness matrix determinant	1.371E+201470		
	Infinity Norm	2.628E+11		
Summary				

4.0 RESULTS - SUMMARY

Description	Value	Unit	Comment
Max. displacement in X	-0.3	mm	CO1, FE Mesh Node No. 3113 (X: 5.190, Y: -1.000, Z: -5.570 m)
Max. displacement in Y	-10.1	mm	LC2, FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
Max. displacement in Z	6.0	mm	CO1, FE Mesh Node No. 1336 (X: 0.000, Y: 0.700, Z: -2.279 m)
Max. vector displacement	10.4	mm	LC2, FE Mesh Node No. 1930 (X: 0.000, Y: -1.750, Z: -5.570 m)
Max. rotation about X	-2.2	mrad	LC2, FE Mesh Node No. 2553 (X: 0.000, Y: -1.750, Z: -1.772 m)
Max. rotation about Y	-0.1	mrad	CO1, FE Mesh Node No. 2899 (X: -5.190, Y: -1.250, Z: -5.570 m)
Max. rotation about Z	-2.1	mrad	CO1, FE Mesh Node No. 1958 (X: -3.460, Y: -1.750, Z: -5.570 m)
Other Settings:			
Number of 1D finite elements	0		
Number of 2D finite elements	3736		
Number of 3D finite elements	0		
Number of FE mesh nodes	3699		
Number of equations	22194		
Internal forces referred to deformed system for...:			
Max. number of iterations	150		
Number of divisions for member results	10		
Division of cable/foundation/tapered members	10		
Number of member divisions for searching maximum values	10		
Subdivisions of FE mesh for graphical results	0		
Percentage of iterations according to Picard method in combination with Newton-Raphson method	5	%	
Activate ineffective supports	☐		
Options:			
Activate shear stiffness of members (Ay, Az)	☒		
Activate member divisions for large deformation or post-critical analysis	☒		
Activate entered stiffness modifications	☒		
Ignore rotational degrees of freedom	☐		
Check of critical forces of members	☒		
Nonsymmetric direct solver if demanded by nonlinear model	☐		
Method for the system of equations	Direct		
Plate bending theory	Mindlin		
Solver version	64-bit		
Precision and Tolerance:			
Change default setting	☐		
Nonlinear effects - Activate:			
Support and elastic foundations	☐		

4.3 LINES - SUPPORT FORCES

Line No.	LC/CO	Node No.	Location x [m]	Support Forces [kN/m] p_X	p_Y	p_Z	Support Moments [kNm/m] m_X	m_Y	m_Z
4	LC1	390	0.000	0.00	2.50	0.00	0.00	0.00	0.00
			0.259	0.00	-0.31	0.00	0.00	0.00	0.00
			0.519	0.00	-0.23	0.00	0.00	0.00	0.00
			0.779	0.00	-0.07	0.00	0.00	0.00	0.00
			1.038	0.00	0.09	0.00	0.00	0.00	0.00
			1.298	0.00	0.18	0.00	0.00	0.00	0.00
			1.557	0.00	0.22	0.00	0.00	0.00	0.00
			1.816	0.00	0.20	0.00	0.00	0.00	0.00
			2.076	0.00	0.18	0.00	0.00	0.00	0.00
			2.336	0.00	0.21	0.00	0.00	0.00	0.00
		393	2.595	0.00	3.26	0.00	0.00	0.00	0.00
	LC2	390	0.000	0.00	427.28	0.00	0.00	0.00	0.00
			0.259	0.00	161.13	0.00	0.00	0.00	0.00
			0.519	0.00	106.42	0.00	0.00	0.00	0.00
			0.779	0.00	64.06	0.00	0.00	0.00	0.00
			1.038	0.00	49.79	0.00	0.00	0.00	0.00
			1.298	0.00	46.59	0.00	0.00	0.00	0.00
			1.557	0.00	50.53	0.00	0.00	0.00	0.00
			1.816	0.00	66.18	0.00	0.00	0.00	0.00
			2.076	0.00	112.16	0.00	0.00	0.00	0.00
			2.336	0.00	171.49	0.00	0.00	0.00	0.00
		393	2.595	0.00	467.26	0.00	0.00	0.00	0.00
	LC3	390	0.000	0.00	17.47	0.00	0.00	0.00	0.00
			0.259	0.00	6.75	0.00	0.00	0.00	0.00
			0.519	0.00	4.58	0.00	0.00	0.00	0.00
			0.779	0.00	2.89	0.00	0.00	0.00	0.00
			1.038	0.00	2.32	0.00	0.00	0.00	0.00
			1.298	0.00	2.20	0.00	0.00	0.00	0.00
			1.557	0.00	2.40	0.00	0.00	0.00	0.00
			1.816	0.00	3.08	0.00	0.00	0.00	0.00
			2.076	0.00	5.00	0.00	0.00	0.00	0.00
			2.336	0.00	7.45	0.00	0.00	0.00	0.00
		393	2.595	0.00	20.01	0.00	0.00	0.00	0.00
	LC4	390	0.000	0.00	0.71	0.00	0.00	0.00	0.00
			0.259	0.00	0.24	0.00	0.00	0.00	0.00
			0.519	0.00	0.14	0.00	0.00	0.00	0.00

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
Load Case LC1 - SW				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	4673.86	kN	
	Sum of support reactions in Z	4673.86	kN	Deviation 0.00%
	Resultant of reactions about X	-0.002	kNm	At center of gravity of model (X:7.603, Y:4.150, Z:-1.142 m)
	Resultant of reactions about Y	0.000	kNm	At center of gravity of model
	Resultant of reactions about Z	0.000	kNm	At center of gravity of model
	Max. displacement in X	-0.1	mm	FE Mesh Node No. 3096 (X: 0.000, Y: 4.024, Z: -4.450 m)
	Max. displacement in Y	-0.0	mm	FE Mesh Node No. 4550 (X: 10.893, Y: 0.000, Z: -4.450 m)
	Max. displacement in Z	1.2	mm	FE Mesh Node No. 3452 (X: 15.200, Y: 4.276, Z: -4.203 m)
	Max. vector displacement	1.2	mm	FE Mesh Node No. 3453 (X: 15.200, Y: 4.276, Z: -4.450 m)
	Max. rotation about X	-0.0	mrad	FE Mesh Node No. 303 (X: 1.850, Y: 0.252, Z: 0.000 m)
	Max. rotation about Y	0.1	mrad	FE Mesh Node No. 34 (X: 0.000, Y: 4.024, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 3301 (X: 0.000, Y: 6.791, Z: -2.719 m)
	Method of analysis	Linear		Geometrically linear analysis
	Reduction of stiffness			Cross-sections, Members, Surfaces
	Number of load increments	1		
	Number of iterations	1		
	Maximum value of element of stiffness matrix on diagonal	8.2E+10		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.660E+397516		
	Infinity Norm	2.015E+11		
Load Case LC2 - FLUID LOAD				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	10597.40	kN	
	Sum of support reactions in Z	10597.40	kN	Deviation 0.00%
	Resultant of reactions about X	0.003	kNm	At center of gravity of model (X:7.603, Y:4.150, Z:-1.142 m)
	Resultant of reactions about Y	28.218	kNm	At center of gravity of model
	Resultant of reactions about Z	0.000	kNm	At center of gravity of model
	Max. displacement in X	-0.9	mm	FE Mesh Node No. 3096 (X: 0.000, Y: 4.024, Z: -4.450 m)
	Max. displacement in Y	-0.1	mm	FE Mesh Node No. 4667 (X: 5.616, Y: 0.000, Z: -1.978 m)
	Max. displacement in Z	2.5	mm	FE Mesh Node No. 34 (X: 0.000, Y: 4.024, Z: 0.000 m)
	Max. vector displacement	2.7	mm	FE Mesh Node No. 3096 (X: 0.000, Y: 4.024, Z: -4.450 m)
	Max. rotation about X	-0.1	mrad	FE Mesh Node No. 4583 (X: 5.641, Y: 0.000, Z: -0.494 m)
	Max. rotation about Y	0.3	mrad	FE Mesh Node No. 3109 (X: 0.000, Y: 4.024, Z: -1.236 m)
	Max. rotation about Z	-0.3	mrad	FE Mesh Node No. 3294 (X: 0.000, Y: 6.791, Z: -4.450 m)
	Method of analysis	Linear		Geometrically linear analysis
	Reduction of stiffness			Cross-sections, Members, Surfaces
	Number of load increments	1		
	Number of iterations	1		
	Maximum value of element of stiffness matrix on diagonal	8.2E+10		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.660E+397516		
	Infinity Norm	2.015E+11		
Load Combination CO1 - 1.4*LC1 + 1.4*LC2				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	21379.80	kN	
	Sum of support reactions in Z	21379.80	kN	Deviation 0.00%
	Resultant of reactions about X	0.0	kNm	At center of gravity of model (X:7.6, Y:4.2, Z:-1.1 m)
	Resultant of reactions about Y	39.6	kNm	At center of gravity of model
	Resultant of reactions about Z	0.0	kNm	At center of gravity of model
	Max. displacement in X	-1.4	mm	FE Mesh Node No. 3096 (X: 0.000, Y: 4.024, Z: -4.450 m)
	Max. displacement in Y	-0.1	mm	FE Mesh Node No. 4667 (X: 5.616, Y: 0.000, Z: -1.978 m)
	Max. displacement in Z	5.2	mm	FE Mesh Node No. 3453 (X: 15.200, Y: 4.276, Z: -4.450 m)
	Max. vector displacement	5.4	mm	FE Mesh Node No. 3096 (X: 0.000, Y: 4.024, Z: -4.450 m)
	Max. rotation about X	-0.1	mrad	FE Mesh Node No. 4583 (X: 5.641, Y: 0.000, Z: -0.494 m)
	Max. rotation about Y	0.5	mrad	FE Mesh Node No. 3110 (X: 0.000, Y: 4.024, Z: -0.989 m)
	Max. rotation about Z	-0.4	mrad	FE Mesh Node No. 3294 (X: 0.000, Y: 6.791, Z: -4.450 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	8.2E+10		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.765E+397516		
	Infinity Norm	2.015E+11		
Load Combination CO4 - 1.2*LC1 + 0.8*LC3				
	Sum of loads in X	26.48	kN	
	Sum of support reactions in X	26.48	kN	Deviation -0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Sum of loads in Z	5608.64	kN	
	Sum of support reactions in Z	5608.64	kN	Deviation 0.00%
	Resultant of reactions about X	0.0	kNm	At center of gravity of model (X:7.6, Y:4.2, Z:-1.1 m)
	Resultant of reactions about Y	-28.7	kNm	At center of gravity of model
	Resultant of reactions about Z	0.0	kNm	At center of gravity of model
	Max. displacement in X	0.2	mm	FE Mesh Node No. 3453 (X: 15.200, Y: 4.276, Z: -4.450 m)
	Max. displacement in Y	-0.0	mm	FE Mesh Node No. 4546 (X: 9.880, Y: 0.000, Z: -4.450 m)
	Max. displacement in Z	1.4	mm	FE Mesh Node No. 3453 (X: 15.200, Y: 4.276, Z: -4.450 m)
	Max. vector displacement	1.4	mm	FE Mesh Node No. 3453 (X: 15.200, Y: 4.276, Z: -4.450 m)
	Max. rotation about X	-0.0	mrad	FE Mesh Node No. 2565 (X: 13.250, Y: 0.252, Z: 0.000 m)
	Max. rotation about Y	-0.1	mrad	FE Mesh Node No. 2688 (X: 15.200, Y: 4.276, Z: 0.000 m)
	Max. rotation about Z	-0.1	mrad	FE Mesh Node No. 3475 (X: 15.200, Y: 1.509, Z: -4.450 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	8.2E+10		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.656E+397516		
	Infinity Norm	2.015E+11		
Load Combination CO5 - 1.2*LC1 + 0.8*LC4				
	Sum of loads in X	26.48	kN	
	Sum of support reactions in X	26.48	kN	Deviation 0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	5608.64	kN	
	Sum of support reactions in Z	5608.64	kN	Deviation 0.00%
	Resultant of reactions about X	0.0	kNm	At center of gravity of model (X:7.6, Y:4.2, Z:-1.1 m)
	Resultant of reactions about Y	-28.7	kNm	At center of gravity of model
	Resultant of reactions about Z	0.0	kNm	At center of gravity of model
	Max. displacement in X	0.1	mm	FE Mesh Node No. 3096 (X: 0.000, Y: 4.024, Z: -4.450 m)
	Max. displacement in Y	-0.0	mm	FE Mesh Node No. 4550 (X: 10.893, Y: 0.000, Z: -4.450 m)
	Max. displacement in Z	1.4	mm	FE Mesh Node No. 3452 (X: 15.200, Y: 4.276, Z: -4.203 m)
	Max. vector displacement	1.4	mm	FE Mesh Node No. 3452 (X: 15.200, Y: 4.276, Z: -4.203 m)
	Max. rotation about X	-0.0	mrad	FE Mesh Node No. 2565 (X: 13.250, Y: 0.252, Z: 0.000 m)
	Max. rotation about Y	-0.1	mrad	FE Mesh Node No. 2687 (X: 14.964, Y: 4.276, Z: 0.000 m)
	Max. rotation about Z	-0.0	mrad	FE Mesh Node No. 2862 (X: 0.000, Y: 0.755, Z: -4.450 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	8.2E+10		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.648E+397516		
	Infinity Norm	2.015E+11		
Load Combination CO6 - 1.2*LC1 + 0.8*LC5				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	55.89	kN	
	Sum of support reactions in Y	55.89	kN	Deviation -0.00%
	Sum of loads in Z	5608.64	kN	
	Sum of support reactions in Z	5608.64	kN	Deviation 0.00%
	Resultant of reactions about X	60.5	kNm	At center of gravity of model (X:7.6, Y:4.2, Z:-1.1 m)
	Resultant of reactions about Y	0.0	kNm	At center of gravity of model
	Resultant of reactions about Z	-0.1	kNm	At center of gravity of model
	Max. displacement in X	-0.2	mm	FE Mesh Node No. 3096 (X: 0.000, Y: 4.024, Z: -4.450 m)
	Max. displacement in Y	0.1	mm	FE Mesh Node No. 5847 (X: 5.320, Y: 8.300, Z: -4.450 m)
	Max. displacement in Z	1.4	mm	FE Mesh Node No. 39 (X: 15.200, Y: 8.300, Z: 0.000 m)
	Max. vector displacement	1.4	mm	FE Mesh Node No. 3449 (X: 15.200, Y: 4.779, Z: -4.450 m)
	Max. rotation about X	0.1	mrad	FE Mesh Node No. 239 (X: 1.850, Y: 8.048, Z: 0.000 m)
	Max. rotation about Y	0.1	mrad	FE Mesh Node No. 36 (X: 0.000, Y: 4.276, Z: 0.000 m)
	Max. rotation about Z	0.1	mrad	FE Mesh Node No. 2898 (X: 0.000, Y: 1.258, Z: -4.450 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	8.2E+10		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.656E+397516		
	Infinity Norm	2.015E+11		
Load Combination CO7 - 1.2*LC1 + 0.8*LC6				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Sum of loads in Y	55.89	kN	
	Sum of support reactions in Y	55.89	kN	Deviation 0.00%
	Sum of loads in Z	5608.64	kN	
	Sum of support reactions in Z	5608.64	kN	Deviation 0.00%
	Resultant of reactions about X	60.5	kNm	At center of gravity of model (X:7.6, Y:4.2, Z:-1.1 m)
	Resultant of reactions about Y	0.0	kNm	At center of gravity of model
	Resultant of reactions about Z	-0.1	kNm	At center of gravity of model
	Max. displacement in X	-0.1	mm	FE Mesh Node No. 3096 (X: 0.000, Y: 4.024, Z: -4.450 m)
	Max. displacement in Y	0.0	mm	FE Mesh Node No. 5591 (X: 10.893, Y: 8.300, Z: -4.450 m)
	Max. displacement in Z	1.4	mm	FE Mesh Node No. 39 (X: 15.200, Y: 8.300, Z: 0.000 m)
	Max. vector displacement	1.4	mm	FE Mesh Node No. 39 (X: 15.200, Y: 8.300, Z: 0.000 m)
	Max. rotation about X	0.1	mrad	FE Mesh Node No. 239 (X: 1.850, Y: 8.048, Z: 0.000 m)
	Max. rotation about Y	0.1	mrad	FE Mesh Node No. 35 (X: 0.258, Y: 4.276, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 2898 (X: 0.000, Y: 1.258, Z: -4.450 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	8.2E+10		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.648E+397516		
	Infinity Norm	2.015E+11		
Load Combination CO8 - 0.9*LC1 + 1.6*LC3				
	Sum of loads in X	52.96	kN	
	Sum of support reactions in X	52.96	kN	Deviation -0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	4206.48	kN	
	Sum of support reactions in Z	4206.48	kN	Deviation 0.00%
	Resultant of reactions about X	0.0	kNm	At center of gravity of model (X:7.6, Y:4.2, Z:-1.1 m)
	Resultant of reactions about Y	-57.4	kNm	At center of gravity of model
	Resultant of reactions about Z	0.0	kNm	At center of gravity of model
	Max. displacement in X	0.3	mm	FE Mesh Node No. 3453 (X: 15.200, Y: 4.276, Z: -4.450 m)
	Max. displacement in Y	-0.0	mm	FE Mesh Node No. 4545 (X: 9.627, Y: 0.000, Z: -4.450 m)
	Max. displacement in Z	1.1	mm	FE Mesh Node No. 3453 (X: 15.200, Y: 4.276, Z: -4.450 m)
	Max. vector displacement	1.1	mm	FE Mesh Node No. 3453 (X: 15.200, Y: 4.276, Z: -4.450 m)
	Max. rotation about X	-0.0	mrad	FE Mesh Node No. 2565 (X: 13.250, Y: 0.252, Z: 0.000 m)
	Max. rotation about Y	-0.1	mrad	FE Mesh Node No. 3912 (X: 15.200, Y: 4.276, Z: -0.989 m)
	Max. rotation about Z	-0.1	mrad	FE Mesh Node No. 3475 (X: 15.200, Y: 1.509, Z: -4.450 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	8.2E+10		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.665E+397516		
	Infinity Norm	2.015E+11		
Load Combination CO9 - 0.9*LC1 + 1.6*LC4				
	Sum of loads in X	52.96	kN	
	Sum of support reactions in X	52.96	kN	Deviation 0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	4206.48	kN	
	Sum of support reactions in Z	4206.48	kN	Deviation 0.00%
	Resultant of reactions about X	0.0	kNm	At center of gravity of model (X:7.6, Y:4.2, Z:-1.1 m)
	Resultant of reactions about Y	-57.4	kNm	At center of gravity of model
	Resultant of reactions about Z	0.0	kNm	At center of gravity of model
	Max. displacement in X	0.3	mm	FE Mesh Node No. 3096 (X: 0.000, Y: 4.024, Z: -4.450 m)
	Max. displacement in Y	-0.0	mm	FE Mesh Node No. 5064 (X: 1.520, Y: 0.000, Z: -4.450 m)
	Max. displacement in Z	1.1	mm	FE Mesh Node No. 3452 (X: 15.200, Y: 4.276, Z: -4.203 m)
	Max. vector displacement	1.1	mm	FE Mesh Node No. 3096 (X: 0.000, Y: 4.024, Z: -4.450 m)
	Max. rotation about X	-0.0	mrad	FE Mesh Node No. 2565 (X: 13.250, Y: 0.252, Z: 0.000 m)
	Max. rotation about Y	-0.1	mrad	FE Mesh Node No. 3104 (X: 0.000, Y: 4.024, Z: -2.472 m)
	Max. rotation about Z	-0.1	mrad	FE Mesh Node No. 2880 (X: 0.000, Y: 1.006, Z: -4.450 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	8.2E+10		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.648E+397516		
	Infinity Norm	2.015E+11		
Load Combination CO10 - 0.9*LC1 + 1.6*LC5				

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	111.78	kN	
	Sum of support reactions in Y	111.78	kN	Deviation 0.00%
	Sum of loads in Z	4206.48	kN	
	Sum of support reactions in Z	4206.48	kN	Deviation 0.00%
	Resultant of reactions about X	121.1	kNm	At center of gravity of model (X:7.6, Y:4.2, Z:-1.1 m)
	Resultant of reactions about Y	0.0	kNm	At center of gravity of model
	Resultant of reactions about Z	-0.3	kNm	At center of gravity of model
	Max. displacement in X	-0.4	mm	FE Mesh Node No. 3096 (X: 0.000, Y: 4.024, Z: -4.450 m)
	Max. displacement in Y	0.1	mm	FE Mesh Node No. 5063 (X: 1.267, Y: 0.000, Z: -4.450 m)
	Max. displacement in Z	1.1	mm	FE Mesh Node No. 39 (X: 15.200, Y: 8.300, Z: 0.000 m)
	Max. vector displacement	1.1	mm	FE Mesh Node No. 3451 (X: 15.200, Y: 4.527, Z: -4.450 m)
	Max. rotation about X	0.0	mrad	FE Mesh Node No. 239 (X: 1.850, Y: 8.048, Z: 0.000 m)
	Max. rotation about Y	0.1	mrad	FE Mesh Node No. 3109 (X: 0.000, Y: 4.024, Z: -1.236 m)
	Max. rotation about Z	0.1	mrad	FE Mesh Node No. 2898 (X: 0.000, Y: 1.258, Z: -4.450 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	8.2E+10		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.664E+397516		
	Infinity Norm	2.015E+11		
Load Combination CO11 - 0.9*LC1 + 1.6*LC6				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	111.78	kN	
	Sum of support reactions in Y	111.78	kN	Deviation 0.00%
	Sum of loads in Z	4206.48	kN	
	Sum of support reactions in Z	4206.48	kN	Deviation 0.00%
	Resultant of reactions about X	121.1	kNm	At center of gravity of model (X:7.6, Y:4.2, Z:-1.1 m)
	Resultant of reactions about Y	0.0	kNm	At center of gravity of model
	Resultant of reactions about Z	-0.3	kNm	At center of gravity of model
	Max. displacement in X	-0.1	mm	FE Mesh Node No. 3096 (X: 0.000, Y: 4.024, Z: -4.450 m)
	Max. displacement in Y	0.1	mm	FE Mesh Node No. 5064 (X: 1.520, Y: 0.000, Z: -4.450 m)
	Max. displacement in Z	1.1	mm	FE Mesh Node No. 39 (X: 15.200, Y: 8.300, Z: 0.000 m)
	Max. vector displacement	1.1	mm	FE Mesh Node No. 47 (X: 15.200, Y: 8.300, Z: -4.450 m)
	Max. rotation about X	0.0	mrad	FE Mesh Node No. 239 (X: 1.850, Y: 8.048, Z: 0.000 m)
	Max. rotation about Y	0.1	mrad	FE Mesh Node No. 36 (X: 0.000, Y: 4.276, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 2809 (X: 0.000, Y: 0.252, Z: -4.450 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	8.2E+10		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.648E+397516		
	Infinity Norm	2.015E+11		
Load Combination CO12 - LC1 + LC2				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	15271.30	kN	
	Sum of support reactions in Z	15271.30	kN	Deviation 0.00%
	Resultant of reactions about X	0.0	kNm	At center of gravity of model (X:7.6, Y:4.2, Z:-1.1 m)
	Resultant of reactions about Y	28.2	kNm	At center of gravity of model
	Resultant of reactions about Z	0.0	kNm	At center of gravity of model
	Max. displacement in X	-1.0	mm	FE Mesh Node No. 3096 (X: 0.000, Y: 4.024, Z: -4.450 m)
	Max. displacement in Y	-0.1	mm	FE Mesh Node No. 4667 (X: 5.616, Y: 0.000, Z: -1.978 m)
	Max. displacement in Z	3.7	mm	FE Mesh Node No. 3453 (X: 15.200, Y: 4.276, Z: -4.450 m)
	Max. vector displacement	3.8	mm	FE Mesh Node No. 3096 (X: 0.000, Y: 4.024, Z: -4.450 m)
	Max. rotation about X	-0.1	mrad	FE Mesh Node No. 4583 (X: 5.641, Y: 0.000, Z: -0.494 m)
	Max. rotation about Y	0.3	mrad	FE Mesh Node No. 3110 (X: 0.000, Y: 4.024, Z: -0.989 m)
	Max. rotation about Z	-0.3	mrad	FE Mesh Node No. 3294 (X: 0.000, Y: 6.791, Z: -4.450 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	8.2E+10		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.734E+397516		
	Infinity Norm	2.015E+11		

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
Load Combination CO15 - LC1 + LC2 + LC3				
	Sum of loads in X	33.10	kN	
	Sum of support reactions in X	33.10	kN	Deviation -0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	15271.30	kN	
	Sum of support reactions in Z	15271.30	kN	Deviation 0.00%
	Resultant of reactions about X	0.0	kNm	At center of gravity of model (X:7.6, Y:4.2, Z:-1.1 m)
	Resultant of reactions about Y	-7.6	kNm	At center of gravity of model
	Resultant of reactions about Z	0.0	kNm	At center of gravity of model
	Max. displacement in X	1.1	mm	FE Mesh Node No. 3453 (X: 15.200, Y: 4.276, Z: -4.450 m)
	Max. displacement in Y	-0.1	mm	FE Mesh Node No. 4404 (X: 9.416, Y: 0.000, Z: -1.978 m)
	Max. displacement in Z	3.7	mm	FE Mesh Node No. 3453 (X: 15.200, Y: 4.276, Z: -4.450 m)
	Max. vector displacement	3.9	mm	FE Mesh Node No. 3453 (X: 15.200, Y: 4.276, Z: -4.450 m)
	Max. rotation about X	-0.1	mrad	FE Mesh Node No. 4320 (X: 9.441, Y: 0.000, Z: -0.494 m)
	Max. rotation about Y	-0.4	mrad	FE Mesh Node No. 3912 (X: 15.200, Y: 4.276, Z: -0.989 m)
	Max. rotation about Z	-0.4	mrad	FE Mesh Node No. 3475 (X: 15.200, Y: 1.509, Z: -4.450 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	8.2E+10		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.742E+397516		
	Infinity Norm	2.015E+11		
Load Combination CO16 - LC1 + LC2 + LC4				
	Sum of loads in X	33.10	kN	
	Sum of support reactions in X	33.10	kN	Deviation 0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	15271.30	kN	
	Sum of support reactions in Z	15271.30	kN	Deviation 0.00%
	Resultant of reactions about X	0.0	kNm	At center of gravity of model (X:7.6, Y:4.2, Z:-1.1 m)
	Resultant of reactions about Y	-7.6	kNm	At center of gravity of model
	Resultant of reactions about Z	0.0	kNm	At center of gravity of model
	Max. displacement in X	0.9	mm	FE Mesh Node No. 3453 (X: 15.200, Y: 4.276, Z: -4.450 m)
	Max. displacement in Y	-0.1	mm	FE Mesh Node No. 4404 (X: 9.416, Y: 0.000, Z: -1.978 m)
	Max. displacement in Z	3.7	mm	FE Mesh Node No. 3453 (X: 15.200, Y: 4.276, Z: -4.450 m)
	Max. vector displacement	3.8	mm	FE Mesh Node No. 3453 (X: 15.200, Y: 4.276, Z: -4.450 m)
	Max. rotation about X	-0.1	mrad	FE Mesh Node No. 4320 (X: 9.441, Y: 0.000, Z: -0.494 m)
	Max. rotation about Y	-0.3	mrad	FE Mesh Node No. 3912 (X: 15.200, Y: 4.276, Z: -0.989 m)
	Max. rotation about Z	-0.3	mrad	FE Mesh Node No. 3475 (X: 15.200, Y: 1.509, Z: -4.450 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	8.2E+10		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.731E+397516		
	Infinity Norm	2.015E+11		
Load Combination CO17 - LC1 + LC2 + LC5				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	69.87	kN	
	Sum of support reactions in Y	69.87	kN	Deviation -0.00%
	Sum of loads in Z	15271.30	kN	
	Sum of support reactions in Z	15271.30	kN	Deviation 0.00%
	Resultant of reactions about X	75.7	kNm	At center of gravity of model (X:7.6, Y:4.2, Z:-1.1 m)
	Resultant of reactions about Y	28.2	kNm	At center of gravity of model
	Resultant of reactions about Z	-0.2	kNm	At center of gravity of model
	Max. displacement in X	-1.2	mm	FE Mesh Node No. 3096 (X: 0.000, Y: 4.024, Z: -4.450 m)
	Max. displacement in Y	0.1	mm	FE Mesh Node No. 5062 (X: 1.013, Y: 0.000, Z: -4.450 m)
	Max. displacement in Z	3.7	mm	FE Mesh Node No. 39 (X: 15.200, Y: 8.300, Z: 0.000 m)
	Max. vector displacement	3.9	mm	FE Mesh Node No. 3114 (X: 0.000, Y: 4.276, Z: -4.450 m)
	Max. rotation about X	0.1	mrad	FE Mesh Node No. 5624 (X: 5.641, Y: 8.300, Z: -0.494 m)
	Max. rotation about Y	0.4	mrad	FE Mesh Node No. 3110 (X: 0.000, Y: 4.024, Z: -0.989 m)
	Max. rotation about Z	0.4	mrad	FE Mesh Node No. 2916 (X: 0.000, Y: 1.509, Z: -4.450 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	8.2E+10		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.742E+39751		

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
		16		
	Infinity Norm	2.015E+11		
Load Combination CO18 - LC1 + LC2 + LC6				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	69.87	kN	
	Sum of support reactions in Y	69.87	kN	Deviation 0.00%
	Sum of loads in Z	15271.30	kN	
	Sum of support reactions in Z	15271.30	kN	Deviation 0.00%
	Resultant of reactions about X	75.7	kNm	At center of gravity of model (X:7.6, Y:4.2, Z:-1.1 m)
	Resultant of reactions about Y	28.2	kNm	At center of gravity of model
	Resultant of reactions about Z	-0.2	kNm	At center of gravity of model
	Max. displacement in X	-1.0	mm	FE Mesh Node No. 3096 (X: 0.000, Y: 4.024, Z: -4.450 m)
	Max. displacement in Y	0.1	mm	FE Mesh Node No. 5063 (X: 1.267, Y: 0.000, Z: -4.450 m)
	Max. displacement in Z	3.7	mm	FE Mesh Node No. 39 (X: 15.200, Y: 8.300, Z: 0.000 m)
	Max. vector displacement	3.8	mm	FE Mesh Node No. 3114 (X: 0.000, Y: 4.276, Z: -4.450 m)
	Max. rotation about X	0.1	mrad	FE Mesh Node No. 5624 (X: 5.641, Y: 8.300, Z: -0.494 m)
	Max. rotation about Y	0.3	mrad	FE Mesh Node No. 3110 (X: 0.000, Y: 4.024, Z: -0.989 m)
	Max. rotation about Z	0.3	mrad	FE Mesh Node No. 2916 (X: 0.000, Y: 1.509, Z: -4.450 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	8.2E+10		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.731E+3975 16		
	Infinity Norm	2.015E+11		
Load Combination CO19 - 0.6*LC1 + LC3				
	Sum of loads in X	33.10	kN	
	Sum of support reactions in X	33.10	kN	Deviation 0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	2804.32	kN	
	Sum of support reactions in Z	2804.32	kN	Deviation 0.00%
	Resultant of reactions about X	0.0	kNm	At center of gravity of model (X:7.6, Y:4.2, Z:-1.1 m)
	Resultant of reactions about Y	-35.8	kNm	At center of gravity of model
	Resultant of reactions about Z	0.0	kNm	At center of gravity of model
	Max. displacement in X	0.2	mm	FE Mesh Node No. 3453 (X: 15.200, Y: 4.276, Z: -4.450 m)
	Max. displacement in Y	-0.0	mm	FE Mesh Node No. 4545 (X: 9.627, Y: 0.000, Z: -4.450 m)
	Max. displacement in Z	0.7	mm	FE Mesh Node No. 3453 (X: 15.200, Y: 4.276, Z: -4.450 m)
	Max. vector displacement	0.7	mm	FE Mesh Node No. 3453 (X: 15.200, Y: 4.276, Z: -4.450 m)
	Max. rotation about X	-0.0	mrad	FE Mesh Node No. 2565 (X: 13.250, Y: 0.252, Z: 0.000 m)
	Max. rotation about Y	-0.0	mrad	FE Mesh Node No. 3946 (X: 15.200, Y: 4.276, Z: -0.742 m)
	Max. rotation about Z	-0.1	mrad	FE Mesh Node No. 3475 (X: 15.200, Y: 1.509, Z: -4.450 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	8.2E+10		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.663E+3975 16		
	Infinity Norm	2.015E+11		
Load Combination CO20 - 0.6*LC1 + LC4				
	Sum of loads in X	33.10	kN	
	Sum of support reactions in X	33.10	kN	Deviation 0.00%
	Sum of loads in Y	0.00	kN	
	Sum of support reactions in Y	0.00	kN	
	Sum of loads in Z	2804.32	kN	
	Sum of support reactions in Z	2804.32	kN	Deviation 0.00%
	Resultant of reactions about X	0.0	kNm	At center of gravity of model (X:7.6, Y:4.2, Z:-1.1 m)
	Resultant of reactions about Y	-35.8	kNm	At center of gravity of model
	Resultant of reactions about Z	0.0	kNm	At center of gravity of model
	Max. displacement in X	0.2	mm	FE Mesh Node No. 3096 (X: 0.000, Y: 4.024, Z: -4.450 m)
	Max. displacement in Y	-0.0	mm	FE Mesh Node No. 5064 (X: 1.520, Y: 0.000, Z: -4.450 m)
	Max. displacement in Z	0.7	mm	FE Mesh Node No. 3452 (X: 15.200, Y: 4.276, Z: -4.203 m)
	Max. vector displacement	0.7	mm	FE Mesh Node No. 3096 (X: 0.000, Y: 4.024, Z: -4.450 m)
	Max. rotation about X	-0.0	mrad	FE Mesh Node No. 2565 (X: 13.250, Y: 0.252, Z: 0.000 m)
	Max. rotation about Y	-0.0	mrad	FE Mesh Node No. 3104 (X: 0.000, Y: 4.024, Z: -2.472 m)
	Max. rotation about Z	-0.1	mrad	FE Mesh Node No. 2880 (X: 0.000, Y: 1.006, Z: -4.450 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	8.2E+10		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		

4.0 RESULTS - SUMMARY

	Description	Value	Unit	Comment
	Stiffness matrix determinant	1.652E+397516		
	Infinity Norm	2.015E+11		
Load Combination CO21 - 0.6*LC1 + LC5				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	69.87	kN	
	Sum of support reactions in Y	69.87	kN	Deviation -0.00%
	Sum of loads in Z	2804.32	kN	
	Sum of support reactions in Z	2804.32	kN	Deviation 0.00%
	Resultant of reactions about X	75.6	kNm	At center of gravity of model (X:7.6, Y:4.2, Z:-1.1 m)
	Resultant of reactions about Y	0.0	kNm	At center of gravity of model
	Resultant of reactions about Z	-0.2	kNm	At center of gravity of model
	Max. displacement in X	-0.2	mm	FE Mesh Node No. 3096 (X: 0.000, Y: 4.024, Z: -4.450 m)
	Max. displacement in Y	0.1	mm	FE Mesh Node No. 5063 (X: 1.267, Y: 0.000, Z: -4.450 m)
	Max. displacement in Z	0.7	mm	FE Mesh Node No. 39 (X: 15.200, Y: 8.300, Z: 0.000 m)
	Max. vector displacement	0.7	mm	FE Mesh Node No. 3451 (X: 15.200, Y: 4.527, Z: -4.450 m)
	Max. rotation about X	0.0	mrad	FE Mesh Node No. 239 (X: 1.850, Y: 8.048, Z: 0.000 m)
	Max. rotation about Y	0.1	mrad	FE Mesh Node No. 3109 (X: 0.000, Y: 4.024, Z: -1.236 m)
	Max. rotation about Z	0.1	mrad	FE Mesh Node No. 2898 (X: 0.000, Y: 1.258, Z: -4.450 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	8.2E+10		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.662E+397516		
	Infinity Norm	2.015E+11		
Load Combination CO22 - 0.6*LC1 + LC6				
	Sum of loads in X	0.00	kN	
	Sum of support reactions in X	0.00	kN	
	Sum of loads in Y	69.87	kN	
	Sum of support reactions in Y	69.87	kN	Deviation -0.00%
	Sum of loads in Z	2804.32	kN	
	Sum of support reactions in Z	2804.32	kN	Deviation 0.00%
	Resultant of reactions about X	75.7	kNm	At center of gravity of model (X:7.6, Y:4.2, Z:-1.1 m)
	Resultant of reactions about Y	0.0	kNm	At center of gravity of model
	Resultant of reactions about Z	-0.2	kNm	At center of gravity of model
	Max. displacement in X	-0.1	mm	FE Mesh Node No. 3096 (X: 0.000, Y: 4.024, Z: -4.450 m)
	Max. displacement in Y	0.0	mm	FE Mesh Node No. 5064 (X: 1.520, Y: 0.000, Z: -4.450 m)
	Max. displacement in Z	0.7	mm	FE Mesh Node No. 39 (X: 15.200, Y: 8.300, Z: 0.000 m)
	Max. vector displacement	0.7	mm	FE Mesh Node No. 47 (X: 15.200, Y: 8.300, Z: -4.450 m)
	Max. rotation about X	0.0	mrad	FE Mesh Node No. 239 (X: 1.850, Y: 8.048, Z: 0.000 m)
	Max. rotation about Y	0.0	mrad	FE Mesh Node No. 36 (X: 0.000, Y: 4.276, Z: 0.000 m)
	Max. rotation about Z	0.0	mrad	FE Mesh Node No. 2844 (X: 0.000, Y: 0.503, Z: -4.450 m)
	Method of analysis	2nd Order		Second order analysis (Nonlinear, Timoshenko)
	Internal forces referred to deformed system for...	☒		N, V_y, V_z, M_y, M_z, M_T
	Reduction of stiffness			Materials, Cross-sections, Members, Surfaces
	Consider favorable effects of tensile forces	☒		
	Divide results by CO factor	☐		
	Number of load increments	1		
	Number of iterations	2		
	Maximum value of element of stiffness matrix on diagonal	8.2E+10		
	Minimum value of element of stiffness matrix on diagonal	1.0E+04		
	Stiffness matrix determinant	1.652E+397516		
	Infinity Norm	2.015E+11		
Summary				
	Max. displacement in X	-1.4	mm	CO1, FE Mesh Node No. 3096 (X: 0.000, Y: 4.024, Z: -4.450 m)
	Max. displacement in Y	-0.1	mm	CO1, FE Mesh Node No. 4667 (X: 5.616, Y: 0.000, Z: -1.978 m)
	Max. displacement in Z	5.2	mm	CO1, FE Mesh Node No. 3453 (X: 15.200, Y: 4.276, Z: -4.450 m)
	Max. vector displacement	5.4	mm	CO1, FE Mesh Node No. 3096 (X: 0.000, Y: 4.024, Z: -4.450 m)
	Max. rotation about X	-0.1	mrad	CO1, FE Mesh Node No. 4583 (X: 5.641, Y: 0.000, Z: -0.494 m)
	Max. rotation about Y	0.5	mrad	CO1, FE Mesh Node No. 3110 (X: 0.000, Y: 4.024, Z: -0.989 m)
	Max. rotation about Z	-0.4	mrad	CO1, FE Mesh Node No. 3294 (X: 0.000, Y: 6.791, Z: -4.450 m)
	Other Settings:			
	Number of 1D finite elements	0		
	Number of 2D finite elements	7132		
	Number of 3D finite elements	0		
	Number of FE mesh nodes	7169		
	Number of equations	43014		
	Internal forces referred to deformed system for...:			
	Max. number of iterations	100		
	Number of divisions for member results	10		
	Division of cable/foundation/tapered members	10		
	Number of member divisions for searching maximum values	10		
	Subdivisions of FE mesh for graphical results	0		
	Percentage of iterations according to Picard method in combination with Newton-Raphson method	5	%	
	Activate ineffective supports	☐		

4.0 RESULTS - SUMMARY

Options:			
Activate shear stiffness of members (Ay, Az)	☒		
Activate member divisions for large deformation or post-critical analysis	☒		
Activate entered stiffness modifications	☒		
Ignore rotational degrees of freedom	☐		
Check of critical forces of members	☒		
Nonsymmetric direct solver if demanded by nonlinear model	☐		
Method for the system of equations	Direct		
Plate bending theory	Mindlin		
Solver version	64-bit		
Precision and Tolerance:			
Change default setting	☐		
Nonlinear effects - Activate:			
Support and elastic foundations	☐		

4.3 LINES - SUPPORT FORCES

Line No.	LC/CO	Node No.	Location x [m]	Support Forces [kN/m] p_X	p_Y	p_Z	Support Moments [kNm/m] m_X	m_Y	m_Z
1	LC1	59	0.000	0.00	14.03	0.00	0.00	0.00	0.00
			0.258	0.00	3.59	0.00	0.00	0.00	0.00
			0.517	0.00	-0.90	0.00	0.00	0.00	0.00
			0.775	0.00	-6.33	0.00	0.00	0.00	0.00
			1.033	0.00	-13.41	0.00	0.00	0.00	0.00
			1.292	0.00	-20.89	0.00	0.00	0.00	0.00
		37	1.550	0.00	-37.96	0.00	0.00	0.00	0.00
	LC2	59	0.000	0.00	-39.94	0.00	0.00	0.00	0.00
			0.258	0.00	-38.29	0.00	0.00	0.00	0.00
			0.517	0.00	-36.34	0.00	0.00	0.00	0.00
			0.775	0.00	-34.37	0.00	0.00	0.00	0.00
			1.033	0.00	-36.21	0.00	0.00	0.00	0.00
			1.292	0.00	-41.32	0.00	0.00	0.00	0.00
		37	1.550	0.00	-74.27	0.00	0.00	0.00	0.00
	CO1	59	0.000	0.00	-36.27	0.00	0.00	0.00	0.00
			0.258	0.00	-48.57	0.00	0.00	0.00	0.00
			0.517	0.00	-52.13	0.00	0.00	0.00	0.00
			0.775	0.00	-56.98	0.00	0.00	0.00	0.00
			1.033	0.00	-69.48	0.00	0.00	0.00	0.00
			1.292	0.00	-87.11	0.00	0.00	0.00	0.00
		37	1.550	0.00	-157.12	0.00	0.00	0.00	0.00
	CO4	59	0.000	0.00	14.97	0.00	0.00	0.00	0.00
			0.258	0.00	2.41	0.00	0.00	0.00	0.00
			0.517	0.00	-2.72	0.00	0.00	0.00	0.00
			0.775	0.00	-8.87	0.00	0.00	0.00	0.00
			1.033	0.00	-16.85	0.00	0.00	0.00	0.00
			1.292	0.00	-25.33	0.00	0.00	0.00	0.00
		37	1.550	0.00	-44.05	0.00	0.00	0.00	0.00
	CO5	59	0.000	0.00	15.48	0.00	0.00	0.00	0.00
			0.258	0.00	3.14	0.00	0.00	0.00	0.00
			0.517	0.00	-1.91	0.00	0.00	0.00	0.00
			0.775	0.00	-7.88	0.00	0.00	0.00	0.00
			1.033	0.00	-15.34	0.00	0.00	0.00	0.00
			1.292	0.00	-23.10	0.00	0.00	0.00	0.00
		37	1.550	0.00	-38.98	0.00	0.00	0.00	0.00
	CO6	59	0.000	0.00	19.20	0.00	0.00	0.00	0.00
			0.258	0.00	6.50	0.00	0.00	0.00	0.00
			0.517	0.00	0.81	0.00	0.00	0.00	0.00
			0.775	0.00	-6.17	0.00	0.00	0.00	0.00
			1.033	0.00	-15.55	0.00	0.00	0.00	0.00
			1.292	0.00	-25.58	0.00	0.00	0.00	0.00
		37	1.550	0.00	-49.77	0.00	0.00	0.00	0.00
	CO7	59	0.000	0.00	19.71	0.00	0.00	0.00	0.00
			0.258	0.00	7.23	0.00	0.00	0.00	0.00
			0.517	0.00	1.63	0.00	0.00	0.00	0.00
			0.775	0.00	-5.18	0.00	0.00	0.00	0.00
			1.033	0.00	-14.04	0.00	0.00	0.00	0.00
			1.292	0.00	-23.34	0.00	0.00	0.00	0.00
		37	1.550	0.00	-44.70	0.00	0.00	0.00	0.00
	CO8	59	0.000	0.00	8.89	0.00	0.00	0.00	0.00
			0.258	0.00	-0.57	0.00	0.00	0.00	0.00
			0.517	0.00	-4.10	0.00	0.00	0.00	0.00
			0.775	0.00	-8.24	0.00	0.00	0.00	0.00
			1.033	0.00	-13.59	0.00	0.00	0.00	0.00
			1.292	0.00	-19.32	0.00	0.00	0.00	0.00
		37	1.550	0.00	-31.17	0.00	0.00	0.00	0.00
	CO9	59	0.000	0.00	9.91	0.00	0.00	0.00	0.00
			0.258	0.00	0.89	0.00	0.00	0.00	0.00
			0.517	0.00	-2.47	0.00	0.00	0.00	0.00
			0.775	0.00	-6.26	0.00	0.00	0.00	0.00
			1.033	0.00	-10.56	0.00	0.00	0.00	0.00
			1.292	0.00	-14.85	0.00	0.00	0.00	0.00
		37	1.550	0.00	-21.02	0.00	0.00	0.00	0.00
	CO10	59	0.000	0.00	17.35	0.00	0.00	0.00	0.00
			0.258	0.00	7.61	0.00	0.00	0.00	0.00
			0.517	0.00	2.96	0.00	0.00	0.00	0.00
			0.775	0.00	-2.85	0.00	0.00	0.00	0.00
			1.033	0.00	-10.98	0.00	0.00	0.00	0.00
			1.292	0.00	-19.81	0.00	0.00	0.00	0.00
		37	1.550	0.00	-42.60	0.00	0.00	0.00	0.00
	CO11	59	0.000	0.00	18.37	0.00	0.00	0.00	0.00